

IQ and Aptitude Test

1. Verbal reasoning (series, analogy, classification, coding-decoding, insert the missing character, common property, matrices, direction and distance sense test, ranking order, assertion and reason, induction, deduction, courses of action, etc) $6 \times 1 = 6$ Marks
2. Non-verbal reasoning (series, analogy, classification, Venn diagrams, matrices, construction of squares and triangles, and figure formation and analysis, dot situation, water images, mirror images, embedded figures, etc.) $6 \times 1 = 6$ Marks
3. Quantitative aptitude (arithmetical reasoning / operation, percentage, fraction, decimal, ratio, average, loss, profit, date/calendar, time and work, etc.) $6 \times 1 = 6$ Marks
4. General mental ability, logical reasoning and analytical ability $6 \times 1 = 6$ Marks



Chapter 1 - Verbal reasoning

Sub-Chapter 1 - Sequence and Series

Sub-Chapter 2 - Analogy/ common property/ similarity

Sub-Chapter 3 - Classification/ Odd one out

Sub-Chapter 4 - Coding/ Decoding

Sub-Chapter 5 - Insert missing character

Sub-Chapter 6 - Matrix

Sub-Chapter 7 - Distance and directions

Sub-Chapter 8 - Ranking order, Seating Arrangement and Alphabet Test

Sub-chapter 9 - Clock, Date and calendar

Sub-chapter 10 - Mensuration and Area, Volume

Sub-chapter 11 - Arithmetical reasoning and operation

Sub-Chapter 12- Assertion and reason and Syllogism

Sub-Chapter 13 - Induction and deduction

Sub-Chapter 14 - Statements and Course of action

Sub-Chapter 15 - Statement and Assumptions

Sub-Chapter 16 - Statement and Conclusion

Sub-Chapter 17 - Situation Judgements Test/ Making Judgements

Sub-Chapter 18 - Cause and Effects

Chapter 2 - Non-verbal reasoning

Sub-Chapter 1 - Sequence and Series

Sub-Chapter 2 - Analogy

Sub-Chapter 3 - Classification

Sub-Chapter 4 - Venn diagram

Sub-Chapter 5 - Matrix

Sub-chapter 6 - Construction of squares and triangles

Sub-chapter 7 - Figure Completion, formation, analysis, Grouping figure and Rule detection

Sub-chapter 8 - Dot situation/ Shaded parts

Sub-chapter 9 - Water image/ Mirror image

Sub-chapter 10 - Embedded figures

Sub-chapter 11 - Paper folding and cutting

Sub-chapter 12 - Cubes and Dices

Sub-chapter 13 - Analytical Reasoning

Sub-chapter 14 - Fault-Finding Test

Chapter 3 - Quantitative aptitude

Sub-chapter 1 - Percentage

Sub-chapter 2 - Decimal and Fraction

Sub-chapter 3 - Ratio and proportion

Sub-chapter 4 - Alligation or Mixture
Sub-chapter 5 - Average
Sub-chapter 6 - Profit and loss
Sub-chapter 7 - Simple Interest
Sub-chapter 8 - Compound Interest
Sub-chapter 9 - Population Growth and depression
Sub-chapter 10 - Partnership
Sub-chapter 11 - Number system Natural number and Prime Number
Sub-chapter 12 - Square Number/Root and Cube Number/Root
Sub-chapter 13 - Algebra numerical system
Sub-chapter 14 - Time and work
Sub-chapter 15 - Chain Rule
Sub-chapter 16 - Problems on Pipes and Cistern
Sub-chapter 17 - Problems on Ages
Sub-chapter 18 - Permutation and Combination
Sub-chapter 19 - Probability
Sub-chapter 20 - Game and Race or Time and Distance
Sub-chapter 21 - Height and Distance
Sub-chapter 22 - Problems on Trains and speed
Sub-chapter 23 - Boats and Streams

Chapter 4 - General mental ability, logical reasoning and analytical ability

Sub-chapter 1 - Data Verification, Checking and Truth
Sub-chapter 2 - Substitution
Sub-chapter 3 - Confusing Instruction Test
Sub-chapter 4 - Jumble words/ Ordering of words
Sub-chapter 5 - Logical Sequence of Words
Sub-chapter 6 - Essential Parts
Sub-chapter 7 - Word Formation Test
Sub-chapter 8 - Blood relations

Chapter 1 - Verbal reasoning

Sub-Chapter 1 - Sequence and Series

- **Natural number** $N = 1, 2, 3, 4, \dots$
- **Whole Number** $W = 0, 1, 2, 3, 4, \dots$
- **Integers** = All positive + Negative numbers
- **Rational number** = Can be represent in fraction of two whole number, $\frac{p}{q}$ where $q \neq 0$
- **Irrational number** = Cannot be represent in fraction of two whole number
- **Real number** = All rational + Irrational number

AP	GP	HP
$t_n = a + (n - 1)d$	$t_n = ar^{n-1}$	
$S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} (a + b)$	$S_n = \frac{ar^n - 1}{r - 1}$	
$S_n = \frac{n}{2} (n + 1)$ For natural number		
$S_n = n^2$ For odd number		
$S_n = n(n + 1)$ For even number		
$S_n = \frac{n(n+1)(2n+1)}{6}$ For square number		
$S_n = \left[\frac{n(n+1)}{2}\right]^2$ For cube number		

- $x^2 + 1$ (10, 17, 26, 37, 50, 65)
- $x^2 - 1$ (0, 3, 8, 15, 24, 35, 48)
- $x^2 + x$ (12, 20, 30, 42, 56, 72)
- $x^2 - x$ (43, 30, 20, 12, 6)
- $x^3 - 1$ (7, 26, 63, 124, 215, 342, 511)
- **Square of prime number** (4, 9, 25, 49, 121, 169, 289, 361)
- **Arranging words in meaningful order** (May be in function/ process/ nature/ size/ hierarchy)
 - Eg. - 1. Death 2. Marriage 3. Education 4. Birth 5. Funeral (4-3-2-1-5)
- **Ascending or descending order** मा **Logical letters/ words** हुनसक्ने
- **Words** मा कुनै अक्षर हटाउन सकिने
 - Eg. - UMBRELLA - MBRELLA-BRELLA-RELLA
- **Repetition series** : N^{th} letter in A, AB, ABC, ABCD
 - First of all - know the pattern (Eg. - 1, 2, 3, 4,)
 - Set formula as pattern -
- **For (Eg. - 1, 2, 3, 4,) pattern**
 - 50th letter : Nearest N^{th} number below 50 ($9^{\text{th}} = 45$), Remaining number = 5, So ABCDE
 - 100th letter : Nearest N^{th} number below 100 ($13^{\text{th}} = 91$), Remaining number = 9, So ABCDEFGHI
 - $N^{\text{th}} = 9$ भएको बेला अब बाँकी १० वटा सम्म अक्षर लेख्न सकिने हुन्छ
- **We can solve all pattern by the formula** : If difference = 1, $S_n = \frac{n}{2} (n + 1)$ If difference = 2, $S_n = n (n + 1)$ If difference = 3, $S_n = \frac{3n}{2} (n + 1)$
- **Sequence of place value of letters in regular or opposite order**

A	B	C	D	E	F	G	H	I	J
Z	Y	X	W	V	U	T	S	R	Q
1 or 26	2 or 25	3 or 24	4 or 23	5 or 22	6 or 21	7 or 20	8 or 19	9 or 18	10 or 17

- Eg. - B - D - F - H (+2, +2, +2)
- Eg. - B - D - G - K - P (+2, +3, +4, +5)
- $N \pm 1, N \pm N, N^2 \pm 1, N^2 \pm N, N^3 \pm 1, N^3 \pm N$ आदीको स्वरूपमा हुनसक्ने
- May be series of first letters of numbers, day, week, month name
- **समान अन्तर हुने बढ्दो वा घट्दो क्रममा** हुनसक्ने
- **समान अन्तर नहुने बढ्दो वा घट्दो क्रममा** हुनसक्ने

- ज्यामितीय सम्बन्ध हुने बढ्दो वा घट्दो क्रममा हुनसक्ने
- Series of prime or composit number
- Series of square numbers (1, 4, 9, 16, 25, 49, 64, 81, 100, 121, 144, 169)
- Series of cube numbers (1, 8, 27, 64, 125)
- The Fibonacci series (The next number is found by adding up the two numbers before it)
- Eg. - 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ... (next number in the sequence above is $21+34 = 55$)
- Repeated and continuous series/ Letter series -
- Two line number series.

Sub-Chapter 2 - Analogy/ common property/ similarity

- Choosing pair of words with correct maning
- Find relation of pair of given words/ numbers in 5W+H pattern
- Find positive and negative relation of pair of given words
- Be careful in given relation of pair of words/numbers with Necessary pair of words/numbers
- Relation with habitat/ special couple/ measurements/ units/ small pieces/ capital city/ offsprings/ groups/ male-female/ national signs/ inventors/ disease and cause/ logy/ religions/ sounds/ games/ authors
- May be specific name (Jupiter : Planet → Mt. Everest : Himalaya) - Biggest/ Highest
- No necessary to calculation in all situations
- Eg. - $\underline{565} \rightarrow 55$, Then $865 \rightarrow 85$
- Relation of sum of all digits - Eg. $17231 : 14 :: 29428 : (?)$ (25)
- Relation of multiplication of all digits(432 → 24)
- Sum of all digit will be same in both parts
- May be Fraction or decimal of consecutive prime number.($\frac{7}{11} \rightarrow \frac{13}{17}$)
- Difference of multiplication and addition.(97 → 47 where $(9 \times 7) - (9 + 7) = 47$)
- Division of self digit(183 → 6)
- Reverse after multiplication(7 → 41)
- Square of digit(132 → 194)
- Sum of square of digit.(14 → 17, 72 → 81)
- Difference/ sum of reverse of digit
- Jumping sum of digit. (1864 → 712 where, $1 + 6 = 7$, $8 + 4 = 12$)
- Multiplication and making new digit. (9386 → 2748 where, $9 \times 3 = 24$ and $8 \times 6 = 48$)
- Multiplication to all digit by any number. (2748 → 414816)
- Sum/ Difference with each other will be equal in both parts ... (7854 - 2222 = 5362)
- Certain forms

Normal	Square	Cube	Mixed
$x : x \pm 1$ (5 → 6)	$x : x^2$ (6 → 36)	$x : x^3$ (6 → 216)	$x^2 : x^3$ (36 → 216)
$x : x \pm n$ (10 → 15)	$x^2 : x \pm 1$ (121 → 12)	$x^3 : x \pm 1$ (27 → 4)	$x : x^2 \pm x^3$ (2 → 12)
$x : \frac{2x}{5}$ (60 → 24)	$x : x^2 \pm n$ (6 → 39)	$x : x^3 \pm n$ (4 → 68)	$x^2 : (x \pm 1)^3$ (9 → 8)
	$x : (\frac{x}{10})^2$ (60 → 36)	$x : p^3 + p^3$ (23 → 35)	$x^2 : x^3 + 1$ (9 → 28)
	$x : \frac{x^2}{2}$ (12 → 72)	$x : \frac{x^3}{2}$ (8 → 256)	$x : 2x^2$ (9 → 162)
	$x : p^2 + p^3$ (23 → 13)	$x^3 - x = p^3 - p$ (24 → 60)	
	$x : x^2 + x$ (3 → 12)		
	$(x + y)^2 : n^2$ (34 → 49)		

- Finding similar group like given number group.

Sub-Chapter 3 - Classification/ Odd one out

- Find the odd man out pattern
 - Cancel the unmatched alternatives consecutively
 - Check prime number/ consecutive prime
 - Check odd or even number
 - Square or cube numbers ($\pm n$)
 - Divisibility by certain number.
 - Sum/multiplication of all digit may be equal
 - Vowel sounds/ consonant sounds
 - Mirror image/ water image
 - Rotation matching
 - Alphabet gaping/ place value of alphabet/ addition
 - Same letter in first and last
 - Birds/ mammals/
 - Month of equal days/ metal-nonmetals
 - May be in table relations
- Eg. - A, B, C, D passed in physics, chemistry, drama, english

	A	B	C	D
Physics	√	×	×	√
Chemistry	×	√	×	√
Drama	√	√		√
English	√	×	√	×

❖ **Find the odd man out.**

- 3, 5, 11, 14, 17, 21 → 14 (Even number)
- 8, 27, 64, 100, 125, 216, 343 → 100 (Cube number)
- 10, 25, 45, 54, 60, 75, 80 → 54 (54 isn't Multiples of 5)
- 396, 462, 572, 427, 671, 264 → 427 ($4 + 7 \neq 2$)
- 6, 9, 15, 21, 24, 28, 30 → 28 (28, isn't a multiple of 3)
- 2, 5, 10, 17, 26, 37, 50, 64 → $((1*1)+1, (2*2)+1, (3*3)+1, (4*4)+1, (5*5)+1, (6*6)+1, \dots)$
- 331, 482, 551, 263, 383, 362, 284 → 383 ($3 \times 3 \neq 8$)

Sub-Chapter 4 - Coding/ Decoding

- **One on one letters/ numbers system coding-decoding**
 - Step forward/ backward/ Reverse
 - Ascending/ descending
- Coding/ Decoding after making reverse order.
- Letters/ words jumping with taking certain letters/ group
- Making letters opposite and In real form and vice-versa
- Matching letters value by forward/ backward
- Pattern may be in $\frac{\text{Vowel}}{\text{Consonant}}$ or $\frac{\text{Vowel}}{\text{Total}}$.
- **Deciphering message coding** - Eg. (common in two or more column)

Sun shine brightly → <u>ba</u> lo sul	Then, Sun = lo
Houses are brightly lit → kado ula ari <u>ba</u>	brightly = ba
Light comes from Sun → dopi kup <u>lo</u> nro	

- **Coding by sum of letters value**
Eg. - G = 7, BAG = $2 + 1 + 7 = 10$, GIFT = $7 + 9 + 6 + 20 = 42$
- **Summation or multiplication of value of letters and dividing by number of letters in words**

Eg. - $E = 5/1 = 1$

HOTEL = $8 + 15 + 20 + 5 + 12 = 60/5 = 12$

- **Matrix type coding-decoding -----First (↓) -----See Left then Right.**

↓	0	1	2	3	4		↓	5	6	7	8	9
0	F	O	M	S	R		5	A	T	D	I	P
1	S	R	F	O	M		6	I	P	A	T	D
2	O	M	S	R	F		7	T	D	I	P	A
3	R	F	O	M	S		8	P	A	T	D	I
4	M	S	R	F	O		9	D	I	P	A	T

- Be careful that watch by row first then column. (Style may be mentioned)
- Search by first letter, then second consecutively.
- Eg. - MOST : 02, 13, 34, 56
- **Coding-decoding may be by noticing friends, card partners etc. – All are equal valued.**
Eg. In a game of cards, GEORGE partnered MARY. Then TED partnered whom?
Form Given, TED = ANN
GEORGE = $7 + 5 + 15 + 18 + 7 + 5 = 57 \rightarrow$ MARY = $13 + 1 + 18 + 25 = 57$
TED = $20 + 5 + 4 = 29 \rightarrow$ ANN = $1 + 14 + 14 = 29$

- **MUSCLE [SNORED] ROTUND**

DILUTE [?] VALUES $\rightarrow$ LEAVES (Finding third row from fourth row)

M	U	S	C	L	E	S	N	O	R	E	D	R	O	T	U	N	D
		1			5	1	2	3	4	5	6	4	3			2	6
D	I	L	U	T	E	L	E	A	V	E	S	V	A	L	U	E	S
		1			5							4	3			2	6

- **ILEUM [AUTOMOBILE] TABOO**
ELITE [?] PORCH $\rightarrow$ HELICOPTER (From rearranging)
- **GOSPEL [MILES] IMPOSE**
PSEUDO [?] TARTAN $\rightarrow$ ATONE (Finding third row from fourth row method)
- **If there is given two or more then answer must be search from given too.**
- **If Z = 2197 and R = 729. How would J be written in that code? $\rightarrow 125$**
Z Code $\Rightarrow 26 \Rightarrow 26/2 \Rightarrow 13^3 = 2197$ R Code $\Rightarrow 18 \Rightarrow 18/2 \Rightarrow 9^3 = 729$
Similarly, J Code $\Rightarrow 10 \Rightarrow 10/2 \Rightarrow 5^3 = 125$
- **If E = 5 and READ is coded as 7, then what is the code of 'DEAR' ? $\rightarrow 7$**
Here $E = 5 = 5/1 = 5$
 $\Rightarrow$ READ = $18 + 5 + 1 + 4 = 28/4 = 7$
 $\Rightarrow$ DEAR = $4 + 5 + 1 + 18 = 28/4 = 7$
- In this question, there is a word written in capital letter, with one letter underlined. For each letter in that word there is a code written. That code is denoted by either 1, 2, 3, 4 or 5 not in the same order. you have to find out the exact code for the underlined letter in the word. The number of that code is the answer. $\rightarrow$ C
E L I E S
- If $7\Omega 6 = 84$, $8\Omega 7 = 112$ and $8\Omega 4 = 64$, then find the value of $3\Omega 4 = ? \rightarrow 24$

Sub-Chapter 5 - Insert missing character

- In triangular, straight line, polygonal vertex, space matrix structure
- In Puzzles solution : Problems based on restriction मा विकल्पलाई प्रतिबन्धात्मक नियमहरुसँग तुलना गरेर हेर्ने। हरेक एक विकल्पको हेरेर मात्रै अर्को विकल्पमा जाने। (Donot break the restrictions.)

What number comes inside the circle? → 6 $(4 + 2) / 2 = 3$	→ 9 $(7 + 9) - 4 = 12$	→ 19 Prime Numbers.	→ 39 $3 \times 2 - 1 = 5$ (1, 2, 3 etc.)
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→ 6 $9 + 4 + 2 = 15$ (All)	→ 9 All diagram sum = 22	→ 16 decreases by 1, 2, 3.....	→ 7
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→ 13 Increasing same....	→ 1 Row = $17^2, 18^2, 19^2$	→ 8, 1	Odd Out : A:58 B:86
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→ 1 $36, 49, 64, 81$	→ 253 $(1 \times 2) + 3 = 5$ so on...	→ 23 Prime Numbers 2-23	→ 10 $(11+9+2)/2 = (4+1+6)$
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→ 10 $(4 + 2 + 7) - (3+1) = 9$	→ 1143 $1, 3, 5, 7, 9, 11$ $9, 8, 7, 6, 5, 4$ So on..	→ 6
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→ 5	→ 15	→ 7
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Sub-Chapter 6 - Matrix

- Sum of Row and column may be equal.
- Sum of Row and column any group may be equal with any other group.
- Multiplication/ Division by any number.
- Digital non-recurring.

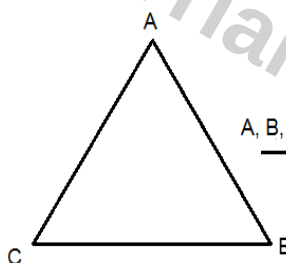
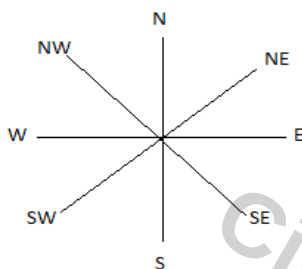
1	3	7
5	12	14
25	?	28

→ 8281

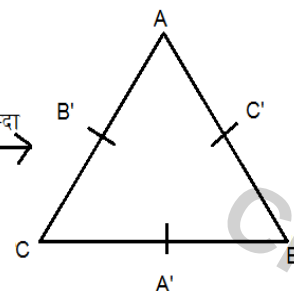
<table border="1"> <tr><td>31</td><td>17</td><td>58</td><td>87</td></tr> <tr><td>68</td><td>19</td><td>61</td><td>56</td></tr> <tr><td>91</td><td>22</td><td>70</td><td>50</td></tr> <tr><td>10</td><td>142</td><td>11</td><td>?</td></tr> </table> → 7 (31+68+91+10=200)	31	17	58	87	68	19	61	56	91	22	70	50	10	142	11	?	<table border="1"> <tr><td>7</td><td>2</td><td>36</td><td></td></tr> <tr><td>5</td><td>6</td><td>2</td><td>8</td></tr> <tr><td>3</td><td>8</td><td>1</td><td>9</td></tr> <tr><td>8</td><td>4</td><td>4</td><td>?</td></tr> </table> → 2 72/2 = 36	7	2	36		5	6	2	8	3	8	1	9	8	4	4	?	<table border="1"> <tr><td>7</td><td>3</td><td>6</td><td>2</td></tr> <tr><td>2</td><td>8</td><td>5</td><td>4</td></tr> <tr><td>1</td><td>1</td><td>2</td><td>4</td></tr> <tr><td>4</td><td>2</td><td>1</td><td>?</td></tr> </table> → 4 7 + 2 + 1 + 4 = 14 (All)	7	3	6	2	2	8	5	4	1	1	2	4	4	2	1	?	
31	17	58	87																																																
68	19	61	56																																																
91	22	70	50																																																
10	142	11	?																																																
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3	8	1	9																																																
8	4	4	?																																																
7	3	6	2																																																
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1	1	2	4																																																
4	2	1	?																																																

- Sum of Row and column any group may be equal with average of these group.
- Sum of Row and column any group may be square or cube number.

Sub-Chapter 7 - Distance and directions



A, B, C भुजा Clockwise 1.5 घुम्दा



- Changing the direction → Substitution the direction
- Regular rotate in rotation in degree
- Regular rotate in clockwise or anticlockwise direction
- Use $h^2 = p^2 + b^2$ for the expression : Man walking
- For playing cards, Position of player is always in E-W-N-S (opposite) direction
- If turning 45° in right or left make angle with forward direction.
- Shadow making distance and direction (Morning has always R-S)
- कुनै दिशामा फर्केकोमा क्रमशः घुम्दा - कुनै दिशाबाट लिइएका दायाँ र बायाँ घुमाईको संख्या बारबर भएमा व्यक्ति पुनः शुरुकै दिशातर्फ हुन्छ।

विहान (छायाँ सधैं पश्चिमतर्फ पर्छ)	साँझ (छायाँ सधैं पूर्वतर्फ पर्छ)
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मानिस फर्केको	छाँया	मानिस फर्केको	छाँया
पूर्व	पछाडी/ पश्चिम	पूर्व	अगाडी/पूर्व
पश्चिम	अगाडि/पश्चिम	पश्चिम	पछाडि/पूर्व
उत्तर	वायाँ	उत्तर	दायाँ
दक्षिण	दायाँ	दक्षिण	वायाँ

Sub-Chapter 8 - Ranking order, Seating Arrangement and Alphabet Test

❖ Ranking order

- Dropped - One step backward
- कुनै दिइएको संख्यालाई वर्गमा राख्दा -
- दिएको संख्यामा ४ घटाउने (यो वर्गको चार कुनामा राख्ने संख्या हो)
- बाँकी संख्यालाई ४ ले बराबर भाग लगाउने
- **In a group of 3 member out of group of Ram, Shyam, Raju, Amrit, Rohit, Dinesh**
- Ram - Shyam एक समूहमा नहुने → Ram - Shyam (×)
- Raju - Amrit एकसाथ अनिवार्य हुने → Raju - Amrit (✓)
- Rohit - Dinesh उही समूहमा नहुने → Rohit - Dinesh (×)
त्यसैकारण - समूहमा हुनुपर्ने व्यक्ति - Raju - Amrit (Care from option)
- **Conform the overlapping number first.**
- Finding by total number counting.
- यदि कुनै लाइनमा व्यक्तिको न्युनतम संख्या पत्ता लगाउनु परेको रहेछ भने overlapping भएको हुन्छ
- overlapping भएको निश्चित भएमा दिइएको भन्दा अर्को पक्षको समेत स्थिति पत्ता लगाउने
- After merging the section, new ranking = Sum of number of students of all class before him/her.
- Total number members in ranking order = a + X + b + Y + c + Z + d



❖ Alphabet Test

- **Preceded → Given → Followed**
- Sunday → Monday → Tuesday (Monday is preceded by Sunday and followed by Tuesday)
- **Pair of letter** in the given words, each of which has as many letters many letters between them in the words as in the English alphabet. Eg.- **STORM**

Alphabet	S	T	O	R	M
Positional value	19	20	15	18	13

From left side - ST

From right side - MO (i.e. 2 Pairs)

- As far away from the beginning means - In the same position of letters
- **Question** : Which letter will come 3rd to the left of 11th from left

3 rd to the left - 3L	Since, Both have L/L	Same/ same letters = subtraction
11 th from left - 11 L	11L - 3L = 8L = H (Value of A to Z)	Different latters = addition
If in a result L then count - A to Z Otherwise in a result R then count - Z to A)		

- **Changing the place of letters in the words**
- **If Same position of letters is concerned - then, Count alphabetically/ Put alphabetical value**

❖ Analytical logical order

- Write in order to the probable event
- Make the natural and meaningful order of letters which are given in disorder

- Make appropriate order (Eg.- first of all buy a cow and then cut grass)

Sub-chapter 9 - Clock, Date and calendar

❖ Clock

- Angle made by hour hand in 1 min. = 0.5°
- Angle made by minute hand in 1 min. = 6°
- Minute hand rotate 55 min. faster than hour hand in 1 hour. So, minute hand gains 55 minute on hour hand in 1 hour.
- Minute hand and hour hand **can't be Concide** in between **12:00 - 1:00**
- Minute hand and hour hand **can't be Opposite with each other** in between **6:00 - 7:00**
- Minute hand and hour hand **can't be Right angled with each other** in between **3:00 - 4:00 and 9:00 - 10:00**.

Minute and hour hands			
	In 1 hour	In 12 hours	In 24 hours
Concide	1 time	11 times	22 times
Opposite with each other	1 time	11 times	22 times
Right angled with each other	2 times	22 times	44 times

- When two hand are in right angled they are in 15 minute apart from each others.
- When two hand are in opposite direction they are in 30 minute apart from each others.
- Angle between two hands in h hours and m minutes. = $\frac{60h-11m}{2}$ degree
Eg. for 5 : 30 o'clock = $\frac{60 \times 5 - 11 \times 30}{2} = 15$ degree
- **Reflex angle = 360° - Angle between them.**
- The reflex angle between the hands of a clock at 10.25 is = 360° - Angle between them. = $260^\circ - 162.5^\circ = 197.5^\circ$
- **Angle traced by hour hand = Travelling hour $\times$ 30**
- A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:
Angle traced by hour hand in 5 hrs 10 min or $(\frac{31}{6} \text{ hr.}) = \frac{31}{6} \times 30 = 155^\circ$
- **Multiply the calculated minute by $\frac{60}{55}$ to find out the Minute to become concide/ right angle/ opposite direction/ certain degree.** (Make watch and count necessary minute)
Eg. Let hour hand be in 5:00 and minute hand be in 12:00, To concide minute hand must travel 25 min. So, two hands between 5 o'clock and 6 o'clock. = $25 \text{ minute} \times \frac{60}{55} = 27 \frac{3}{11}$
The become concide in 5 past $27 \frac{3}{11}$ minutes
- **Time of Distance between minute hand and hour hand (M) = $\frac{60}{55} \times (5x \pm m)$** , X and y are Two given time.
Eg. **At what time between 2 and 3 o'clock, will the hands of clock be 4 miute apart ?**

$\frac{60}{55} \times (5 \times 2 + 4) = 15 \frac{3}{11}$ minutes.	$\frac{60}{55} \times (5 \times 2 - 4) = 6 \frac{6}{11}$ minutes.
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- **How much angle minute hand rotate at 02:20 ? (Angle = Minute $\times$ 6°)**
Angle = $140 \times 6 = 540^\circ$
- **At what time between 5.30 and 6 will the hands of a clock be at right angles? $\rightarrow (43 \frac{5}{11})$**
- **At what angle the hands of a clock are inclined at 15 minutes past 5? $\rightarrow (67 \frac{1}{2})$**
- **How many times in a day, the hands of a clock are *straight*?** (Coincide + opposite = 44)
- **A watch which gains 5 seconds in 3 minutes was set right at 7 a.m. In the afternoon of the same day, when the watch indicated quarter past 4 o'clock, the true time is: (4:00)**
- **A watch which gains uniformly is 2 minutes low at noon on Monday and is 4 min. 48 sec fast at 2 p.m. on the following Monday. When was it correct? (2 p.m. on Wednesday)**

Correct clock	False clock	Calculation
3 min. = 180sec.	3min. 5sec. = 185 sec.	7 am to 4 : 15 = 9 hr. 15 minute = 555 min
x	555 min.	$x = \frac{180\text{sec.} \times 555 \text{ min.}}{185 \text{ sec.}} \rightarrow x = 9 \text{ hr. (7 to 4 = 9 hr)}$

- **How much does a watch lose per day, if its hands coincide every 64 minutes?**

55 min. spaces are covered in 60 min.

1 min. spaces are covered in $\frac{60}{55}$ min.

60 min. spaces are covered in $\frac{60}{55} \times 60 \text{ min.} = 65\frac{5}{11} \text{ min.}$

Loss in 64 min. = $65\frac{5}{11} \text{ min} - 64 \text{ min.} = \frac{16}{11} \text{ min.}$

Loss in 24 hr. = $\frac{16}{11} \times \frac{1}{64} \times 24 \times 60 = 32\frac{8}{11} \text{ min.}$

❖ **Date and calendar**

- Make circular disc for finding weeks/ days.
- Ordinary years = 52 weeks + 1 day $\rightarrow$ 1 odd day
- Leap years = 52 weeks + 2 days (February must be in 29 days) $\rightarrow$ 2 odd days
- Leap year must be divisible by 4 for last two digit or 400 for century year.
- First day of century cannot be : Sunday $\rightarrow$ Wednesday $\rightarrow$ Friday
- Last day of century cannot be : Tuesday $\rightarrow$ Thursday $\rightarrow$ Saturday (T)
- First/ Last day of century can be : Monday
- **Finding the days from given date**

Century	Century code	• Month code : 144 - 025 - 036 - 146 (From January to December) • Leap year (January = 0 and february = 3) when month crossed • Leap year = Year divided by 4 • Five point must be known to find days from date. 1. Century code number 2. Year number 3. Leap year (Year number divided by 4) 4. Month code number 5. day number
1500-1500	0	
1600-1699	6	
1700-1799	4	
1800-1899	2	
1900-1999	0	
2000-2099	6	
2100-2199	4	
2200-2299	2	

- **Eg. - 2020 January 1**
 1. Century code number = 6 (For 2000-2099)
 2. Year number = 20
 3. Leap year = 5
 4. Month code number = 0 (for January in leap year)
 5. day number = 1
 6. Sum of total data = $6 + 20 + 5 + 0 + 1 = 32$ (Then $32/7 = 4$ and remainder = 4, therefore Wednesday)
 7. Value for sunday = 1, Monday = 2....., Saturday = 0
- Day of **same year and same month codeed month** is same days of : **(No same days in May/ June/ October) in ordinary year (Eg. 2019)**

January - October	February - March - November	April - July	Septmber - December
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- Day of **same year and same month codeed month** is same days of : **(No same days in May/ June/ August) in leap year (Eg. 2016)**

January - April - July	February - August	March - November	Septmber - December
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• **Odd days**

- 100 years = 5 odd days
- 200 years = 3 odd days
- 300 years = 1 odd days
- 400 years = 0 odd days

- **Same Calendars (28 → 6 → 11 → 11)**
- **In leap years** = Add 28 years (Eg. 2004, 2032, 2060)
- **After first leap year** = Add 6 (Eg. 2005, 2011)
- **After second/ third leap year** = Add 11 (Eg. 2006, 2017)
- If any day is Monday and then, 100th day of that day = $100/7 = 14$ and remainder = 2, Therefore Monday +1 = Tuesday (Taking 2-1)
- **Today is Monday. After 61 days, it will be :**
 $\frac{61}{7} \rightarrow$ Remainder = 5, So Tuesday, Wednesday, Thursday, Friday, **Saturday**
- **On what dates of April, 2001 did Wednesday fall?** (First find the day on 1st April, 2001)
- **How many days are there in x weeks x days?** ($7x + x = 8x$)
- **On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?** → Sunday (Before February 29 we take 2 odd days in leap year)

Sub-chapter 10 - Mensuration and Area, Volume

- The point where the three medians of a triangle meet, is called centroid. The centroid divided each of the medians in the ratio 2 : 1.
- The area of the triangle formed by joining the mid-points of the sides of a given triangle is one-fourth of the area of the given triangle.
- **Volume of cube** = $(l \times b \times h)$ cubic units.
- **Surface area of cube** = $6a^2$ sq. units and diagonal = $\sqrt{3}a$ units.
- **For cylinder**, Let radius of base = r and Height (or length) = h . Then,
 Volume = $(\pi r^2 h)$ cubic units.
 Curved surface area = $(2\pi rh)$ sq. units.
 Total surface area = $2\pi r(h + r)$ sq. units.
- **For cone**, Let radius of base = r and Height = h . Then,
 Slant height, $l^2 = h^2 + r^2$
 Volume = $(\frac{1}{3}\pi r^2 h)$ cubic units.
 Curved surface area = (πrl) sq. units.
 Total surface area = $(\pi rl + \pi r^2)$ sq. units.
- **For sphere**, Let the radius of a hemisphere be r . Then,
 Volume = $(\frac{4}{3}\pi r^3)$ cubic units.
 Total surface area = $(4\pi r^2)$ sq. units.
- **For hemisphere**
 Volume = $(\frac{2}{3}\pi r^3)$ cubic units.
 Curved surface area = $(2\pi r^2)$ sq. units.
 Total surface area = $(3\pi r^2)$ sq. units.
- **For room painting Surface Area of room (A) = $2(lb + bh + lh)$**
- **If any formula Satisfies the condition $A = B \times C$,
 By taking constant to A.
 Then Increasing/ decreasing of B by, We must increase/ decrease C by $\frac{R}{R \pm 100}$**
- **If any formula Satisfies the condition $A = B \times C$,
 Change in area (%) = $[(x + y) + \frac{xy}{100}] \%$ [Put (+) for increasing and (-) for decreasing]**

Conditions

- | | |
|--|---------------------------------------|
| 1. आयतको क्षेत्रफल = लम्बाई $\times$ चौडाई | 4. राजश्व = मूल्य $\times$ विक्री |
| 2. बर्ग = लम्बाई $\times$ चौडाई | 5. राजश्व = खरिद $\times$ वस्तुको खपत |
| 3. खर्च = मूल्य $\times$ उपभोग | 6. दुरी = गति $\times$ समय |

7. कार्य = बल $\times$ दूरी

8. कार्य = कामदारको क्षमता $\times$ समय

9. कार्य = कामदारको संख्या $\times$ कामदारको क्षमता

10. कार्य = कामदारको संख्या $\times$ काम गर्ने दिन

11. आयतन = क्षेत्रफल $\times$ उचाई

12. विक्री वापत आय = मूल्य $\times$ विक्री परिणाम

Note :

- Increase or decrease one side by $x\%$: Change in area (%) Always by same or $x\%$
- **With Taking same area** : In squares/ Rectangles **decreasing** any length or breadth by $x\%$, Then, other length or breadth will **increases** by $(\frac{100x}{100-x})\%$
- **With Taking same area** : In squares/ Rectangles **Increasing** any length or breadth by $x\%$, Then, other length or breadth will **decreases** by $(\frac{100x}{100+x})\%$
- If a square is divided into four rectangles, Then $(\frac{x^2}{100} + 2x)\%$
- If all sides of rectangle increases/decreases by $x\%$, then perimeter increases/decreases by $x\%$
- If all radius of circle increases/decreases by $x\%$, then perimeter increases/decreases by $x\%$
- If perimeter of square and circle is same. If area of square is x , then area of circle will be $\frac{4x}{\pi}$
- If ratio of diagonal of two squares is $x : y$, then Ratio of area of these squares = $x^2 : y^2$
- If ratio of area of square is $x : y$, then Ratio of perimeter of square = $\sqrt{x} : \sqrt{y}$
- **If any formula Satisfies the condition $A = B \times C \times D$,**
Change in volume (%) = $[(x + y + z) + \frac{xy+yz+zx}{100} + \frac{xyz}{(100)^2}] \%$
[Put (+) for increasing and (-) for decreasing]

[Put (+) for increasing and (-) for decreasing]

Conditions

1. आयतन = लम्बाई $\times$ चौडाई $\times$ उचाई

• Sum of interior angles of polygon = $180(n-2)$, Where, n = sides of polygon

	Area of B = 15 cm^2 (3×5) Area of C = 48 cm^2 (8×6) So, Area of A = $6 \times 3 = 18 \text{ cm}^2$ So, Area of D = $8 \times 5 = 40 \text{ cm}^2$ (Note : Manage multiples with adding 11, Eg. - $3 + 8 = 11$)
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- **In square walking diagonally across the percent saved by not walking along the edges is 29.28 % or 30% (approx.)** (Hypoteneus of right angle is about 30 % smaller than sum of p and b.)
- **The diagonal of the floor of a rectangular closet is $7\frac{1}{2}$ feet. The shorter side of the closet is $4\frac{1}{2}$ feet. What is the area of the closet in square feet?** (27 sq. feet) ($h^2 = p^2 + b^2$)
- **The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then the area of the park (in sq. m) is:** $P = 2(l + b)$
Area = $(480 \times 320) \text{ m}^2 = 153600 \text{ m}^2$.
- **An error 2% in excess is made while measuring the side of a square. The percentage of error in the calculated area of the square is:** 4.04 %
Difference in area = $(102)^2 - (100)^2 = (102 + 100) \times (102 - 100) = 404 \text{ cm}^2$
Percentage error = $\frac{404}{100 \times 100} \times 100\% = 4.04\%$
- **Area of cross road between garden (A) = $w(l + b - w)$** Where, w = width of crossroad, l = length of garden and b = breadth of garden,

- A rectangular park 60 m long and 40 m wide has two concrete crossroads running in the middle of the park and rest of the park has been used as a lawn. If the area of the lawn is 2109 sq. m, then what is the width of the road? (3 m.)
- What is the least number of squares tiles required to pave the floor of a room 15 m 17 cm long and 9 m 2 cm broad? (814)
- The difference between the length and breadth of a rectangle is 23 m. If its perimeter is 206 m, then its area is: (2520 m²) (Let $l = x$ and $b = x - 23$)
- The length of a rectangle is halved, while its breadth is tripled. What is the percentage change in area? (50%)
- The length of a rectangular plot is 20 metres more than its breadth. If the cost of fencing the plot @ 26.50 per metre is Rs. 5300, what is the length of the plot in metres? (60 m.) let, $l = x$, $b = x - 20$ Then, perimeter = $\frac{5300}{26.50} = 200$ m. Now, Use $P = 2(l + b)$
- A rectangular field is to be fenced on three sides leaving a side of 20 feet uncovered. If the area of the field is 680 sq. feet, how many feet of fencing will be required? (88 m.)
- A tank is 25 m long, 12 m wide and 6 m deep. The cost of plastering its walls and bottom at 75 paise per sq. m, is: (Rs. 558) Area = $lb + 2h(l + b)$
- A right triangle with sides 3 cm, 4 cm and 5 cm is rotated the side of 3 cm to form a cone. The volume of the cone so formed is: (12π cm³)
- In a shower, 5 cm of rain falls. The volume of water that falls on 1.5 hectares of ground is: (750 m³)
- A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of four walls, the volume of the hall is: (1200 m³)
 $15 \times 12 + 15 \times 12 = 15 \times h + 15 \times h + 12 \times h + 12 \times h \rightarrow 2(15 \times 12) = 2h(15 + 12)$
 $\rightarrow h = 20/3$ m So, Volume = 1200 m³
- 66 cubic centimetres of silver is drawn into a wire 1 mm in diameter. The length of the wire in metres will be: (84 m.) - Cylindrical body
Let the length of the wire be h . Radius = $\frac{1}{2}$ mm = $\frac{1}{20}$ cm
Volume = ($\pi r^2 h$) = 66 $\rightarrow h = \frac{7 \times 20 \times 20 \times 66}{22 \times 1 \times 1} \rightarrow h = 84$ m.
- A hollow iron pipe is 21 cm long and its external diameter is 8 cm. If the thickness of the pipe is 1 cm and iron weighs 8 g/cm³, then the weight of the pipe is: (3.696 kg.)
- A boat having a length 3 m and breadth 2 m is floating on a lake. The boat sinks by 1 cm when a man gets on it. The mass of the man is: (60 kg)
Volume of water displaced = $(3 \times 2 \times 0.01)$ m³ = 0.06 m³
Mass of man = Volume of water displaced x Density of water = (0.06×1000) kg = 60 kg.
- 50 men took a dip in a water tank 40 m long and 20 m broad on a religious day. If the average displacement of water by a man is 4 m³, then the rise in the water level in the tank will be: (25 cm.)
Total volume of water displaced = (4×50) m³ = 200 m³
 $\rightarrow$ Rise in water level = $(\frac{200}{40 \times 20})$ m. = 0.25 m. = 25cm (Note : divide all displaced water by area)
- A cistern 6m long and 4 m wide contains water up to a depth of 1 m 25 cm. The total area of the wet surface is: (49 m²)
- A metallic sheet is of rectangular shape with dimensions 48 m x 36 m. From each of its corners, a square is cut off so as to make an open box. If the length of the square is 8 m, the volume of the box (in m³) is: (5120 m³.)
Clearly, $l = (48 - 16)$ m = 32 m (Two sides cut), $b = (36 - 16)$ m = 20 m, $h = 8$ m
Volume of the box = $(32 \times 20 \times 8)$ m³ = 5120 m³.

- The curved surface area of a cylindrical pillar is 264 m² and its volume is 924 m³. Find the ratio of its diameter to its height. (7 : 3)
- A cistern of capacity 8000 litres measures externally 3.3 m by 2.6 m by 1.1 m and its walls are 5 cm thick. The thickness of the bottom is: (1 dm)
- What is the total surface area of a right circular cone of height 14 cm and base radius 7 cm? (498.35 cm²)
- A large cube is formed from the material obtained by melting three smaller cubes of 3, 4 and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube? (25 : 18)
- How many bricks, each measuring 25 cm x 11.25 cm x 6 cm, will be needed to build a wall of 8 m x 6 m x 22.5 cm? (6400)

$$\text{Number of bricks} = \frac{\text{Volume of the wall}}{\text{Volume of 1 brick}} = \frac{800 \times 600 \times 22.5}{25 \times 11.25 \times 6} = 6400$$

Sub-chapter 10.1 - Data Sufficiency from Mensuration and Area, Volume

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- I alone sufficient while II alone not sufficient to answer
 - II alone sufficient while I alone not sufficient to answer
 - Either I or II alone sufficient to answer
 - Both I and II are not sufficient to answer
 - Both I and II are necessary to answer
- **The area of a rectangle is equal to the area of right-angles triangle. What is the length of the rectangle? → D**
 - The base of the triangle is 40 cm.
 - The height of the triangle is 50 cm.
 - **The area of playground is 1600 m². What is the perimeter? → C**
 - It is a perfect square playground.
 - It costs Rs. 3200 to put a fence around the playground at the rate of Rs. 20 per metre.
 - **What is the height of the triangle? → A**
 - The area of the triangle is 20 times its base.
 - The perimeter of the triangle is equal to the perimeter of a square of side 10 cm.
 - **What will be the cost of painting the inner walls of a room if the rate of painting is Rs. 20 per square foot? → E (2R = 44, H = 12, A = 2RH = 44 x 12 → Cost of painting = Rs. (44 x 12 x 20).**
 - Circumference of the floor is 44 feet.
 - The height of the wall of the room is 12 feet.
 - **What is the volume of 32 metre high cylindrical tank? → C**
 - The area of its base is 154 m².
 - The diameter of the base is 14 m.
 - **Is a given rectangular block, a cube? → D**
 - At least 2 faces of the rectangular block are squares.
 - The volume of the block is 64.
 - **What is the capacity of a cylindrical tank? → C**
 - Radius of the base is half of its height which is 28 metres.
 - Area of the base is 616 sq. metres and its height is 28 metres.
 - **What is the height of a circular cone? → D**
 - The area of that cone is equal to the area of a rectangle whose length is 33 cm.
 - The area of the base of that cone is 154 sq. cm.
 - **What is the volume of a cube? → C**
 - The area of each face of the cube is 64 square metres.
 - The length of one side of the cube is 8 metres.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

• **What is the area of the hall? → A**

Material cost of flooring per square metre is Rs. 2.50

Labour cost of flooring the hall is Rs. 3500

Total cost of flooring the hall is Rs. 14,500.

- A. I and II only
- B. II and III only
- C. All I, II and III
- D. Any two of the three
- E. None of these

• **What is the area of a right-angled triangle? → A**

1. The perimeter of the triangle is 30 cm
2. The ratio between the base and the height of the triangle is 5 : 12.
3. The area of the triangle is equal to the area of a rectangle of length 10 cm.

- A. I and II only
- B. II and III only
- C. I and III only
- D. III, and either I or II only
- E. None of these

(Let base = $5x$ and height = $12x$. So, $h = 13x$, $5x + 12x + 13x = 30 \rightarrow x = 1$, So, $A = 30\text{cm}^2$)

• **What is the area of rectangular field? → B**

1. The perimeter of the field is 110 metres.
2. The length is 5 metres more than the width.
3. The ratio between length and width is 6 : 5 respectively.

- A. I and II only
- B. Any two of the three
- C. All I, II and III
- D. I, and either II or III only
- E. None of these

• **What is the area of the given rectangle? → E**

1. Perimeter of the rectangle is 60 cm.
2. Breadth of the rectangle is 12 cm.
3. Sum of two adjacent sides is 30 cm.

- A. I only
- B. II only
- C. I and II only
- D. II and III only
- E. II and either I or III

• **What is the cost painting the two adjacent walls of a hall at Rs. 5 per m^2 which has no windows or doors? → C**

1. The area of the hall is 24 sq. m.
2. The breadth, length and height of the hall are in the ratio of 4 : 6 : 5 respectively.
3. Area of one wall is 30 sq. m.

- A. I only
- B. II only
- C. III only
- D. Either I or III
- E. All I, II and III are required.

• **What is the capacity of the cylindrical tank? → E**

1. The area of the base is 61,600 sq. cm.
2. The height of the tank is 1.5 times the radius.
3. The circumference of base is 880 cm.

- A. Only I and II
- B. Only II and III
- C. Only I and III

- D. Any two of the three
E. Only II and either I or III

Sub-chapter 11 - Arithmetical reasoning and operation

❖ Arithmetical reasoning

• Climbing Polls Problems

Day of reach in top of the poll by ant/ monkey with day to reach top (d) = $\frac{h (\text{Height}) - d (\text{Down})}{u (\text{Up}) - d (\text{Down})}$	(d) = $\frac{h - d}{u - d}$ (Note : If result is 8.4 then, we consider 9 days)
Wall climbing with rest time (d) = $\frac{h - d}{u - d}$ Now, day to reach top (dt) = d + R(d-1)	Where, day to reach top = dt Rest time = R (Unit change is necessary at last)

- A squarrel is climbing up a cylindrical post spirally, making the circuit in 4 feet. If the post is 16 feet high and 3 feet in circumference, how many feet does it travel to the post?

Width/ circuit of path (b) = 4 feet,

Circumference (p) = 3 feet

So, distance traveled in climbing 4 feet (h) = $\sqrt{3^2 + 4^2} = 5$ feet

Distance traveled in climbing 1 feet = $\frac{5}{4}$

Distance traveled in climbing 16 feet = $\frac{5}{4} \times 16 = 20$ feet

- David gets on the elevator at the 11th floor of a building and rides up at the rate of 57 floors per minute. At the same time, Albert gets on an elevator at the 51st floor of the same building and rides down at the rate of 63 floors per minute. If they continue travelling at these rates, then at which floor will their paths cross ?

Total speed = 57 + 63 = 120 floor/minutes

Total rest floor = 51 - 11 = 40 floors

Time to cover 1 floor = $\frac{40}{120} = \frac{1}{3}$ minutes

Number of rest floors crossed by David = $57 \times \frac{1}{3} = 19$, Already passed by him = 11,

So crossed total floor = 19 + 11 = 30th floor

- Total number of hand checking and playing games : $S_n = \frac{n(n-1)}{2}$
- Total number of playing games in knock out system : $S_n = n(n-1)$
- Total number of namaskar : $S_n = n(n-1)$

❖ Sequence and series formula

- Sum of natural number : $S_n = \frac{n(n+1)}{2}$
- Sum of natural odd number : $S_n = n(n+1)$
- Sum of natural even number : $S_n = n^2$

❖ Page number of books

• Page 1 - 9 : 9 digits • Page 10 - 99 : 180 digits • Page 100 - 999 : $900 \times 3 = 2700$ digits • Page 1000 - 9999 : $9000 \times 4 = 36000$ digits	– Book of 300 pages - – Page 1 - 9 : 9 digits – Page 10 - 99 : 180 digits – Page 100 - 300 : $201 \times 3 = 603$ digits – Then, Total digits = 792
• Digit in any number (For 1) – 1 - 9 : 1 – 10 - 19 : 11 – 20 - 100 : 9 Total = 21	• Digit in any number (For 2 and son on) – 1 - 9 : 1 – 10 - 20 : 2 – 21 - 30 : 11 – 30 - 100 : 7 Total = 20

❖ For four-legged or three-legged

- Number of Four-legged ($4L$) = $\frac{L}{dL} - H$, - Number of Two-legged ($2L$) = $H - (4L)$	L = Total legs $4L$ = Number of four-legged dL = Difference of legs H = Number of heads
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❖ **Payment Of hotel**

$Ex = \frac{T}{N-n} - \frac{T}{n}$	Ex = Extra cost T = Total cost N = number of total friends n = Number of friends without purse
For handshakes and league game = $\frac{n(n+1)}{2}$	

- **A university library budget committee must reduce exactly five of eight areas of expenditure—I, J, K, L, M, N, O and P—in accordance with the following conditions:**

- If both I and O are reduced, P is also reduced.
- If L is reduced, neither N nor O is reduced.
- If M is reduced, J is not reduced.
- Of the three areas J, K, and N exactly two are reduced.

If both K and N are reduced, which one of the following is a pair of areas neither of which could be reduced? → J and L

- What do you understand by - 'If K is there L has to be there'** → k is there, then L will also be there
- What is Latus Rectum of Parabola?** → line segment
- What does this symbol mean '&'?** → Ampersand
- A girl counted in the following way on the fingers of her left hand : She started by calling the thumb 1, the index finger 2, middle finger 3, ring finger 4, little finger 5 and then reversed direction calling the ring finger 6, middle finger 7 and so on. She counted upto 1994. She ended counting on which finger ?** → Index finger.

Thumb	Index Finger	Middle finger	Ring Finger	Little finger
1 / 9	2 / 8 / 0	3 / 7	4 / 6	5

- Repetition after 8 counting, $8n + 1$ satisfies : Find by finger value or remainder value first - Divide by 8**
- For week count** like finger counting (**Repetition after 12** counting, $12n + 1$ satisfies : Find by week value or remainder value first - Divide by 12)
- For month count** like finger counting (**Repetition after 22** counting, $22n + 1$ satisfies : Find by week value or remainder value first - Divide by 22)
- If p, q, r, s, t are numbers such that $p+q < r+s$; $q+r < s+t$; $r+s < t+p$; $s+t < p+q$, then the largest and the smallest numbers are respectively.**
Let $p+q < r+s$ (i) $q+r < s+t$ (ii) $r+s < t+p$ (iii) $s+t < p+q$ (iv)
From equation (i) and (iii); $q < t$
From equation (ii) and (iv); $r < p$
From equation (i) and (iv); $t < r$
Thus $q < t < r < p$ From (i) $p+q < r+s$, also $r < p \Rightarrow q < s$ From (iv) $s+t < p+q \Rightarrow s < p$
Hence, largest and smallest numbers are p and q respectively.
- If $P+Q > S+T$, $Q+S > P+T$, Then,**
 $P+Q > S+T$ (i) $Q+S > P+T$ (ii)
Adding, $P+S+2Q > P+S+2T \rightarrow Q > T$
- One page is torn from a booklet whose pages are numbered in the usual manner starting from the first page as 1. The sum of the numbers on the remaining pages is 195. The torn page contains which of the following numbers?**

Sum of natural number **above** (Because - One page is torn) $195 : \frac{n(n+1)}{2} = 210$, For $n = 20$

Now rest number = $210 - 195 = 15$ (So, Option must be $x + y = 15$ for two pages. For Eg. 7 and 8)

- In an examination a student scores 3 marks for every correct answer and loses 2 mark for every wrong answer. If he attempts all 150 questions and scores 400 marks, what is the number of questions he attempts correctly is? → 140 Questions.

Right × Right value – Wrong × Wrong value = Obtained marks

$$\text{Or, } R \times 3 - W \times 2 = 400$$

$$\text{Or, } 3R - 2(150 - R) = 400 \quad (\text{Where, Wrong} = 150 - \text{Right}) \quad \text{So, } R = 140$$

- For Percentage gives like 25% of wrong answer will be deducted ; Wrong value = $\frac{25}{100} = \frac{1}{4}$
- 21 mango trees, 42 apple trees and 56 orange trees have to be planted in rows such that each row contains the same number of trees of one variety only. What is the minimum number of rows in which the above trees may be planted?

$$\text{Number of rows} = \frac{\text{Total}}{\text{HCF}} = \frac{21+42+56}{7} = 17$$

- चकलट खाने समस्या : सुरुको चकलेट = X र बाँकी चकलेट = Y मानौं

$$Y = \dots\dots(X-1) \left(1 - \frac{1}{a}\right)^3 - \left(1 - \frac{1}{a}\right)^2 - \left(1 - \frac{1}{a}\right)$$

- A pineapple costs Rs. 7 each. A watermelon costs Rs. 5 each. Ram spends Rs. 38 on these fruits. The number of pineapples and watermelons purchased are ?

Let no. of pineapples = x

Let no. Of watermelons = y

From information, $7x + 5y = 38$ (Now trying $x = 1$ to 5 to test for zero remainder. $x = 4, y = 2$)

Therefore, no. Of pineapples = 4 and watermelon is 2.

- There are students and benches in a classroom . If 4 students sit in each bench, 3 benches are left vacant. But if 3 students sit in a bench, 3 students are left standing. What are the total no. of students ?

Let there be x students in the class.

$$\text{When 4 students sit on each bench, number of benches} = \frac{x}{4} + 3 \dots\dots\dots(i)$$

$$\text{When 3 students sit on each bench number of benches} = \frac{x-3}{3} \dots\dots\dots(ii) \quad \frac{x}{4} + 3 = \frac{x-3}{3} \rightarrow x = 48$$

Sub-Chapter 12 - Assertion and reason and Syllogism

Trick : Both A and R may be correct but be careful for correct explanation

❖ Assertion and reason

- Two statements will be given - A : Assertion B : Reason

There will be four or five possibilities related to given assertion and reason-

1. Both A and R are true and R is correct explanation of A
2. Both A and R are true and R isn't correct explanation of A
3. A is true but R is false
4. A is false but R is true
5. Both A and R are false

❖ Syllogism

- There will be given certain statements with relations.
- Write all relations in veen diagram
- Be clear in possibilities and conclusions (Note - Possibility cannot follow, Only conclusion follow)
- There will be four, five or more conditions.
 1. If only conclusion (I) follows
 2. If only conclusion (II) follows
 3. If either conclusion (I) or (II) follows
 4. If neither conclusion (I) or (II) follows
 5. If both (I) and (II) follow

- **Statements:** No women teacher can play. Some women teachers are athletes. → D
Conclusions:
I. Male athletes can play.
II. Some athletes can play.
A. Only conclusion I follows B. Only conclusion II follows C. Either I or II follows
D. Neither I nor II follows E. Both I and II follow
- **Statements:** All bags are cakes. All lamps are cakes. → C
Conclusions:
I. Some lamps are bags.
II. No lamp is bag.
A. Only conclusion I follows B. Only conclusion II follows C. Either I or II follows
D. Neither I nor II follows E. Both I and II follow
- **Statements:** All mangoes are golden in colour. No golden-coloured things are cheap. → B
Conclusions:
I. All mangoes are cheap.
II. Golden-coloured mangoes are not cheap.
A. Only conclusion I follows B. Only conclusion II follows C. Either I or II follows
D. Neither I nor II follows E. Both I and II follow
- **Statements:** Some kings are queens. All queens are beautiful. → D
Conclusions:
I. All kings are beautiful.
II. All queens are kings.
A. Only conclusion I follows B. Only conclusion II follows C. Either I or II follows
D. Neither I nor II follows E. Both I and II follow
- **Statements:** Some doctors are fools. Some fools are rich. → D
Conclusions:
I. Some doctors are rich
II. Some rich are doctors.
A. Only conclusion I follows B. Only conclusion II follows C. Either I or II follows
D. Neither I nor II follows E. Both I and II follow

Sub-Chapter 13 - Induction and deduction

❖ Sub-Chapter 13.1 - Induction

- Moving from Specific to general. Promise to conclusion.
- Example of making general rules of laws.
- Finding appropriate conclusion from two statements.
Eg.- a) Dog is carnivorous, it eats flesh. b) Cat is carnivorous, it eats flesh
Conclusion - Carnivorous eat flesh.

❖ Sub-Chapter 13.2 - Deduction

- Moving from General specific. Conclusion to promise
- Theory/ rule/ laws to examples
- Finding appropriate theory/ rules/ laws from two specific statements.
Eg. - a) Stars twinkle in the sky. b) A comet is a star.
Conclusion - A comet twinkle in the sky

Sub-Chapter 14 - Statements and Course of action

- Finding the fact way in contemporary conditions.
- Based on logical precautions.
A. Only I follows
B. B. Only II follows
C. C. Either I or II follows
D. D. Neither I nor II follows
E. E. Both I and II follow
- **Statement:** Severe drought is reported to have set in several parts of the country. → B

Courses of Action:

1. Government should immediately make arrangement for providing financial assistance to those affected.
2. Food, water and fodder should immediately be sent to all these areas to save the people and cattle.
- **Statement:** Since its launching in 1981, Vayudoot has so far accumulated losses amounting to Rs 153 crore. → A

Courses of Action:

1. Vayudoot should be directed to reduce wasteful expenditure and to increase passenger fare.
2. An amount of about Rs 300 crore should be provided to Vayudoot to make the airliner economically viable.
- **Statement:** A large number of people die every year due to drinking polluted water during the summer. → E

Courses of Action:

1. The government should make adequate arrangements to provide safe drinking water to all its citizens.
2. The people should be educated about the dangers of drinking polluted water.
- **Statement:** There is an unprecedented increase in migration of villagers to urban areas as repeated crop failure has put them into precarious financial situation.

Courses of Action:

1. The villagers should be provided with alternate source of income in their villages which will make them stay put.
2. The migrated villagers should be provided with jobs in the urban areas to help them survive
- **Statement:** The sale of a particular product has gone down considerably causing great concern to the company. → A

Courses of Action:

1. The company should make a proper study of rival products in the market.
2. The price of the product should be reduced and quality improved.
- **Statement:** A recent study shows that children below five die in the cities of the developing countries mainly from diarrhoea and parasitic intestinal worms. E

Courses of Action:

1. Governments of the developing countries should take adequate measures to improve the hygienic conditions in the cities.
2. Children below five years in the cities of the developing countries need to be kept under periodic medical check-up
- **Statement:** The Government has decided not to provide financial support to voluntary organisations from next Five Year Plan and has communicated that all such organisations should raise funds to meet their financial needs. → B

Courses of Action:

1. Voluntary organisations should collaborate with foreign agencies.
2. They should explore other sources of financial support.
- **Statement:** There has been an unprecedented increase in the number of successful candidates in this year's School Leaving Certificate Examination. → A

Courses of Action:

1. The government should make arrangements to increase number of seats of intermediate courses in existing colleges.
2. The government should take active steps to open new colleges to accommodate all these successful candidates.
- **Statement:** On an average, about twenty people are run over by trains and die every day while crossing the railway tracks through the level crossing. → B

Courses of Action:

1. The railway authorities should be instructed to close all the level crossings.
2. Those who are found crossing the tracks, when the gates are closed, should be fined heavily

- **Statement:** In spite of the Principal's repeated warnings, a child was caught exploding crackers secretly in the school. → B

Courses of Action:

1. All the crackers should be taken away from the child and he should be threatened not to do it again.
2. The child should be severely punished for his wrong act.

- **Statement:** A large number of engineering graduates in the country are not in a position to have gainful employment at present and the number of such engineers is likely to grow in the future. → B

Courses of Action:

1. The government should launch attractive employment generation schemes and encourage these graduates to opt for such schemes to use their expertise and knowledge effectively.
2. This happened due to proliferation of engineering colleges in the country and thereby lowered the quality of the engineering graduates. Those colleges which are not equipped to impart quality education should be closed down.

- **Statement:** Every year, at the beginning or at the end of the monsoons, we have some cases of conjunctivitis, but this year, it seems to be a major epidemic, witnessed after nearly four years. → B

Courses of Action:

1. Precautionary measures should be taken after every four years to check this epidemic.
2. People should be advised to drink boiled water during rainy season.

- **Statement:** Every year large number of devotees dies due to severe cold on their way to the shrine located at the top of the mountain range. → A

Courses of Action:

1. The devotees should be discouraged to visit the shrine without having proper warm clothes and other amenities.
2. The government should provide warm clothes and shelter to all the devotees visiting the shrine.

- **Statement:** There has been an unprecedented increase in the number of requests for berths in most of the long distance trains during the current holiday season. → A

Courses of Action:

1. The railway authority should immediately increase the capacity in each of these trains by attaching additional coaches.
2. The people seeking accommodation should be advised to make their travel plan after the holiday.

- **Statement:** There has been less than forty percent voter turnout in the recent assembly elections. → D

Courses of Action:

1. The election commission should cancel the entire election process as the votes cast are not adequate to represent people
2. The election commission should take away the voting rights of those who did not exercise their rights.

- **Statement:** The vegetable traders feel that the prices of onion will again go up shortly in the State 'P' → E

Courses of Action:

1. The 'P' State Government should purchase and store sufficient quantity of onion in advance to control prices.
2. The 'P' State Government should make available network of fair price shops for the sale of onions during the period of shortage.

- **Statement:** Youngsters are often found staring at obscene posters. → B

Courses of Action:

1. Children should be punished and penalized if they are found doing so.
2. Any display of such material should be banned.

- **Statement:** One of the problems facing the food processing industry is the irregular supply of raw material. The producers of raw material are not getting a reasonable price. → B

Courses of Action:

1. The government should regulate the supply of raw material to other industries also.
2. The government should announce an attractive package to ensure regular supply of raw material for food processing industry.

- **Statement:** A large number of students have been caught using unfair means during the final-year degree examination. → D

Courses of Action:

1. All these students should be debarred permanently from appearing for any examination conducted by the authority.
2. The guardians of these students should be called by the authority to inform them that any such behaviour in future will not be tolerated.

- **Statement:** There is a substantial increase in the number of accidents causing deaths and severe injuries due to malfunctioning of the traffic signals. → C

Courses of Action:

1. The traffic police should immediately post traffic personnel at the junctions.
2. The signal system should immediately be repaired or replaced

- **Statement:** Mr. X, an active member of the Union, often insults his superiors in the office with his rude behaviour. → D

Courses of Action:

1. He should be transferred to some other department.
2. The matter should be referred to the Union.

In each question below is given a statement followed by three courses of action numbered I, II and III. You have to assume everything in the statement to be true, then decide which of the three given suggested courses of action logically follows for pursuing.

- **Statement:** The weather bureau has through a recent bulletin forecast heavy rainfall during the next week which may cause water logging in several parts of the city. → D

Courses of Action:

1. The bulletin should be given wide publicity through the mass media.
2. The civic authority should keep in readiness the pumping system for removal of water from these parts.
3. The people should be advised to stay indoors during the period.

- A. None follows
- B. Only I and II follow
- C. Only II follows
- D. Only II and III follow
- E. None of these

- **Statement:** A train derailed near a station while moving over a bridge and fell into a river, resulting in the death of 65 people. → C

Courses of Action:

1. The Railway Authorities should clarify the reason of the accident to the Government.
2. The Government should allocate funds to compensate the destruction caused.
3. The protection walls of the bridge should be made strong enough to avoid such accidents.

- A. Only I and II follow
- B. Only II and III follow
- C. Only III follows
- D. All follow
- E. None of these

- **Statement:** Poverty is increasing because the people, who are deciding how to tackle it, know absolutely nothing about the poor. → C

Courses of Action:

1. The decision makers should go to the grass root levels.
2. The decision makers should come from the poorer sections of the society.
3. A new set of decision makers should replace the existing one.

- A. Only I follows
- B. Only II follows
- C. Only either I or III follows
- D. All follow

- **Statement:** Poverty is increasing because the people, who are deciding how to tackle it, know absolutely nothing about the poor. → A

Courses of Action:

- 1. The decision makers should go to the grass root levels.
- 2. The decision makers should come from the poorer sections of the society.
- 3. A new set of decision makers should replace the existing one.

- A. Only I follows
- B. Only II follows
- C. Only either I or III follows
- D. All follow

- **Statement:** There is an unprecedented increase in the production of wheat this kharif season in most parts of the country. → E

Courses of Action:

- 1. The government should immediately lower down the procurement price of wheat.
- 2. The farmers should be asked to store the excess produces with themselves to be used for future
- 3. The government should make its best efforts to export wheat to augment its presence in international market.

- A. Only I and II follow
- B. Only II and III follow
- C. Only I and III follow
- D. All follow
- E. None of these

- **Statement:** Any further increase in the pollution level in the city by way of industrial effluents and automobile exhaustions would pose a severe threat to the inhabitants. → C

Courses of Action:

- 1. All the factories in the city should immediately be closed down.
- 2. The automobiles should not be allowed to ply on the road for more than four hours a day.
- 3. The Government should restrict the issue of fresh licences to factories and automobiles.

- A. None follows
- B. Only II follows
- C. C. Only III follows
- D. All follows
- E. None of these

Sub-Chapter 15 - Statement and Assumptions

Trick :

- 1. Negative form cannot be accept

In each question below is given a statement followed by two assumptions numbered I and II. You have to consider the statement and the following assumptions and decide which of the assumptions is implicit in the statement. Give answer

- A. Only assumption I is implicit
- B. Only assumption II is implicit
- C. Either I or II is implicit
- D. Neither I nor II is implicit
- E. Both I and II are implicit

- **Statement:** "You are hereby appointed as a programmer with a probation period of one year and your performance will be reviewed at the end of the period for confirmation." - A line in an appointment letter. → E

Assumptions:

- 1. The performance of an individual generally is not known at the time of appointment offer.

2. Generally an individual tries to prove his worth in the probation period.
 - **Statement:** It is desirable to put the child in school at the age of 5 or so. → A
 - Assumptions:**
 1. At that age the child reaches appropriate level of development and is ready to learn.
 2. The schools do not admit children after six years of age.
 - **Statement:** "In order to bring punctuality in our office, we must provide conveyance allowance to our employees." - In charge of a company tells Personnel Manager. → B
 - Assumptions:**
 1. Conveyance allowance will not help in bringing punctuality.
 2. Discipline and reward should always go hand in hand.
 - **Statement:** Unemployment allowance should be given to all unemployed Indian youth above 18 years of age. → A
 - Assumptions:**
 1. There are unemployed youth in India who needs monetary support.
 2. The government has sufficient funds to provide allowance to all unemployed youth.
 - **Statement:** "If you trouble me, I will slap you." - A mother warns her child. → A
 - Assumptions:**
 1. With the warning, the child may stop troubling her.
 2. All children are basically naughty.
 - **Statement:** The State government has decided to appoint four thousand primary school teachers during the next financial year. → A
 - Assumptions:**
 1. There are enough schools in the state to accommodate four thousand additional primary school teachers.
 2. The eligible candidates may not be interested to apply as the government may not finally appoint such a large number of primary school teachers.
 - **Statement:** A warning in a train compartment - "To stop train, pull chain. Penalty for improper use Rs. 500." → E
 - Assumptions:**
 1. Some people misuse the alarm chain.
 2. On certain occasions, people may want to stop a running train.
 - **Statement:** If it is easy to become an engineer, I don't want to be an engineer. → B
 - Assumptions:**
 1. An individual aspires to be professional.
 2. One desires to achieve a thing which is hard earned.
 - **Statement:** The concession in rail fares for the journey to hill stations has been cancelled because it is not needed for people who can spend their holidays there. → A
 - Assumptions:**
 1. Railways should give concession only to needy persons.
 2. Railways should not encourage people to spend their holidays at hill stations.
 - **Statement:** "The bridge was built at the cost of Rs. 128 crores and even civil bus service is not utilizing it, what a pity to see it grossly underutilized." - A citizen's view on a new flyover linking east and west sides of a suburb. → B
 - Assumptions:**

The building of such bridges does not serve any public objective.

There has to be some accountability and utility of money spent on public projects.
 - **Statement:** The Government has decided to levy 2 percent on the tax amount payable for funding drought relief programmes. → E
 - Assumptions:**

1. The Government does not have sufficient money to fund drought relief programmes.
2. The amount collected by way of surcharge may be adequate to fund these drought relief programmes.
- **Statement:** Detergents should be used to clean clothes. → B
Assumptions:
 1. Detergents form more lather.
 2. Detergents help to dislodge grease and dirt.
- **Statement:** It will be a substantial achievement in the field of education if one provides one school for every village in our country and enforce attendance. → E
Assumptions:
 1. Children in villages do not attend school regularly.
 2. Providing school to every village is desirable.
- **Statement:** The government has decided to disinvest large chunk of its equity in select public sector undertakings for a better fiscal management. → A
Assumptions:
 1. The amount generated out of the disinvestment process may reduce substantially the mounting fiscal deficits.
 2. There will be enough demand in the market for the shares of these undertakings.
- **Statement:** Never before such a lucid book was available on the topic. → A
Assumptions:
 1. Some other books were available on this topic.
 2. You can write lucid books on very few topics.
- **Statement:** Please do not use lift while going down - an instruction on the top floor of a five-storey building. → B
Assumptions:
 1. While going down, the lift is unable to carry any load.
 2. Provision of lift is a matter of facility and not of right.
- **Statement:** "I have not received telephone bills for nine months inspite of several complaints" - A telephone customer's letter to the editor of a daily → E
Assumptions:
 1. Every customer has a right to get bills regularly from the telephone company.
 2. The customer's complaints point to defect in the services which are expected to be corrected.
- **Statement:** "This drink can be had either as it is, or after adding ice to it." - An advertisement. → E
Assumptions:
 1. People differ in their preferences.
 2. Some people will get attracted to the drink as it can be had as it is.
- **Statement:** Government has permitted unaided colleges to increase their fees. → A
Assumptions:
 1. Unaided colleges are in financial difficulties.
 2. Aided colleges do not need to increase fees.
- **Statement:** Be humble even after being victorious. → B
Assumptions:
 1. Many people are humble after being victorious.
 2. Generally people are not humble.
- **Statement:** The government has decided to pay compensation to the tune of Rs. 1 lakh to the family members of those who are killed in railway accidents. → A
Assumptions:
 1. The government has enough funds to meet the expenses due to compensation.
 2. There may be reduction in incidents of railway accidents in near future.

- **Statement:** Films have become indispensable for the entertainment of people. → B

Assumptions:

1. Films are the only media of entertainment.
2. People enjoy films.

- **Statement:** Of all the newspapers published in Mumbai, readership of the "Times" is the largest in the Metropolis. → D

Assumptions:

1. 'Times' is not popular in mofussil areas.
2. 'Times' has the popular feature of cartoons on burning social and political issues.

- **Statement:** Apart from the entertainment value of television, its educational value cannot be ignored. → E

Assumptions:

1. People take television to be a means of entertainment only.
2. The educational value of television is not realised properly.

- **Statement:** Children are influenced more by their teachers nowadays. → A

Assumptions:

1. The children consider teachers as their models.
2. A large amount of children's time is spent in school.

- **Statement:** You know that your suit is excellent when people ask about your tailor who tailored the suit. → A

Assumptions:

1. People do not ask about your tailor if your suit is not good.
2. The people want to know the criterion of an excellent suit.

Sub-Chapter 16 - Statement and Conclusion

Each question presents a situation and asks you to make a judgment regarding that particular circumstance. Choose an answer based on given information.

- A. Only conclusion (1) follows
- B. Only conclusion (2) follows
- C. Either (I) or (II) follows
- D. Neither (I) nor (II) follows
- E. Both (I) and (II) follow

- **Statements:** In a one day cricket match, the total runs made by a team were 200. Out of these 160 runs were made by spinners. → D

Conclusions:

1. 80% of the team consists of spinners.
2. The opening batsmen were spinners.

- **Statements:** Government has spoiled many top ranking financial institutions by appointing bureaucrats as Directors of these institutions. → E

Conclusions:

1. Government should appoint Directors of the financial institutes taking into consideration the expertise of the person in the area of finance.
2. The Director of the financial institute should have expertise commensurate with the financial work carried out by the institute.

- **Statements:** Population increase coupled with depleting resources is going to be the scenario of many developing countries in days to come. → B

Conclusions:

1. The population of developing countries will not continue to increase in future.
2. It will be very difficult for the governments of developing countries to provide its people decent quality of life.

- **Statements:** The T.V. programmes, telecast specially for women are packed with a variety of recipes and household hints. A major portion of magazines for women also contains the items mentioned above. → B

Conclusions:

1. Women are not interested in other things.
2. An average woman's primary interest lies in home and specially in the kitchen.
- Statements: The distance of 900 km by road between Bombay and Jafra will be reduced to 280 km by sea. This will lead to a saving of Rs. 7.92 crores per annum on fuel. → B

Conclusions:

1. Transportation by sea is cheaper than that by road.
2. Fuel must be saved to the greatest extent

Sub-Chapter 17 - Situation Judgements Test/ Making Judgements

Each question presents a situation and asks you to make a judgment regarding that particular circumstance. Choose an answer based on given information.

Sub-Chapter 18 - Cause and Effects**Trick :**

1. If there is "Therefore" Combine it.

In each of the following questions, two statements numbered I and II are given. There may be cause and effect relationship between the two statements.

- A. Statement (I) is the cause and statement (II) is its effect
- B. Statement (II) is the cause and statement (I) is its effect
- C. Both the statements (I) and (II) are independent causes
- D. Both the statements (I) and (II) are effects of independent causes
- E. Both the statements (I) and (II) are effects of some common cause

• Read both the statements in each question and mark your answer → D

1. The prices of petrol and diesel in the domestic market have remained unchanged for the past few months.
2. The crude oil prices in the international market have gone up substantially in the last few months. →

• Read both the statements in each question and mark your answer → B

1. The government has recently fixed the fees for professional courses offered by the unaided institutions which are much lower than the fees charged last year.
2. The parents of the aspiring students launched a severe agitation last year protesting against the high fees charged by the unaided institutions.

• Read both the statements in each question and mark your answer → B

1. The Reserve Bank of India has recently put restrictions on few small banks in the country.
2. The small banks in the private and co-operative sector in India are not in a position to withstand the competitions of the bigger in the public sector.

• Read both the statements in each question and mark your answer → D

1. All the schools in the area had to be kept closed for most part of the week.
2. Many parents have withdrawn their children from the local schools.

• Read both the statements in each question and mark your answer → C

1. India has surpassed the value of tea exports this year over all the earlier years due to an increase in demand for quality tea in the European market.
2. There is an increase in demand of coffee in the domestic market during the last two years.

Chapter 2 - Non-verbal reasoning

Sub-Chapter 1 - Sequence and Series

- Rotation in clockwise or anticlockwise direction within fixed or different degree.
- Jumping forward or backward a small picture/ Arrows
- Adding line segment/ dots/ shapes/ Arrows

Sub-Chapter 2 - Analogy

- Choosing pair relations as series.
- Increasing size/ Enlargements.

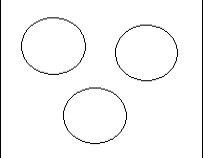
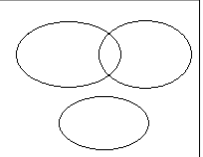
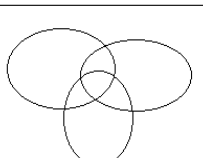
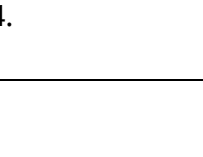
Sub-Chapter 3 - Classification

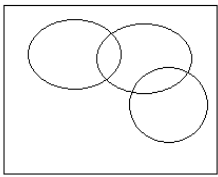
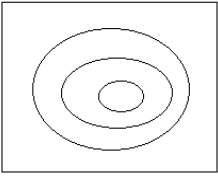
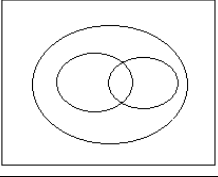
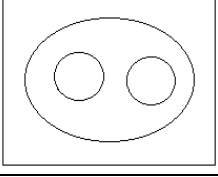
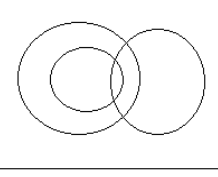
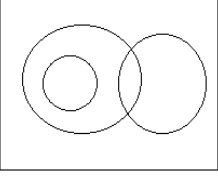
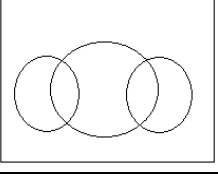
- Odd man out in series.

Sub-Chapter 4 - Veen diagram

❖ Types of veen diagrams

- At first, match the relation between two veen diagrams then match relation with third one.
- Veen diagram may be in different shapes.

1. 	• Liquids, Solids, Gases • Star, Moon, Earth	•
2. 	• Ripe, Mangoes, Baskets	•
3. 	• Tall, Men, Educated • Smart, Women Engineers	•
4. 	• Liquids, Water, Solids • Criminal, Bandits, Lawyers • Penalty, Award, Extra cooperation	•
4. 	• Ink, Red, Crayons • Exam, Question, Exercise	•

		
5. 	• Grammar, Preposition, From • Canies, Dogs, Pups	•
6. 	• Mother, Sister, Female	•
7. 	• Vegetables, Brinjals, Potatoes • Father, son, daughter	•
8. 	• Sweet, Sugar, Tea • Men, Bachelors, Swimmers • Woman, Mother, Criminal	•
9. 	•	•
10. 	• Cloths, Red, Flower	•

Sub-Chapter 5 - Matrix

- Figure can be set in mathematical matrix form.

- Only matrix rule can be followed.

Sub-chapter 6 - Construction of squares and triangles

- Arranging different parts of triangles and squares for making full figure.
- First of all match all pieces with 90° shapes.

Sub-chapter 7 - Figure Completion, formation, analysis, Grouping figure and Rule detection

❖ Completion of incompleting pattern

❖ Figure formation and analysis

- Fill the incomplete figure by choosing from alternatives.
- Match identical face and choose the alternatives.

❖ Grouping figure

- समान गुण भएका चित्रलाई एक साथमा राख्ने
- दुई, तीन समूह वा कुनै एकल नै पनि हुनसक्छ
- समान गुण हुने समूह अगाडि वा पछाडि पनि हुनसक्ने

❖ Rule detection

- Finding the figure series in order to given law.
- Figure pattern opening and closing series

Sub-chapter 8 - Dot situation/ Shaded parts

- Two, three or more geometrical shapes having possible dots or shaded parts.
- Option must be determined first.
- For multiple dots, verify from one, two consecutively and then reject the non-appropriate alternatives.

Sub-chapter 9 - Mirror image /Water image

❖ Mirror image

- Left-right reflection
- 11 English capital letters have identical shape in mirror image - A, H, I, M, O, T, U, V, W, X, Y
- 9 English small letters have identical shape in mirror image - i, l, m, n, o, u, v, w, x
- Mirror image time = 12:00 - Real time
- Final image in mirror when using **more than one mirror** :
 - If even mirror used (Time is same)
 - If odd mirror used (12:00 - Real time)

❖ Water image

- Up-down reflection
- 8 English capital letters have identical shape in water image - C, D, E, H, I, K, O, X
- 4 English small letters have identical shape in water image - c, l, o, x
- Undetermined image shape will be made in water image of watch.
- घडीमा समय विकल्पहरुमा 'अनिश्चित' नदिई वास्तविक वा अवास्तविक दुवै समय निकाल्नु परेका मिनेटमा 30 भन्दा कम वा 30 मिनेट नै भए १८:30 बाट घटाउने
- घडीमा समय 30 मिनेट भन्दा बढी भएमा १८:३० बाट घटाउने
Eg. - 01 : 15 in watch (18:30 - 04:15 = 17:15 in water image)
02 : 45 in watch (18:30 - 02:45 = 16:45 in water image)

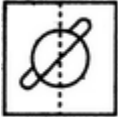
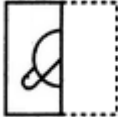
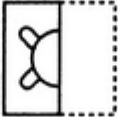
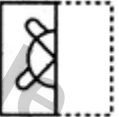
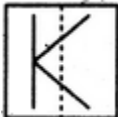
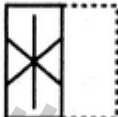
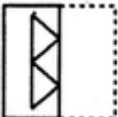
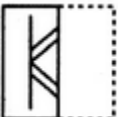
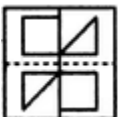
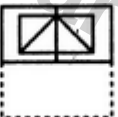
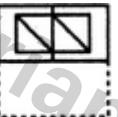
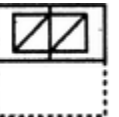
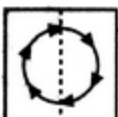
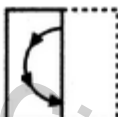
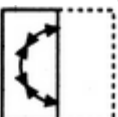
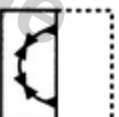
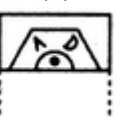
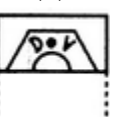
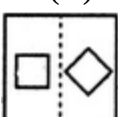
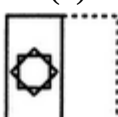
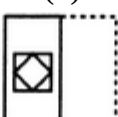
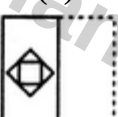
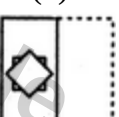
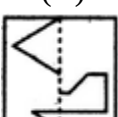
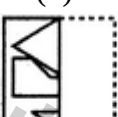
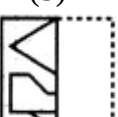
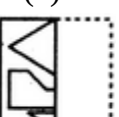
Sub-chapter 10 - Embedded figures

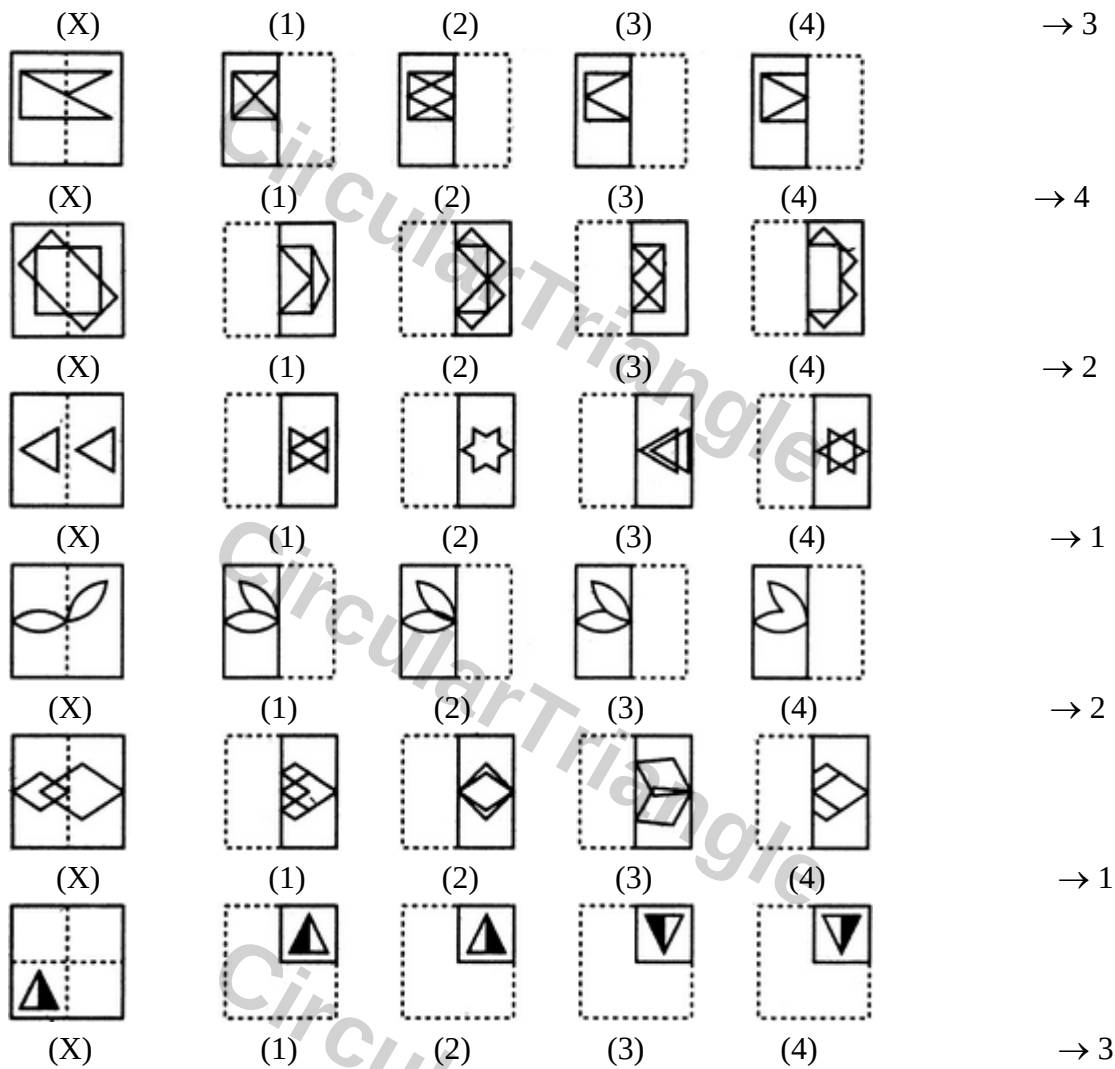
- Finding overlapped part or figure in another figure
- **Two types**
 1. Any part of the figure will embed in one of the given alternatives.
 2. Full figure will be given and possible embeded parts pieces can be in alternatives.
 3. Embeded figure can be choosen in opposite or straight style.
 4. Embeded figure must be equal with each other.

Sub-chapter 11 - Paper folding and cutting

❖ Paper folding

- Types - Verticle/ Horizontal/ Diagonal
- Choosing alternatives in the transparent sheet from folded imaginary figure.
- Folded figure cannot shade the real figure of the pieces of imaginary figure in the transparent sheet.
- Verticle folding/ Diagonal folding = Mirror image and Horizontal folding = Water image
- Making image towards real figure from imaginary figure
- **Note :**
 1. Simple overlaping but not shading.
 2. Careful to image boundry, where it is kept.
- Find out from amongst the four alternatives as to how the pattern would appear when the transparent sheet is folded at the dotted line.

					
(X)	(1)	(2)	(3)	(4)	→ 4
					
(X)	(1)	(2)	(3)	(4)	→ 3
					
(X)	(1)	(2)	(3)	(4)	→ 1
					
(X)	(1)	(2)	(3)	(4)	→ 3
					
(X)	(1)	(2)	(3)	(4)	→ 2
					
(X)	(1)	(2)	(3)	(4)	→ 1
					

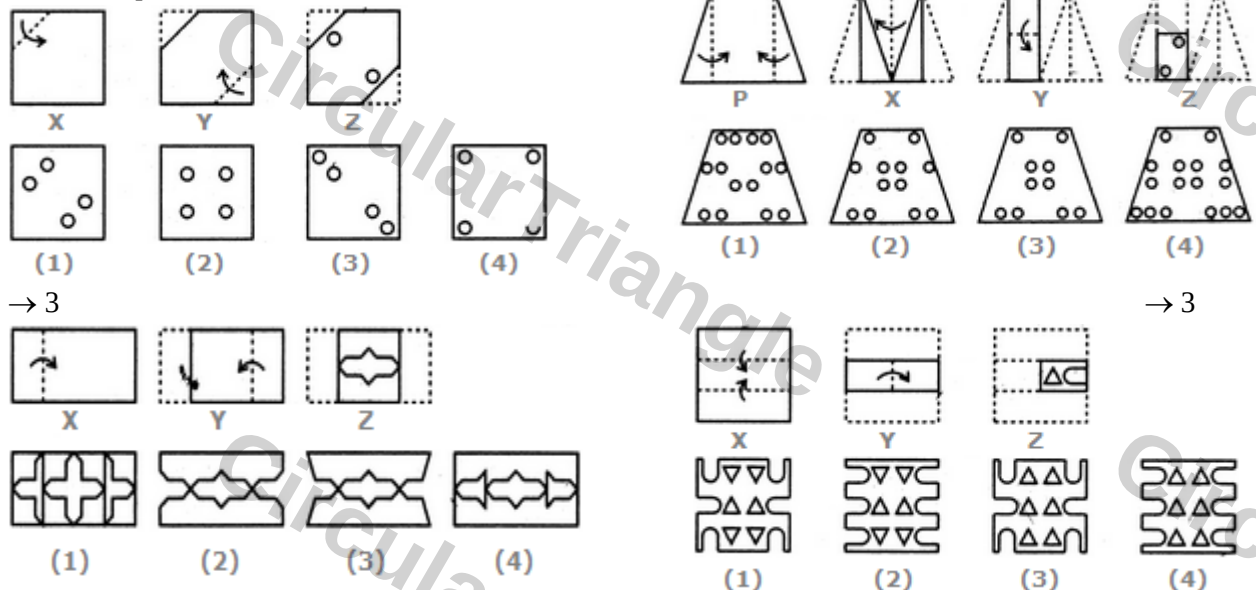


❖ Paper cutting

- Cutting or unfolding to the finally folded figure sheet in certain shape.
- Making image in reverse order to the finally folded figure.
- Only certain portion or part will be folded but not centre.

• Note :

- Watch by back and careful on original folded pattern for whole pattern.
- unfold the pattern in dotted lines times.



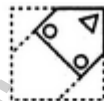
→ 2



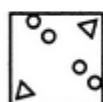
X



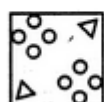
Y



Z



(1)



(2)

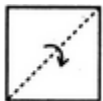


(3)



(4)

→ 2



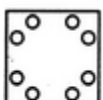
X



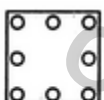
Y



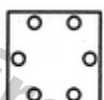
Z



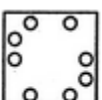
(1)



(2)



(3)



(4)

→ 1



X



Y



Z



(1)



(2)



(3)



(4)

→ 4



X



Y



Z



(1)



(2)



(3)

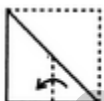


(4)

→ 4



X



Y



Z



(1)



(2)

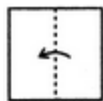


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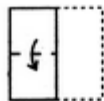


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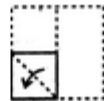
→ 3



P



X



Y



Z



(1)



(2)



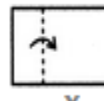
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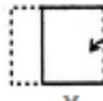
(4)

→ 3

→ 2



X



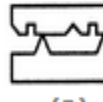
Y



Z



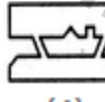
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(2)



(3)



(4)

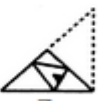
→ 3



X



Y



Z



(1)



(2)

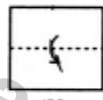


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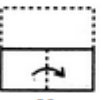


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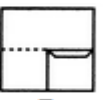
→ 4



X



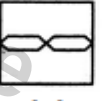
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Z



(1)



(2)

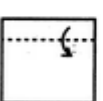


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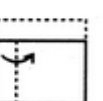


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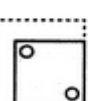
→ 2



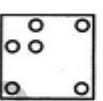
X



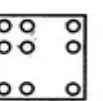
Y



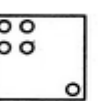
Z



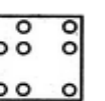
(1)



(2)



(3)



(4)

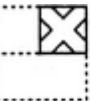
→ 3



X



Y



Z



(1)



(2)



(3)

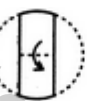


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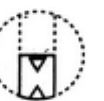
→ 2



X



Y



Z



(1)



(2)

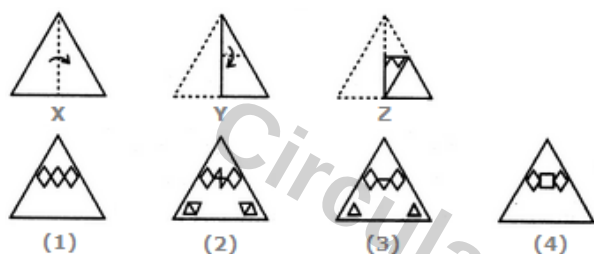


(3)



(4)

→ 4



→ 1

Sub-chapter 12 - Cubes and Dices

❖ Cubes

• Number of cubes made by multiple small cubes

		n = 3	n = 4	n = 5	n = 6
Number of cubes	n^3	27	64	125	216
Three sides painted	8	8	8	8	8
Two sides painted	$12 \times (n-2)$	12	24	36	48
single side painted	$6 \times (n-2)^2$	6	24	36	48
No sides painted	$(n-2)^3$	1	8	27	64
Note - If total cubes = 27 then consider that n = 3 (single line counted)					

• Cutting large cube in small cubes.

Eg. - Cutting **one** larger cube of sized 8 cm × 8 cm × 8 cm into cubes of sized 2 cm × 2 cm × 2 cm.

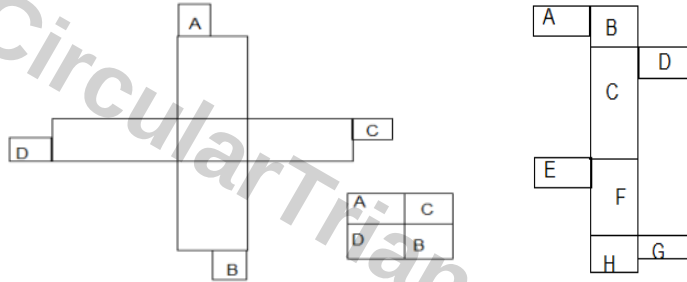
Then, 8 cm = 2 cm × 4 → Total small cubes = 4 × 4 × 4 = 64 (since, n = 4)

❖ Dices

• Types of dices

Standard/ Ordinary/ Real dice	Open dice	Closed dice
- Sum of opposite side = 7 6 + 1 = 7 5 + 2 = 7 4 + 3 = 7		

	- Be careful for which cannot be in the opposite side.(Eg. - With A → B, C) - A, B, C cannot be opposite each other - A, B, C are in adjacent sides. - In a dice there are 12 edges and 8 corners.
(I) (II) If A → D If B → E If C → F • Quarter cutting parts opposite each other	- Member of column (I) and (II) cannot come simultaneously from same row. - Member of column (I) and (II) come from different row.

in full side. Four half cutting	– From two given dice, Start from identical face of both dice and rotate clockwise or anticlockwise direction. 
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- Finding opposite face.
- Filling open dice from closed dice.
- Be care on horizontal and verticle sign.

Sub-chapter 13 - Analytical Resoaning

❖ Counting straight lines

- Verticle lines + Horizontal line + Inclined line

❖ Counting Squares

- If column and row are in equal size :

– E.g. 4×4 Size (i.e.Total number of squares = $4^2 + 3^2 + 2^2 + 1^2 = 30$)

- If column and row are not in equal size :

– E.g. 3×5 Size [i.e.Total number of squares = $(3^2 + 2^2 + 1^2) + (3 + 2 + 1) = 14$]

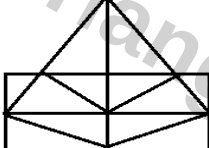
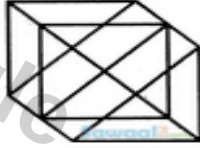
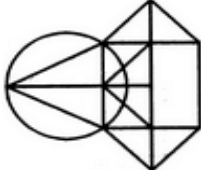
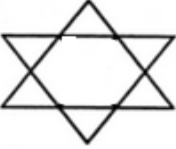
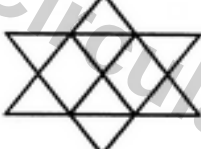
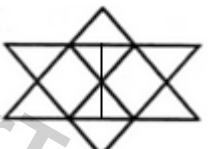
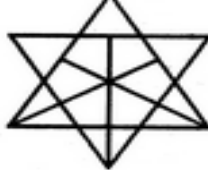
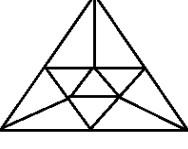
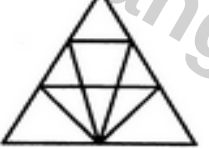
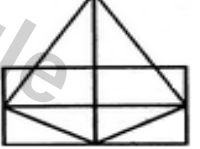
❖ Counting Rectangles/ Parallelograms/ Quardilater

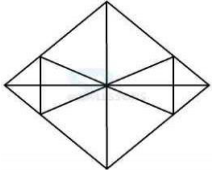
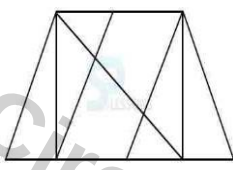
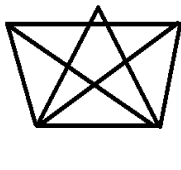
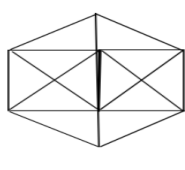
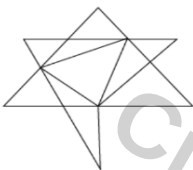
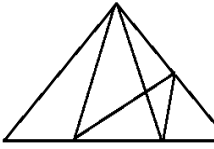
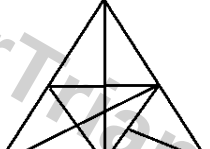
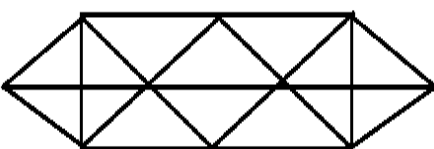
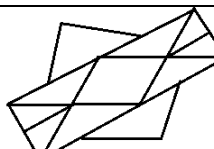
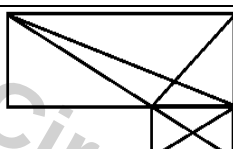
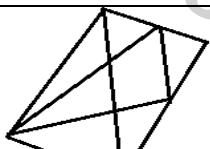
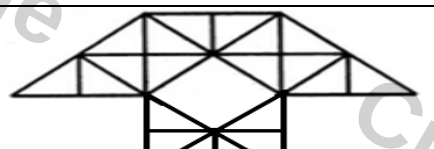
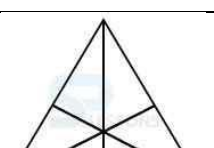
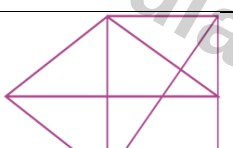
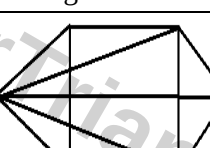
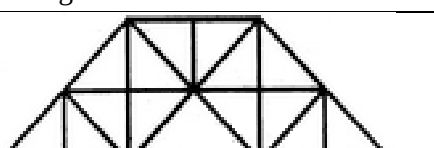
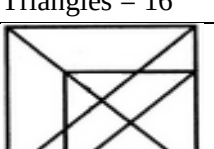
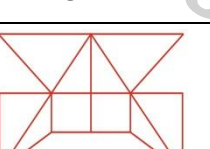
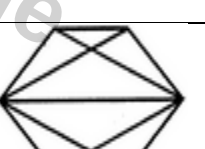
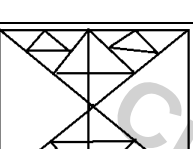
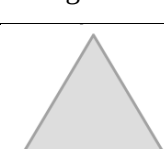
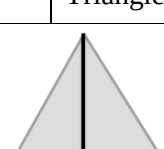
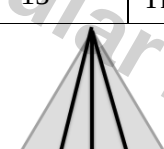
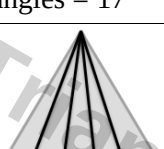
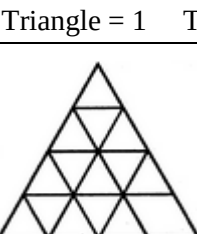
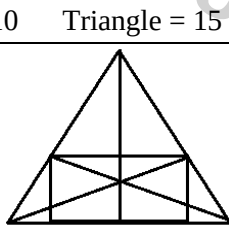
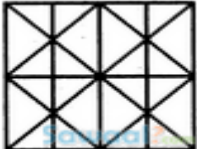
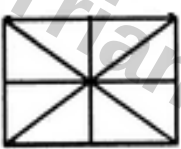
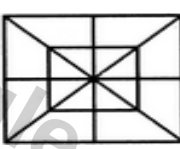
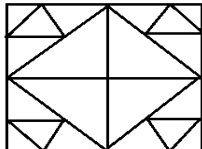
- For all Rectangles/ Parallelograms/ Quardilater in all sizes :

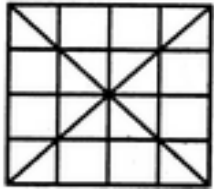
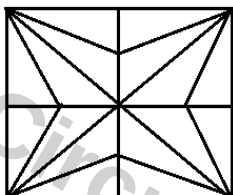
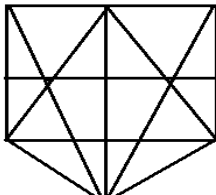
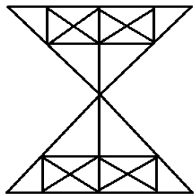
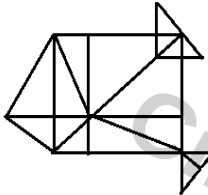
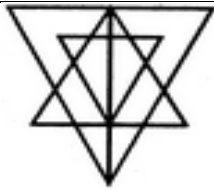
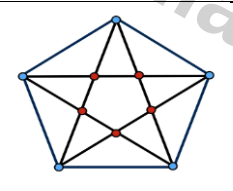
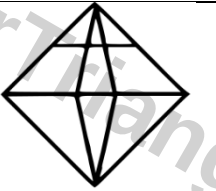
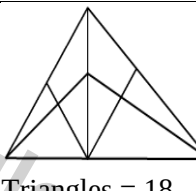
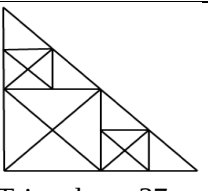
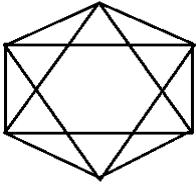
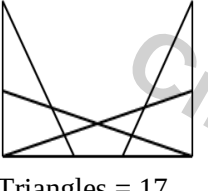
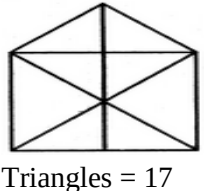
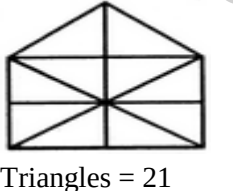
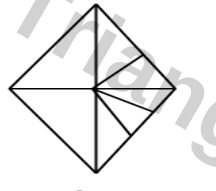
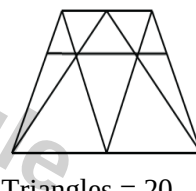
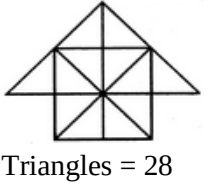
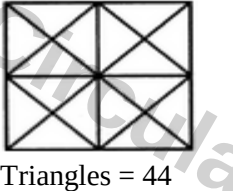
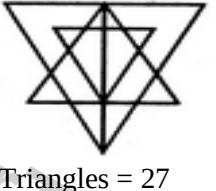
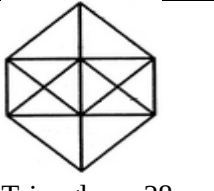
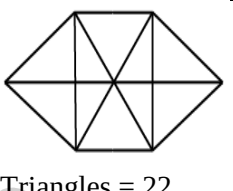
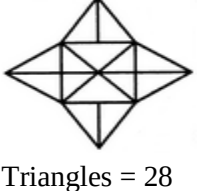
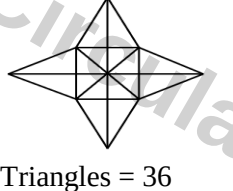
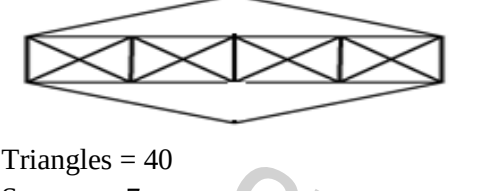
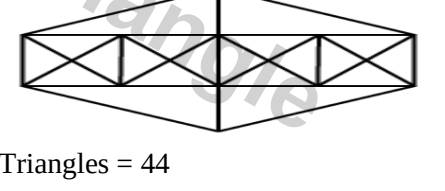
$$\text{Total number of given shape} = \frac{C \times (C+1)}{2} \times \frac{R \times (R+1)}{2}$$

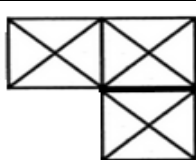
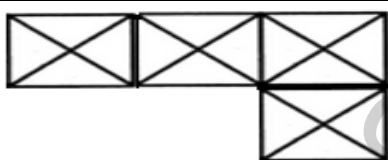
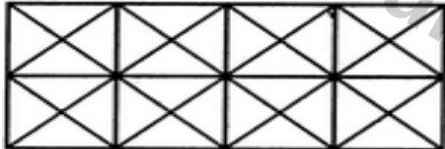
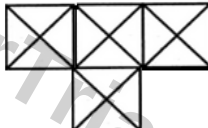
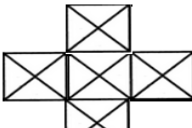
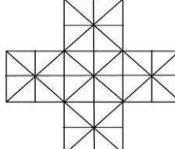
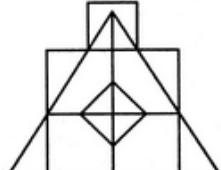
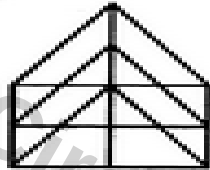
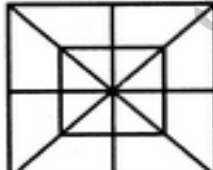
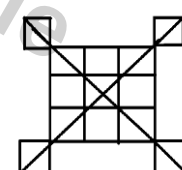
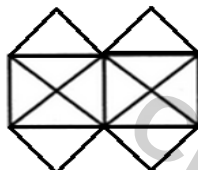
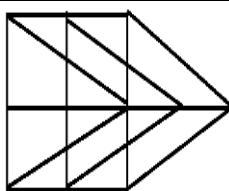
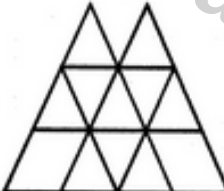
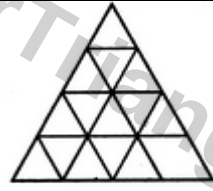
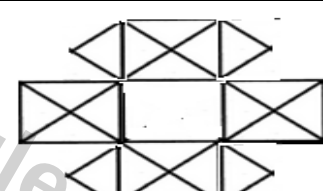
– E.g. 3×5 Size [i.e. Total number of given shape = $\frac{5 \times (5+1)}{2} \times \frac{3 \times (3+1)}{2} = 30$]

❖ Counting Triangles

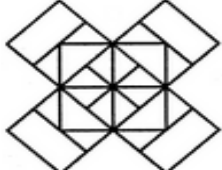
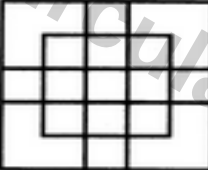
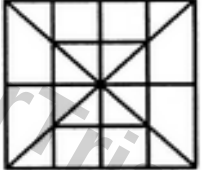
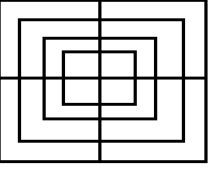
 Triangles = 10	 Triangles = 36	 Triangles = 21	 Triangles = 24	 Triangles = 14
 Triangles = 8	 Triangles = 15	 Triangles = 18	 Triangles = 27	
 Triangles = 15	 Triangles = 28	 Triangles = 18	 Triangles = 15	 Triangles = 10

				
Triangles = 28	Triangles = 14	Triangles = 25	Triangles = 28	Triangles = 17
				
Triangles = 15	Triangles = 21	Triangles = 30	Triangles = 28	
				
Triangles = 18	Triangles = 25	Triangles = 13	Triangles = 40	
				
Triangles = 16	Triangles = 15	Triangles = 22	Triangles = 29	
				
Triangles = 21	Triangles = 15	Triangles = 17	Triangles = 16	Triangles = 48
				Formula Number of triangles = $\frac{T(T+1)}{2}$ Where T = Total single triangle
Fig - 5 Triangle = 1	Fig - 6 Triangle = 3	Fig - 7 Triangle = 10	Fig - 8 Triangle = 15	
	All n vertex of a side If Even : $\frac{2n^3 - n^2 - 2n}{8}$ If Odd : $\frac{2n^3 - n^2 - 2n + 1}{8}$ If n = 5, Then Triangles = 27			Formula Number of triangles = $\frac{T(T+1)}{2} \times L$ Where T = Total single triangle, L = Line stages Triangles = 75
			Triangles = 36	
				
Triangles = 36	Triangles = 14	Triangles = 16	Triangles = 32	Triangles = 28

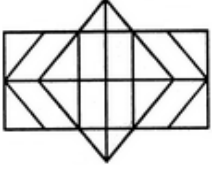
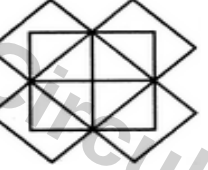
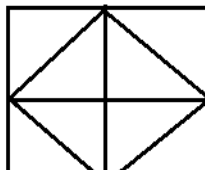
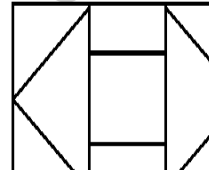
				
Triangles = 48	Triangles = 36	Triangles = 40	Triangles = 60	Triangles = 29
				
Triangles = 27	Triangles = 35	Triangles = 18	Triangles = 18	Triangles = 37
				
			Triangles = 32	Triangles = 17
				
Triangles = 17	Triangles = 21	Triangles = 15	Triangles = 20	
				
Triangles = 28 Square = 6	Triangles = 44 Square = 10	Triangles = 27		
				
Triangles = 28	Triangles = 22			
				
Triangles = 28	Triangles = 36			
				
Triangles = 40 Squares = 7		Triangles = 44		

 8 + 10 + 10 = 28 Triangles = 28 Squares = 5	 Triangles = 29	 Triangles = 39		
 Triangles = 24	 Triangles = 41	 Triangles = 56	 Triangles = 64	
 Triangles = 20 Squares = 7	 Triangles = 21 Parallelogram = 17	 Triangles = 32 Squares = 10	 Triangles = 28 Squares = 18	 Triangles = 20 Squares = 7
 Triangles = 19 Parallelogram = 17	 Triangles = 16 Parallelogram = 22	 Triangles = 27 Parallelogram = 39	 Triangles = 48	

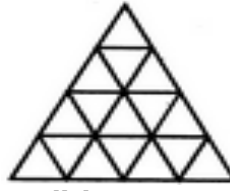
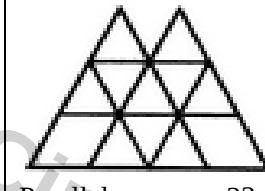
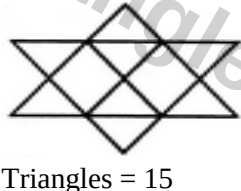
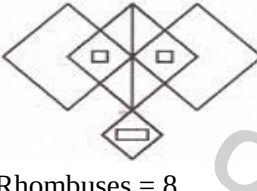
❖ Counting Squares

 Squares = 18	 Squares = 27	 Squares = 16	 Squares = $5 \times 4 = 20$	
---	---	---	---	--

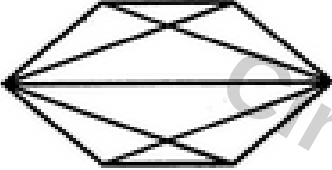
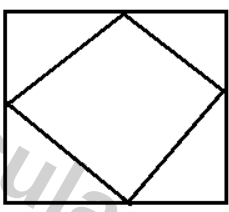
❖ Counting Rectangles/ Pentagone/ Hexagone

 Rectangles = 30 Hexagons = 5	 Rectangles = 20	 Pentagone = 12	 Rectangles = 11	
--	--	---	---	--

❖ Counting Parallelograms

 Parallelograms = 45	 Parallelograms = 23	 Triangles = 15 Parallelogram = 15	 Rhombuses = 8
--	--	--	--

❖ Counting Quardilater

 Quardilater = 9 Pentagon = 6	 Pentagon = 12		
---	--	--	--

- What is the number of triangles that can be formed whose vertices are the vertices of an octagon but have only one side common with that of octagon? → 32

Sub-chapter 14 - Fault-Finding Test

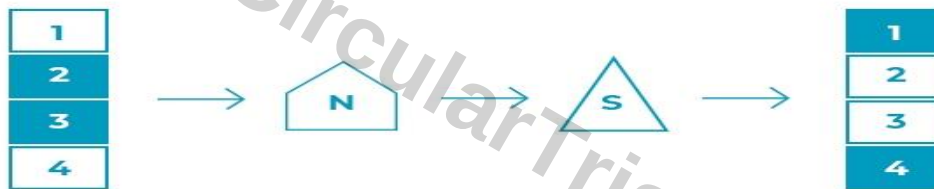
- Select the dysfunctional switch. → B

A. Switch N

B. Switch S

C. Both Switches

D. Neither Switch



— Functional: Inverts digits - turns on digits that are off & turns off digits that are on.

— Dysfunctional: Turns off all digits.



— Functional: Turns on digit 2 and turns off digit 3.

— Dysfunctional: No change.



— On



— Off

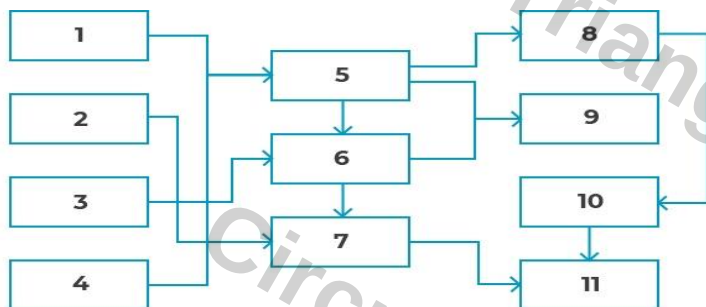
- If switches 1 & 4 are dysfunctional, how many switches will have an output signal? → B

A. 11

B. 6

C. 5

D. 8



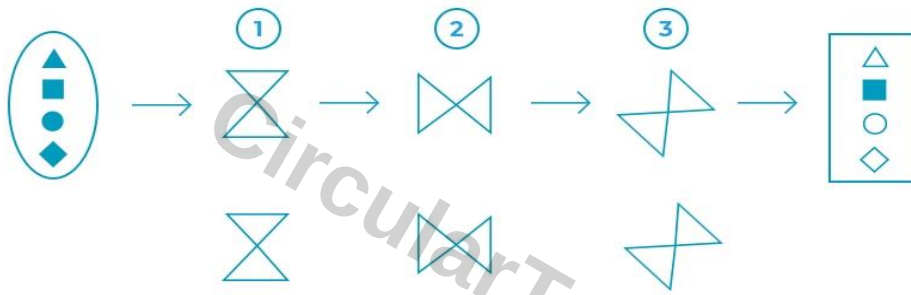
- Select the dysfunctional switch. → B

A. Switch 1

B. Switch 2

C. Switch 3

D. None



Functional	Transfers the  ,  , &  across.	Transfers the shapes on either end.	Transfers all shaded shapes.
Dysfunctional	Transfers all shaded shapes across.	Transfers the shape(s) in the middle.	Transfers first shaded shape.

Chapter 3 - Quantitative aptitude

Sub-chapter 1 - Percentage

❖ Comparing fractional value with respective percentage value -

- $\frac{1}{3} = 50\%$
- $\frac{1}{4} = 25\%$
- $\frac{1}{5} = 20\%$
- $\frac{1}{10} = 10\%$
- $\frac{1}{100} = 100\%$
- $\frac{1}{20} = 5\%$
- $\frac{1}{25} = 4\%$
- $\frac{3}{2} = 75\%$
- $\frac{2}{5} = 40\%$
- $\frac{3}{5} = 60\%$
- $\frac{4}{5} = 80\%$

• Successive increment/decrements of any number, That will be : Satisfies to :

- Number increment/decrements
- Noted number/ discount

$$[(x + y) + \frac{xy}{100}] \%$$

[Put (+) for increasing and (-) for decreasing]

$$[(x + y + z) + \frac{xy+yz+zx}{100} + \frac{xyz}{(100)^2}] \%$$

[Put (+) for increasing and (-) for decreasing]

- कुनै रकम A को x% पहिलो व्यक्तिले लिन्छ, बाँकी रकमको y% दोश्रो व्यक्तिले लिन्छ, बाँकी रकमको z% तेश्रो व्यक्तिले लिएपछि बाँकी रकम R रहन्छ भने शुरूको रकम = $R \left(\frac{100}{100-x} \right) \left(\frac{100}{100-y} \right) \left(\frac{100}{100-z} \right)$

- थप हुने अवस्थामा (+) चिन्हको प्रयोग गर्ने

- If **Pass mark = P%**, Obtained mark = O, Failed by F Then, Full marks = $\frac{100(O+F)}{P}$

- If any number is divided by y in instead of multiplying by x, Then mistake percentage = $\left(\frac{xy-1}{xy} \right) 100 \%$

- Making 2 times/ double : Increasing 100%**

- Making 3 times/ triple : Increasing 200%**

• Voting Problems

Two candidate	A	B
Obtained vote	65 %	35 %
Difference in vote = 768		
Then, 65 % - 35 % = 768 → 30 % = 768		
Let us consider, Total vote = x, So x = 76800/30 = 2560		

- If we write 261 with mistakenly to 216, Wrong percentage = ? Wrong percentage = $\frac{261-216}{261} \times 100 \%$
= 21 %

- If the price of a commodity increases by R%, then the reduction in consumption so as not to increase the expenditure is: $\left(\frac{R}{R+100} \times 100 \right) \%$

- If the price of a commodity decreases by R%, then the increase in consumption so as not to decrease the expenditure is: $\left(\frac{R}{R-100} \times 100 \right) \%$

- If A is R% more than B, then B is less than A by : $\left(\frac{R}{R+100} \times 100 \right) \%$

- If A is R% less than B, then B is more than A by : $\left(\frac{R}{R-100} \times 100 \right) \%$

A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets? → $45\frac{5}{11} \%$

Number of runs made by running = 110 - (3 x 4 + 8 x 6) = 110 - (60) = 50

Required percentage = $\left(\frac{50}{110} \times 100 \right) \%$ = $45\frac{5}{11} \%$

- What percentage of numbers from 1 to 70 have 1 or 9 in the unit's digit?** → $\frac{14}{70} \times 100\% = 20\%$

- If 20% of a = b, then b% of 20 is the same as:** 4% of a

- In a certain school, 20% of students are below 8 years of age. The number of students above 8 years of age is $\frac{2}{3}$ of the number of students of 8 years of age which is 48. What is the total number of students in the school?**

	Cost of Ram/ one	Cost of Shyam/ one	Then, $5x + 3y + 8z = 100$ $6x + 6y + 16z = 150$ $x = 25/2$
For Pen	Rs. 5 (x)	Rs. 6 (x)	
For copy	Rs. 3 (y)	Rs. 6 (y)	
For pencil	Rs. 8 (z)	Rs. 16 (z)	
Let Total cost	Rs. 100	Rs. 150	

Let the number of students be x. Then,

Number of students above 8 years of age = $x \times (100 - 20)\% = x \times 80\%$

$$x \times 80\% = 48 + 48 \times \frac{2}{3} \rightarrow x = 100$$

- **A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?** (Then, error/difference % = $\frac{\text{error/difference}}{\text{Fact}} \times 100\%$)

$$\text{Let the number be } x. \text{ Then, error/difference} = \frac{5x}{3} - \frac{3x}{5} = \frac{16x}{15}$$

$$\text{Then, error/difference \%} = \frac{16x/15}{5x/3} \times 100\% = 64\%$$

- **In an election between two candidates, one got 55% of the total valid votes, 20% of the votes were invalid. If the total number of votes was 7500, the number of valid votes that the other candidate got, was:** $\rightarrow (2700)$ (At first clear that valid and invalid vote)

Number of valid votes = 80% of 7500 = 6000.

Valid votes polled by other candidate = $6000 \times 45\% = 2700$

- **Two tailors X and Y are paid a total of Rs. 550 per week by their employer. If X is paid 120 percent of the sum paid to Y, how much is Y paid per week?** $\rightarrow (250)$
- **Gauri went to the stationers and bought things worth Rs. 25, out of which 30 paise went on sales tax on taxable purchases. If the tax rate was 6%, then what was the cost of the tax free items?** $\rightarrow (\text{Rs. } 19.70)$

- **Elections :**

हुईजना उम्मेदवारमध्ये विजयीले ६०% मत ल्याई १४००० मतले विजय भएमा उसले प्राप्त गरेको मत कति हो ?

विजयी	पराजित	$X \times 20\% = 14000$
६०%	४०%	$X = 70000$
फरक = २०%		विजयीले प्राप्त गरेको मत = ४२०००

Sub-chapter 2 - Decimal and Fraction

- Decimal can be converted into fraction dividing by 100.
- Fraction can be converted into decimal multiplying by 100.

1. Difference Testing in numerator and denominator :

- **For same difference** : Small the numerator is smaller fraction. (Eg. $-\frac{1}{2}, \frac{3}{4}, \frac{5}{6}$, $\rightarrow \frac{1}{2}$ is smallest)
- **For not equal difference** : Larger side is larger in multiplication by making $\frac{a}{b}$ and $\frac{c}{d}$ to ad ---- bc .

$$\text{Eg. } \frac{2}{7}, \frac{1}{3} \rightarrow 2 \times 3 \text{ ---- } 7 \times 1 \rightarrow 6 \text{ ---- } 7 \text{ (Second one is large so larger.)}$$

- **For same denominator** : Small the numerator is smaller fraction. . (Eg. $-\frac{1}{2}, \frac{3}{2}, \frac{5}{2}$, $\rightarrow \frac{1}{2}$ is smallest)
- **For same numerator** : Small the denominator is greater fraction. . (Eg. $-\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$, $\rightarrow \frac{1}{2}$ is greatest)
- **HCF and LCM of fractions** : $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$ $\text{HCF} = \frac{\text{HCF of } a, c, e}{\text{LCM of } b, d, f}$ $\text{LCM} = \frac{\text{LCM of } a, c, e}{\text{HCF of } b, d, f}$
- **Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case. (4)**

Required number = H.C.F. of $(91 - 43)$, $(183 - 91)$ and $(183 - 43)$

= H.C.F. of 48, 92 and 140 = 4

- **Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together ? (16)**
L.C.M. of 2, 4, 6, 8, 10, 12 is 120.
So, the bells will toll together after every 120 seconds (2 minutes).
In 30 minutes, they will toll together $\frac{30}{2} + 1 = 16$ times.
- **The greatest number of four digits which is divisible by 15, 25, 40 and 75 is: (9600)**
Greatest number of 4-digits is 9999.
L.C.M. of 15, 25, 40 and 75 is 600.
On dividing 9999 by 600, the remainder is 399.
Required number $(9999 - 399) = 9600$.
- **The product of two numbers is 4107. If the H.C.F. of these numbers is 37, then the greater number is: (111)**
Let the numbers be $37a$ and $37b$.
Then, $37a \times 37b = 4107 \Rightarrow ab = 3$.
Now, co-primes with product 3 are (1, 3). So, the required numbers are $(37 \times 1, 37 \times 3)$ i.e., (37, 111). Greater number = 111.
- **Three numbers are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is: (40)**
Let the numbers be $3x$, $4x$ and $5x$.
Then, their L.C.M. = $60x$. So, $60x = 2400$ or $x = 40$.
The numbers are (3×40) , (4×40) and (5×40) . Hence, required H.C.F. = 40.
- **The G.C.D. of 1.08, 0.36 and 0.9 is: (0.18)**
Given numbers are 1.08, 0.36 and 0.90. H.C.F. of 108, 36 and 90 is 18,
H.C.F. of given numbers = 0.18.
- **The product of two numbers is 2028 and their H.C.F. is 13. The number of such pairs is: (2)**
Let the numbers be $13a$ and $13b$.
Then, $13a \times 13b = 2028 \Rightarrow ab = 12$
Now, the co-primes with product 12 are (1, 12) and (3, 4).
So, the required numbers are $(13 \times 1, 13 \times 12)$ and $(13 \times 3, 13 \times 4)$.
Clearly, there are 2 such pairs.
- **The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is: (364)**
L.C.M. of 6, 9, 15 and 18 is 90.
Let required number be $90k + 4$, which is multiple of 7.
Least value of k for which $(90k + 4)$ is divisible by 7 is $k = 4$.
Required number = $(90 \times 4) + 4 = 364$.
- **The least number which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4 and 3 is: (23)**
L.C.M. of 5, 6, 4 and 3 = 60.
On dividing 2497 by 60, the remainder is 37.
Number to be added = $(60 - 37) = 23$.
- **The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is: (1683)**
L.C.M. of 5, 6, 7, 8 = 840.
Required number is of the form $840k + 3$
Least value of k for which $(840k + 3)$ is divisible by 9 is $k = 2$.
Required number = $(840 \times 2 + 3) = 1683$.
- **The H.C.F. of two numbers is 11 and their L.C.M. is 7700. If one of the numbers is 275, then the other is: (308)**
Other number = $\frac{11 \times 7700}{275} = 308$
- **The ratio of two numbers is 3 : 4 and their H.C.F. is 4. Their L.C.M. is : (48)**

Let the numbers be $3x$ and $4x$. Then, their H.C.F. = x . So, $x = 4$.
 So, the numbers 12 and 16.
 L.C.M. of 12 and 16 = 48.

- **The smallest number which when diminished by 7, is divisible 12, 16, 18, 21 and 28 is: (1015)**
 Required number = (L.C.M. of 12, 16, 18, 21, 28) + 7 = 1008 + 7 = 1015

- **The least number, which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8 is: (548)**

Required number = (L.C.M. of 12, 15, 20, 54) + 8 = 540 + 8 = 548

- **The greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively, is: (127)**

Required number = H.C.F. of (1657 - 6) and (2037 - 5)
 = H.C.F. of 1651 and 2032 = 127.

- **If the sum of two numbers is 55 and the H.C.F. and L.C.M. of these numbers are 5 and 120 respectively, then the sum of the reciprocals of the numbers is equal to: ($\frac{11}{120}$)**

Let the numbers be a and b .

Then, $a + b = 55$ and $ab = 5 \times 120 = 600$.

The required sum = $\frac{1}{a} + \frac{1}{b} = \frac{11}{120}$

- **What decimal of an hour is a second ? (0.00027)**
- **The price of commodity X increases by 40 paise every year, while the price of commodity Y increases by 15 paise every year. If in 2001, the price of commodity X was Rs. 4.20 and that of Y was Rs. 6.30, in which year commodity X will cost 40 paise more than the commodity Y ? (2011)**

Suppose commodity X will cost 40 paise more than Y after z years.

Then, $(4.20 + 0.40z) - (6.30 + 0.15z) = 0.40 \Rightarrow 0.25z = 0.40 + 2.10$

$$z = \frac{2.50}{0.25} = 10$$

X will cost 40 paise more than Y 10 years after 2001 i.e., 2011.

- **The correct expression of 6.46 (overline) in the fractional form is: $6 + \frac{46}{99} = \frac{640}{99}$**
- **There are two examinations rooms A and B. If 10 students are sent from A to B, then the number of students in each room is the same. If 20 candidates are sent from B to A, then the number of students in A is double the number of students in B. The number of students in room A is: (100)**

Let the number of students in rooms A and B be x and y respectively.

Then, $x - 10 = y + 10$ $x - y = 20$ (i) and $x + 20 = 2(y - 20)$ $x - 2y = -60$ (ii)

Solving (i) and (ii) we get: $x = 100$, $y = 80$.

The required answer A = 100.

- **The price of 10 chairs is equal to that of 4 tables. The price of 15 chairs and 2 tables together is Rs. 4000. The total price of 12 chairs and 3 tables is: (Rs. 3900)**
- **A fires 5 shots to B's 3 but A kills only once in 3 shots while B kills once in 2 shots. When B has missed 27 times, A has killed: (30 Birds)**

Let the total number of shots be x . Then,

$$\text{Shots fired by A} = \frac{5x}{8}$$

$$\text{Shots fired by B} = \frac{3x}{8}$$

$$\text{Killing shots by A} = \frac{5x}{8} \times \frac{1}{3} = \frac{5x}{24}$$

$$\text{Shots missed by B} = \frac{3x}{8} \times \frac{1}{2} = \frac{3x}{16} \quad \text{So, } x = 144$$

$$\text{Birds killed by A} = \frac{5x}{24} = 30$$

- **Eight people are planning to share equally the cost of a rental car. If one person withdraws from the arrangement and the others share equally the entire cost of the car, then the share of each of the remaining persons increased by: ($\frac{1}{7}$)**

$$\text{Original share of 1 person} = \frac{1}{8} \quad \text{New share of 1 person} = \frac{1}{7} \quad \text{Increase} = \frac{1}{56}$$

$$\text{Required fraction} = \frac{\frac{1}{56}}{\frac{1}{8}} = \frac{1}{7}$$

- To fill a tank, 25 buckets of water is required. How many buckets of water will be required to fill the same tank if the capacity of the bucket is reduced to two-fifth of its present ? (

Let the capacity of 1 bucket = x .

Then, the capacity of tank = $25x$.

$$\text{New capacity of bucket} = \frac{2x}{5}$$

$$\text{Required number of buckets} = \frac{25x}{\frac{2x}{5}} = 62.5$$

- In a regular week, there are 5 working days and for each day, the working hours are 8. A man gets Rs. 2.40 per hour for regular work and Rs. 3.20 per hours for overtime. If he earns Rs. 432 in 4 weeks, then how many hours does he work for ? (175)**

Suppose the man works overtime for x hours.

Now, working hours in 4 weeks = $(5 \times 8 \times 4) = 160$.

$$\therefore 160 \times 2.40 + x \times 3.20 = 432 \Rightarrow 3.20x = 432 - 384 = 48 \Rightarrow x = 15.$$

Hence, total hours of work = $(160 + 15) = 175$.

- Free notebooks were distributed equally among children of a class. The number of notebooks each child got was one-eighth of the number of children. Had the number of children been half, each child would have got 16 notebooks. Total how many notebooks were distributed ? (512)**

Let total number of children be x . Then, $x \times \frac{x}{8} = \frac{x}{2} \times 16$ So, $x = 64$

$$\text{Number of notebooks} = \frac{x^2}{8} = 512$$

- David gets on the elevator at the 11th floor of a building and rides up at the rate of 57 floors per minute. At the same time, Albert gets on an elevator at the 51st floor of the same building and rides down at the rate of 63 floors per minute. If they continue travelling at these rates, then at which floor will their paths cross ? (30)**

Suppose their paths cross after x minutes.

$$\text{Then, } 11 + 57x = 51 - 63x \Leftrightarrow 120x = 40 \quad \text{So, } x = \frac{1}{3}$$

$$\text{Number of floors covered by David in } (1/3) \text{ min.} = 57 \times \frac{1}{3} = 19$$

So, their paths cross at $(11 + 19)$ i.e., 30th floor.

- Mixture of milk and water.**

	Milk	Water	Total	Sold after and water, Water = 15 Ltr. + 25 Ltr. Water : Milk = 2:3 (Milk : Water = 3:2)
	80 Ltr.	20 Ltr.	100 Ltr.	
Sold (1/4)	20 Ltr.	5 Ltr.	25 Ltr.	
Remaining	60 Ltr.	15 Ltr.	75 Ltr.	
+ Water	60 Ltr.	40 Ltr.	100 Ltr.	

Sub-chapter 3 - Ratio and proportion

- The ratio of two quantities **a** and **b** in the same units, is the fraction $\frac{a}{b}$
- The ratio **a : b**. Where, **a** is **antecedent** and **b** is **consequent**.
- The equality of two ratios is called proportion. If $a : b = c : d$, we write $a : b :: c : d$
- Here **a** and **d** are called **extremes**, while **b** and **c** are called **mean terms**.
- c** is called **third propotional to a and b** and **d** is **fourth propotional to a, b and c**.
- Product of means = Product of extremes. Thus, $a : b :: c : d \rightarrow (bc) = (ad)$.
- Mean proportional between **a** and **b** is $\sqrt{ab}$.
- Only homologous substance or unit can be calculated in ration and proportion
- The compounded ratio of the ratios: $(a : b)$, $(c : d)$, $(e : f)$ is $(ace : bdf)$.

- **Duplicate Ratios:**

- Duplicate ratio of (a : b) is (a² : b²).

Sub-duplicate ratio of (a : b) is ($\sqrt{a}$: $\sqrt{b}$).

- Triplicate ratio of (a : b) is (a³ : b³).

Sub-triplicate ratio of (a : b) is (a^{1/3} : b^{1/3}).

- If $\frac{a}{b} = \frac{c}{d}$ Then, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

- $X : Y = \frac{\text{Inverse ratio}}{\text{of } x:y} = Y : X$

- $X : Y : Z = \frac{\text{Inverse ratio}}{\text{of } x:y:z} = \frac{1}{x} : \frac{1}{y} : \frac{1}{z}$

- If 2 : 3 : 4 $\frac{\text{Inverse ratio}}{\text{of } 2:3:4} = \frac{1}{2} : \frac{1}{3} : \frac{1}{4} = \frac{12}{2} : \frac{12}{3} : \frac{12}{4} = 6 : 4 : 3$

- If $X : Y : Z = \frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ Then, $X : Y : Z = \frac{12}{2} : \frac{12}{3} : \frac{12}{4} = 6 : 4 : 3$

- If A : B = 3 : 4, A : C = 8 : 9, A : B : C = ?

$\frac{B}{A} \times \frac{A}{C} = \frac{4}{3} \times \frac{8}{9} = \frac{32}{27}$ So, B : C = 32 : 27 Now finding A : B : C by A : B and b : C.

- If sum of two number is x and difference is y, then ratio will be, (x + y) : (x - y)

- If A : B = 3 : 4, B : C = 8 : 9, C : D = 15 : 16 Then, A : B : C : D = ? (Note - Fill in the blanks by nearest number from own row)

A	B	C	D	A : B : C : D
3	4	(4)	(4)	
.....(8)	8	9	(9)	
.....(15)	(15)	15	16	
= 3 × 8 × 15 = 30	= 4 × 8 × 15 = 40	= 4 × 9 × 15 = 45	= 4 × 9 × 16 = 48	30 : 40 : 45 : 48

- If A : B : C : D = 3 : 2 : 5 : 4 and D : E = 3 : 5, Then A : B : C : D : E = ?

A	B	C	D	E
3 × 3	2 × 3	5 × 3	4 × 3	4 × 5
Trick : A, B, C, D are multiplying by numerator of D : E and D is multiplying by both numerator and denominator of D : E.				

- **Coins and ratio :**

Eg. - मोहोर, सुकी र गाई पैसेको अनुपात = 5 : 9 : 4 and Total Rupees is Rs. 206 Then, How many each other ?

- All are converted into homologous quantity as poisa.

- So, 50 × 5x + 25 × 9x + 10 × 4x = 206 × 100 → 515x = 20600 → x = 40

- Hence, Mohor = 5 × 40 = 200, Suki = 9 × 40 = 360 and Gai poise = 4 × 40 = 160

- In a bag, there are coins of 25 p, 10 p and 5 p in the ratio of 1 : 2 : 3. If there is Rs. 30 in all, how many 5 p coins are there? (150)

Let the number of 25 p, 10 p and 5 p coins be x, 2x, 3x respectively.

Then, sum of their values = (25x + 10 × 2x + 5 × 3x)p = Rs. $\frac{60x}{100}$ = Rs. 30 → x = Rs. 50

Hence, the number of 5 p coins = (3 × 50) = 150.

- Someone travelled by Train, Bus and Car in the ratio of distance of journey is 4 : 3 : 2 of which ratio of fare is 1 : 2 : 4, Total fare is Rs. 720 Then, fare of Train, Bus and Car = ?

	Train	Bus	Car	4x+6x+8x=720 So, Ratio of Fare = 2 : 3 : 4
Distance of journey	4	3	2	Therefore,
Fare Ratio	1	2	4	Train = Rs. 160, Bus = 240, Car = 320
Total Fare	4×1=4	3×2=6	2×4=8	

* It satisfies in unitary method ratio.

- **A and B together have Rs. 1210. If $\frac{4}{15}$ of A's amount is equal to $\frac{2}{5}$ of B's amount, how much amount does B have? (Rs. 484)**

$$\frac{4A}{15} = \frac{2B}{5} \rightarrow \frac{A}{B} = \frac{3}{2} \quad \text{Hence, B's share} = \text{Rs. } \frac{2}{5} \times 1210 = 484$$

- **Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is: (4 : 5)**

Let the third number be x. which is 100% itself.

$$\text{Then, first number} = 120\% \text{ of } x = \frac{6x}{5} \quad \text{Second number} = 150\% \text{ of } x = \frac{3x}{2}$$

Ratio of first two numbers = 4 : 5

- **A sum of money is to be distributed among A, B, C, D in the proportion of 5 : 2 : 4 : 3. If C gets Rs. 1000 more than D, what is B's share? (2000)**

$$\text{Let } A = 5x, B = 2x, C = 4x, D = 3x \quad \text{So, } 4x = 3x + 1000 \rightarrow x = 1000 \quad \text{Hence B's share} = 2000$$

- **Seats for Mathematics, Physics and Biology in a school are in the ratio 5 : 7 : 8. There is a proposal to increase these seats by 40%, 50% and 75% respectively. What will be the ratio of increased seats? (2 : 3 : 4)**

Initial ratio of A, B, C = 5 : 7 : 8 = 5x : 7x : 8x

By calculating we find new ratio = 7x : $\frac{21x}{2}$: 14x = 14x : 21x : 28x

Hence, Final ratio = 2 : 3 : 4

- **In a mixture 60 litres, the ratio of milk and water 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is: (From first sight, 60 ltr)**

- In a mixture ratio of milk and water = a : b, Now x ltr. water is added, then ratio of new mixture = a :

$$c. \text{ Quantity of milk} = \frac{ax}{c-b} \text{ ltr. and Quantity of water} = \frac{bx}{c-b}$$

- **The sum of three numbers is 98. If the ratio of the first to second is 2 : 3 and that of the second to the third is 5 : 8, then the second number is: (30) (First finding a : b : c)**

- **If Rs. 782 be divided into three parts, proportional to $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$ then the first part is: (Rs. 204)**

(First break the fractional ratio in $\frac{12}{2} : \frac{24}{3} : \frac{36}{4} = 6 : 8 : 9$)

- **The students in the three classes are in ratio 2 : 3 : 5. If 40 students are increased in each class, the new ratio changes to 4 : 5 : 7. How many students are there in Original ?**

Let, The students in the three classes are = 2x, 3x and 5x.

$$\text{So, } 2x + 40 : 3x + 40 : 5x + 40 = 4 : 5 : 7$$

$$\text{Now, } \frac{2x+40}{3x+40} = \frac{4}{5} \rightarrow 5(2x+40) = 4(3x+40) \rightarrow x = 20 \quad \text{Hence original numbers are, 40, 60, 80}$$

- **The ratio of land to water for the whole earth is 1:2 and 2:3 in the northern hemisphere. What is the ratio of land to water in the southern hemisphere? (Hence Ratio = $\frac{2}{15} : \frac{11}{30} = 4 : 11$)**

	Ratio	Land	Water
For whole Earth	1 : 2	$\frac{1}{3}$	$\frac{2}{3}$
Northern Hemisphere	2 : 3	$\frac{2}{5} \times \frac{1}{2} = \frac{1}{5}$	$\frac{3}{5} \times \frac{1}{2} = \frac{3}{10}$
Remaining (Southern Hemisphere)	?	$\frac{1}{3} - \frac{1}{5} = \frac{2}{15}$	$\frac{2}{3} - \frac{3}{10} = \frac{11}{30}$

Sub-chapter 4 - Alligation or Mixture

- It is the rule that enables us to find the ratio in which two or more ingredients at the given price must be mixed to produce a mixture of desired price
- The cost of a unit quantity of the mixture is called the mean price.

- If two ingredients are mixed, then $\frac{\text{Quantity of cheaper}}{\text{Quantity of dearer}} = \frac{\text{C.P. of dearer} - \text{Mean price}}{\text{Mean price} - \text{C.P. of cheaper}} \rightarrow \frac{QC}{QD} = \frac{CPD - MP}{MP - CPC}$
- Suppose a container contains x of liquid from which y units are taken out and replaced by water. After n times, the quantity of pure liquid = $x \left(1 - \frac{y}{x}\right)^n$ units.

If replacement is in different quantity then, the quantity of pure liquid = $x \left(1 - \frac{y_1}{x}\right) \left(1 - \frac{y_2}{x}\right) \dots$

- A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup? ($\frac{1}{5}$)

Suppose the vessel initially contains 8 litres of liquid.

Let f litres of this liquid be fraction removed.

Left amount = Initial amount (1 - Fraction removed)

$$\rightarrow \frac{1}{2} = \frac{5}{8} (1 - \text{Fraction removed}) \rightarrow f = \frac{1}{5}$$

- Tea worth Rs. 126 per kg and Rs. 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be: (Rs. 175.50)

Tea	T1	T2	T3	mixture is worth Rs. 153 per kg So, $126 + 135 + 2x = 4 \times 153$ $x = \text{Rs. } 175.50$
price per kg	126	135	x	
Ratio	1	1	2	

- A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially? $\rightarrow (21)$

Left amount = Initial amount (1 - Fraction removed)

$$\rightarrow \frac{7}{16} = \frac{7}{14} \left(1 - \frac{9}{x}\right) \rightarrow x = 36 \text{ (Note : A and B = 21 and 16)}$$

- A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each of the containers so as to get 12 litres of milk such that the ratio of water to milk is 3 : 5? (6 litres, 6 litres)

Can 1		Can 2		New mixture		ratio of water to milk is 3 : 5 water milk 25% 50% 37.5% 37.5% 12.5% 12.5%
Water	Milk	Water	Milk	Water	Milk	
25%	75%	50%	50%	$\frac{5}{8}$	$\frac{3}{8}$ or 37.5%	
6 litres, 6 litres						

- In what ratio must a grocer mix two varieties of pulses costing Rs. 15 and Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.50 kg? (7 : 3)

Cost of 1 kg pulses of 1st kind

Rs. 15

Cost of 1 kg pulses of 2nd kind

Rs. 20

Mean price : Rs. 16.50

3.5 or $\frac{7}{2}$

1.5 or $\frac{3}{2}$

Hence, Ratio = 7 : 3

- A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is: (20%)

Let C.P. of 100 litre milk be Re. 100

$$100 \text{ ltr.} + 25 \text{ ltr.} = 125 \text{ ltr.} \quad \text{So, } \frac{25}{125} \times 100 = 20\%$$

- How many kilogram of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10% by selling the mixture at Rs. 9.24 per kg? (63 kg)

- A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container? (29.16 ltr)

- A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is: (2/3)
- In what ratio must water be mixed with milk to gain $16\frac{2}{3}\%$ on selling the mixture at cost price? (1 : 6)
- Find the ratio in which rice at Rs. 7.20 a kg be mixed with rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg. (2 : 3) (By the alligation method)
- In what ratio must a grocer mix two varieties of tea worth Rs. 60 a kg and Rs. 65 a kg so that by selling the mixture at Rs. 68.20 a kg he may gain 10%? (3.2)
- The cost of Type 1 rice is Rs. 15 per kg and Type 2 rice is Rs. 20 per kg. If both Type 1 and Type 2 are mixed in the ratio of 2 : 3, then the price per kg of the mixed variety of rice is: (18)

Cost of 1 kg of Type 1 rice = 15 Cost of 1 kg of Type 2 rice = 20

Mean Price = x

20-x x-15 $\frac{20-x}{x-15} = \frac{2}{3} \rightarrow x = 18$

- 8 litres are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of water is 16 : 65. How much wine did the cask hold originally? (24 ltr)
- A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is: (600 kg)

पहिलो समूहको औसत दोश्रो समूहको औसत कुल औसत अनुपात : पहिलो समूहको संख्या पहिलो समूहको संख्या	• कुनै कक्षामा छात्रको औसत स्कोर 25 र छात्रको औसत स्कोर 40 र कक्षाको औसत 30 र छात्रको संख्या 30 भए छात्रको संख्या कति हुन्छ ? (चित्रअनुसार) अनुपात : 2 : 1 त्यसैले छात्रको संख्या = 60
पुरुषको जनसंख्या वृद्धि प्रतिशत पुरुषको जनसंख्या वृद्धि प्रतिशत कुल जनसंख्या वृद्धि % अनुपात : पुरुषको सुरुको जनसंख्या महिलाको सुरुको जनसंख्या	• कुनै शहरको जनसंख्या 24000 थियो, पुरुष र महिलाको जनसंख्या क्रमशः 11% र 17% ले वृद्धि हुँदा कुल जनसंख्या 15% ले वृद्धि भयो भने सुरुको पुरुष र महिलाको जनसंख्या कति होला ? (चित्रअनुसार) सुरुको अनुपात : 2 : 4 पुरुषको संख्यम = 8000 महिलाको संख्या = 16000
खर्चमा घटवढ % बचतमा घटवढ % आयमा घटवढ % अनुपात : खर्च बचत कुल आय = खर्च + बचत	• कुनै परिवारको आय 6% ले घट्दा र खर्च 8% ले घट्दा बचत 4% ले वृद्धि हुन्छ भने आय र बचतको अपात कति होला ? (चित्रअनुसार) खर्च बचत अनुपात = 5 : 1 त्यसैले आय खर्च अनुपात = 6 : 5

पहिलो वस्तुमा नाफा/नोक्सान % दोश्रो वस्तुमा नाफा/नोक्सान % कुल नाफा/नोक्सान अनुपात : पहिलो वस्तुको क्र.मु. दोश्रो वस्तुको क्र.मु.	- एक व्यापारीले 50 के.जी. चामलको केही भाग 8% नाफामा र बाँकी 18% नाफामा बेचेमा कुल नाफा 14% भए पहिलो विक्रीको चामलको परिणम कति होला ? (चित्रअनुसार) - पहिलो र दोश्रो विक्रीको अनुपात = 2 : 3 - जम्मा परिणाम = 50 के.जी. - पहिलो विक्रीको चामलको परिणम = 20 के.जी.
पहिलो मिश्रणमा वस्तुको मात्रा दोश्रो मिश्रणमा वस्तुको मात्रा नयाँ बने मिश्रणमा वस्तुको मात्रा अनुपात : नयाँ मिश्रण बनाउन पहिलो वस्तुको मात्रा नयाँ मिश्रण बनाउन दोश्रो वस्तुको मात्रा	- If the ratio of milk and water is 5:2 in one pot and the ratio of milk and water is 8:5 in another pot, what ratio of mixture is taken from both pot that the ratio of milk and water will be 9:4? (See only milk.) - Concentration of milk in pot 1 = 5/7 - Concentration of milk in pot 2 = 8/13 - Concentration of milk after mixing = 9/13 - The required ratio is 1/13 : 2/91 = 7/2
पहिलो वस्तुको मूल्य दोश्रो वस्तुको मूल्य मिश्रित वस्तुको नयाँ मूल्य अनुपात : नयाँ वस्तुमा पहिलो र दोश्रो वस्तुको परिमाणको अनुपात	- रु. 15 के.जी. मूल्य भएको पिठो र रु. 30 के.जी. मूल्य भएको पिठोलाई कुन अनुपातमा मिलाउँदा रु. 25 के.जीको पिठो हुन्छ ? (चित्रअनुसार) - अनुपात = 1 : 2

- Two vessels of equal quantities contain milk and water in the ratio 2:3 and 3:4 respectively. What will be the ratio of the milk in the two vessels? (See only milk.)

$$\text{Milk in first vessel} = \frac{2}{5}$$

$$\text{Milk in second vessel} = \frac{3}{7}$$

$$\text{Then, ratio of the milk in two vessels} = \frac{2}{5} : \frac{3}{7} = 14 : 15$$

- Two vessels of equal quantities contain milk and water in the ratio 2:3 and 3:4 respectively. What will be the ratio of the milk and water quantity from both the bases in mixed?

$$\text{Total milk} = \frac{2}{5} + \frac{3}{7} = \frac{29}{35}$$

$$\text{Total water} = \frac{3}{5} + \frac{4}{7} = \frac{41}{35}$$

$$\text{Therefore, New ratio} = 29 : 41 \quad (\text{Trick : Work for three substance mixed})$$

- Ram has to completely put each of the three liquids 120 ltrs of milk, 150 ltrs of jouice, 180 ltrs of water in bottle of equal size without mixing any of the above tyree types of liquids such that each bottle is completely filled.What is the list possible number of bottles required ? → 15

Sub-chapter 5 - Average

- Important : Number × Average = Sum of total number**
Eg. Number of students = x, Average Age = y, Then Sum of age of total students = xy
- Suppose a man covers a **certain distance** at x kmph and an **equal distance** at y kmph,
Then, the average speed druing the whole journey is = $\frac{2xy}{x+y}$ kmph
- Average of first n natural number : $\frac{n(n+1)}{2n}$

Number of students	Average Age	Sum of total Age = 36 × 14 = 504
36	14	Sum of total Age = 37 × 15 = 555 (After inclusion)
Teacher Included = 37	15	Age of teacher = 555 - 504 = 51

Number of students	Average Age	Sum of total Age = 10 × 25 = 250
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10	25	Real Sum of total Age = 250 - Wrong + Right = 230
Age of one is read as 56 instead of 36		Real Average Age of total students = 230/10 = 23

Number of any number	Average	Average of (m + n) = ?	Average of m + n = $\frac{m \times n}{m+n}$
m	n^2	Sum of total = mn^2	Because : $\frac{mn^2+m^2n}{m+n} = mn$
n	m^2	Sum of total = m^2n	

Average marks scored by total students (Let, 100)	68	Note : Consider x for necessary one
Average marks scored by boys (Let, x)	60	$60x + 80(100-x) = 100 \times 68$
Average marks scored by boys (Let, 100-x)	80	boys % (x) = $1200/20 = 60\%$

Speed of car from point A to B = 10km/hrs. (x)	Average rate of total speed of car = $\frac{2xy}{x+y}$ Total Average speed = 300 / 25 = 12 km/hr.
Speed of car from point B to A = 15km/hrs. (y)	
(Note : Satisfied for half/ half straight way.)	

- In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs? (RR = 6.25)
- The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero? (At most = 19, At least = 1)
- The average weight of 8 person's increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person? (85 kg)
Total weight increased = (8×2.5) kg = 20 kg.
Weight of new person = $(65 + 20)$ kg = 85 kg.
- The captain of a cricket team of 11 members is 26 years old and the wicket keeper is 3 years older. If the ages of these two are excluded, the average age of the remaining players is one year less than the average age of the whole team. What is the average age of the team? (23 years)
Let the average age of the whole team by x years.

Member	Captain	WC	9 member	
Average Age	26	29	x-1	$26 \times 1 + 29 \times 1 + 9(x-1) = 11 \times x$ $x = 23$ years

- The average monthly income of P and Q is Rs. 5050. The average monthly income of Q and R is Rs. 6250 and the average monthly income of P and R is Rs. 5200. The monthly income of P is: (Rs. 4000)

P + Q = $(5050 \times 2) = 10100$ (i)	$2(P + Q + R) = 33000 \rightarrow P + Q + R = 16500$
Q + R = $(6250 \times 2) = 12500$ (ii)	So, P = $(P + Q + R) - (Q + R)$
P + R = $(5200 \times 2) = 10400$ (iii)	$= 16500 - 12500 = 4000$

- The average age of husband, wife and their child 3 years ago was 27 years and that of wife and the child 5 years ago was 20 years. The present age of the husband is: (40 years)
Present age of husband, wife and their child = x, y and z
 $\frac{x-3+y-3+z-3}{3} = 27 \rightarrow x+y+z = 90$ i
 $\frac{y-5+z-5}{2} = 20 \rightarrow y+z = 50$ ii, So Present age of husband = 40 years.
- A car owner buys petrol at Rs.7.50, Rs. 8 and Rs. 8.50 per litre for three successive years. What approximately is the average cost per litre of petrol if he spends Rs. 4000 each year? (Rs. 7.98)
Consume of petrol in three years = $\frac{4000}{7.5} + \frac{4000}{8} + \frac{4000}{8.5} = \frac{76700}{51}$
Total amount spent = Rs. $(3 \times 4000) =$ Rs. 12000
Average cost per ltr. = Rs. $\frac{12000 \times 51}{76700} =$ Rs. 7.98
- A pupil's marks were wrongly entered as 83 instead of 63. Due to that the average marks for the class got increased by half ($1/2$). The number of pupils in the class is:
Let there be x pupils in the class.
Total increase in marks = Total increase in average $\rightarrow 20 = \frac{x}{2} \rightarrow x = 40$

- If average of x, y and z is A, then average of x, y, z and A will be also A.

Sub-chapter 5.1 - Data Sufficiency from Average

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- A. I alone sufficient while II alone not sufficient to answer
 - B. II alone sufficient while I alone not sufficient to answer
 - C. Either I or II alone sufficient to answer
 - D. Both I and II are not sufficient to answer
 - E. Both I and II are necessary to answer
- **The average age of P, Q, R and S is 30 years. How old is R? → D**
 1. The sum of ages of P and R is 60 years.
 2. S is 10 years younger than R.
 - **How many candidates were interviewed everyday by the panel A out of the three panels A, B and C? → B**
 1. The three panels on average interview 15 candidates every day.
 2. Out of a total of 45 candidates interviewed everyday by the three panels, the number of candidates interviewed by panel A is more by 2 than the candidates interviewed by panel c and is more by 1 than the candidates interviewed by panel B.
 - **What is the average age of children in the class? → D**
 1. The age of the teacher is as many years as the number of children.
 2. Average age is increased by 1 year if the teacher's age is also included.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- **How many marks did Tarun secure in English? → E**
 1. The average mark obtained by Tarun in four subjects including English is 60.
 2. The total marks obtained by him in English and Mathematics together are 170.
 3. The total marks obtained by him in Mathematics and Science together are 180.
 - A. I and II only
 - B. II and III only
 - C. I and III only
 - D. All I, II and III
 - E. None of these
- **In a cricket team, the average age of eleven players in 28 years. What is the age of the captain? → C**
 1. The captain is eleven years older than the youngest player.
 2. The average age of 10 players, other than the captain is 27.3 years.
 3. Leaving aside the captain and the youngest player, the average ages of three groups of three players each are 25 years, 28 years and 30 years respectively.
 - A. Any two of the three
 - B. All I, II and III
 - C. II only or I and III only
 - D. II and III only
 - E. None of these

Sub-chapter 6 - Profit and loss

- **Relation between CP, SP, MP.**

$$SP = MP \left(1 - \frac{D}{100}\right) = CP \left(1 + \frac{P/L}{100}\right)$$
..... (-) for loss and MP is market price.
- If cost price is no given, We can consider as cost price as 100.
- **If SP of two similar items is same with x% profit/loss and y% profit/loss,**

Then, Gross profit/loss % = $\frac{100x+2xy+100y}{x+200+y}$ % (-) for loss.

- If SP of two similar items is same with x% profit and x% loss, Then, Gross loss % = $(\frac{x}{10})^2$
- If profit and loss % is same for an item but different SP is x and y. Then, CP = $\frac{x+y}{2}$
- If an article is sold by, gain 35%, then C.P. = 100% and SP = 135% of C.P.
- If an article is sold by loss 35% then C.P. = 100% and SP = 65% of C.P.
- Successive discount x% and y%, Then, Final discount = $(x + y + \frac{xy}{100})$ (Use - sign for discount)

$$\text{Final discount} = (x + y + z) + \frac{xy+yz+xz}{100} + \frac{xyz}{10000}$$

- यदि कुनै वस्तु A ले B लाई x% नाफा/घाटामा फेरी B ले C लाई y% नाफा/घाटामा फेरी C ले D लाई z% नाफा/घाटामा बेचेमा

$$A \text{ को क्र.मु.} = D \text{ को क्र.मु.} \times \left(\frac{100}{x+100}\right) \left(\frac{100}{y+100}\right) \left(\frac{100}{z+100}\right)$$

$$D \text{ को क्र.मु.} = A \text{ को क्र.मु.} \times \left(\frac{x+100}{100}\right) \left(\frac{y+100}{100}\right) \left(\frac{z+100}{100}\right) \dots \dots \dots (-) \text{ for loss.}$$

- For same cost price and same two article

If selling price is = S₁ and S₂ with X₁% and X₂% profit respectively. Then

$$\frac{CP}{100} = \frac{S_1}{X_1+100} = \frac{S_2}{X_2+100} \dots \dots \dots \text{Use (+) for profit and (-) for loss. Hence, } \frac{CP}{100} = \frac{S_1-S_2}{X_1-X_2}$$

- Two different discount on same items. (MP and CP are same for same items)

$$\frac{100-D_1}{100-D_2} = \frac{100+P_1}{100+P_2} \dots \dots \dots (-) \text{ for loss.}$$

- Sold in market price means discount = 0%

- Fraud : False scale for selling : Gross profit/ Loss % :

$$\frac{GP+100}{FP+100} = \frac{T}{F} \dots \dots \dots (GP/GL \text{ is Gross profit/ Loss, Fault profit/ Loss (GP/FP) and (-) for loss, } T = \text{True measure, } F = \text{False measure})$$

- Successive fraud : Fraud x% profit/loss in buying and y% profit/loss in selling

$$\text{Total fraud or total profit/Loss} = (x + y + \frac{xy}{100})$$

- Cost price of two respective things are x and y, If selling price of those respective things are y and x, Then, Profit (%) = $(\frac{xy}{y})^2 \times 100\%$

- The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is:

Let C.P. of each article be Re. 1 C.P. of x articles = Rs. x.

S.P. of x articles = Rs. 20.

So, Profit = Rs. (20 - x).

$$\frac{20-x}{x} \times 100 = 20 \rightarrow x = 16$$

- If selling price is doubled, the profit triples. Find the profit percent. → 100%

Let C.P. = x Then, S.P. = y

So, profit = y - xi (Known by formula)

profit or 3(y - x) = 2y - x → y = 2xii (Known by question)

Equating, profit % = 100%

- In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit? (Let, CP = 100, SP = 420, Profit = 320) → 70%

- A vendor bought toffees at 6 for a rupee. How many for a rupee must he sell to gain 20%? → 5

Let C.P. of 6 toffees = Rs. 1i

profit=20% = Rs. $\frac{1}{5}$ ii

$$SP = 1 + \frac{1}{5} = \text{Rs. } \frac{6}{5}$$

$$\text{Toffee sold} = 6 \quad \text{Re. 1, toffees sold} = \frac{5}{6} \times 6 = 5$$

- A shopkeeper expects a gain of 22.5% on his cost price. If in a week, his sale was of Rs. 392, what was his profit? → (Rs. 72) (Let CP = X and Use formula $SP = \frac{100 + \text{Gain \%}}{100} \times CP$)
 - Sam purchased 20 dozens of toys at the rate of Rs. 375 per dozen. He sold each one of them at the rate of Rs. 33. What was his percentage profit? (5.6%)
 Cost Price of 1 toy = $\frac{375}{12}$ = Rs. 31.25 Sp of 1 toy = Rs. 33
 Profit % = $\frac{1.75}{31.25} \times 100\% = 5.6\%$
 - On selling 17 balls at Rs. 720, there is a loss equal to the cost price of 5 balls. The cost price of a ball is → 60%
 C.P. of 12 balls = S.P. of 17 balls = Rs. 720.
 C.P. of 1 ball = Rs. $\frac{720}{12}$ = Rs. 60
 - When a plot is sold for Rs. 18,700, the owner loses 15%. At what price must that plot be sold in order to gain 15%? → Rs. 25,300
 - Some articles were bought at 6 articles for Rs. 5 and sold at 5 articles for Rs. 6. Gain percent is
- | Quantity | Price | Cross multiply | Profit % = $\frac{36 - 25}{25} \times 100\% = 44\%$ |
|----------|-------|----------------|---|
| 6 | Rs. 5 | Cp = 25 | |
| 5 | 6 | Sp = 36 | |
- A shopkeeper sells one transistor for Rs. 840 at a gain of 20% and another for Rs. 960 at a loss of 4%. His total gain or loss percent is: (For same CP)
 Total SP = Rs. 840 + 960 = Rs. 1800 Total CP = $(\frac{100}{100+20} \times 840) + (\frac{100}{100-4} \times 960) = 1700$
 Total Profit = $\frac{100}{1700} \times 1800 = 5 \frac{15}{17}\%$
 - a वटा वस्तु b रुपैयामा खरिद गरी c वटा वस्तु d रुपैयामा विक्री गर्दा हुने नाफा/घाटा % = $\frac{ad-bc}{bc} \times 100\%$
 Eg. 12 वटा अण्डा रु. 10 मा खरिद गरी 5 वटा अण्डा रु. 6 मा विक्री गर्दा हुने नाफा/घाटा % = 44%

Sub-chapter 6.1 - Data Sufficiency from Profit and loss

- Only I is necessary
 - Only II is necessary
 - Either I or II is necessary
 - Both I and II are necessary
 - None of these
- A shopkeeper sells some articles at the profit of 25% on the original price. What is the exact amount of profit? To find the answer, which of the following information given in Statements I and II is/are necessary? → D
 - Sale price of the article
 - Number of articles sold
 - A shopkeeper sells some toys at Rs. 250 each. What percent profit does he make? To find the answer, which of the following information given in Statements I and II is/are necessary? → B
 - Number of toys sold.
 - Cost price of each toy.
- Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.
- I alone sufficient while II alone not sufficient to answer
 - II alone sufficient while I alone not sufficient to answer
 - Either I or II alone sufficient to answer
 - Both I and II are not sufficient to answer
 - Both I and II are necessary to answer
- A man mixes two types of rice (X and Y) and sells the mixture at the rate of Rs. 17 per kg. Find his profit percentage. → D
 - The rate of X is Rs. 20 per kg.
 - The rate of Y is Rs. 13 per kg
 - By selling a product with 20% profit, how much profit was earned? → A

1. The difference between cost and selling price is Rs. 40.
2. The selling price is 120 percent of the cost price.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

• **By selling an article what is the profit percent gained? → A**

1. 5% discount is given on list price.
2. If discount is not given, 20% profit is gained.
3. The cost price of the articles is Rs. 5000.

- A. Only I and II
- B. Only II and III
- C. Only I and III
- D. All I, II and III
- E. None of these

• **What was the percentage of discount given? → E**

1. 23.5% profit was earned by selling an almirah for Rs. 12,350.
2. If there were no discount, the earned profit would have been 30%
3. The cost price of the almirah was Rs. 10,000

- A. Only I and II
- B. Only II and III
- C. Only I and III
- D. Any two of the three
- E. None of these

• **What is the percent profit earned by the shopkeeper on selling the articles in his shop? → C**

1. Labeled price of the articles sold was 130% of the cost price.
2. Cost price of each article was Rs. 550.
3. A discount of 10% on labeled price was offered.

- A. Only I
- B. Only II
- C. I and III
- D. All the three are required
- E. Question cannot be answer even with information in all the three statements.

Sub-chapter 7 - Simple Interest

- The money borrowed or lent out for a certain period is called the **principal** or the **sum**.

• **Sum amount (A)** = $P \left(1 + \frac{TR}{100}\right)$

• **Simple Interest (I)** = $A - P = \frac{PTR}{100}$

• **Simple Interest for half yearly (I)** = $A - P = \frac{PTR}{2 \times 100}$

• **Amount (A)** = $P + I$

- **A sum of money at simple interest amounts to Rs. 815 in 3 years and to Rs. 854 in 4 years. The sum is:**

S.I. for 1 year = Rs. (854 - 815) = Rs. 39.

S.I. for 3 years = Rs.(39 x 3) = Rs. 117.

So, Principal = Rs. (815 - 117) = Rs. 698.

- **Mr. Thomas invested an amount of Rs. 13,900 divided in two different schemes A and B at the simple interest rate of 14% p.a. and 11% p.a. respectively. If the total amount of simple interest earned in 2 years be Rs. 3508, what was the amount invested in Scheme B?**

$P = \text{Rs. } 13,900$

$I_1 + I_2 = \text{Rs. } 3508$

Let, $p_1 = p_1$ and $p_2 = 13900 - p_1$

Schemes A	Schemes B	
$P = p_1 \quad T = 2 \text{ yr.} \quad R = 14\%$	$P = p_2 \quad T = 2 \text{ yr.} \quad R = 11\%$	$I_1 + I_2 = \frac{p_1 \times 2 \times 14}{100} + \frac{(13900 - p_1) \times 2 \times 11}{100}$
$I_1 = \frac{p_1 \times 2 \times 14}{100}$	$I_2 = \frac{(13900 - p_1) \times 2 \times 11}{100}$	$p_1 = 7500 \quad p_2 = 6400$

- **Reena took a loan of Rs. 1200 with simple interest for as many years as the rate of interest. If she paid Rs. 432 as interest at the end of the loan period, what was the rate of interest? (Take care that $T = R$: interest for as many years as the rate)**

- An automobile financier claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest of 10%, the effective rate of interest becomes: (Use half yearly formula)
- A person borrows Rs. 5000 for 2 years at 4% simple interest. He immediately lends it to another person at 6% for 2 years. Find his gain in the transaction per year. (per year = 112.5)

Sub-chapter 7.1 - Data Sufficiency from Simple Interest

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- I alone sufficient while II alone not sufficient to answer
 - II alone sufficient while I alone not sufficient to answer
 - Either I or II alone sufficient to answer
 - Both I and II are not sufficient to answer
 - Both I and II are necessary to answer
- The simple interest on a sum of money is Rs. 50. What is the sum? → E
 - The interest rate is 10% p.a.
 - The sum earned simple interest in 10 years.
 - What is the sum which earned interest? → E
 - The total simple interest was Rs. 7000 after 7 years.
 - The total of sum and simple interest was double of the sum after 5 years.
 - What percentage of simple interest per annum did Anand pay to Deepak? E
 - Anand borrowed Rs. 8000 from Deepak for four years.
 - Anand returned Rs. 8800 to Deepak at the end of two years and settled the loan.
 - What is the rate of simple interest? → D
 - The total interest earned was Rs. 4000.
 - The sum was invested for 4 years.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- I and III only
 - II and III only
 - I and II only
 - I and III only, or II and III only
 - Any two of the three
- What is the principal sum? → E
 - The sum amounts to Rs. 690 in 3 years at S.I.
 - The sum amounts to Rs. 750 in 5 years at S.I.
 - The rate of interest is 5% p.a.

Sub-chapter 8 - Compound Interest

- Yearly compound amount (CA) = $P \left(1 + \frac{R}{100}\right)^T$
- Half-yearly compound amount C(A) = $P \left(1 + \frac{R}{200}\right)^{2T}$
- Yearly compound amount Interest (CI) = $P \left[\left(1 + \frac{R}{100}\right)^T - 1\right]$
- Half-Yearly compound amount Interest (CI) = $P \left[\left(1 + \frac{R}{200}\right)^{2T} - 1\right]$
- Any time xT years compound amount Interest (CI) = $P \left[\left(1 + \frac{R}{x100}\right)^{xT} - 1\right]$
- When Rates are different for different years, say R1%, R2%, R3% for 1st, 2nd and 3rd year respectively. Compound amount (CA) = $P \left(1 + \frac{R1}{100}\right) \left(1 + \frac{R2}{100}\right) \left(1 + \frac{R3}{100}\right)$

- For 2 year difference between C.I. and S.I. $(d) = p\left(\frac{R}{100}\right)^2$
- For 3 year difference between C.I. and S.I. $(d) = p\left(\frac{R}{100}\right)^2 \left(\frac{300+R}{100}\right)$
- Short-cut method for compound interest

Years	Ratios	Eg. What is the compound interest if principal is 5,000, rate 10% in 3 years.						
2 Years	2 : 1	$5000 \times \frac{10}{100} = 500$ $500 \times \frac{10}{100} = 50$ $50 \times \frac{10}{100} = 5$						
3 Years	3 : 3 : 1							
4 years	4 : 6 : 4 : 1	500	50	5	<table><tr><td>1655</td><td>Total = 6655</td></tr></table>		1655	Total = 6655
1655	Total = 6655							
5 Years	5 : 10 : 10 : 5 : 1	3	3	1				
6 Years	6 : 15 : 20 : 15 : 6 : 1	1500	150	5				
7 Years	7 : 21 : 35 : 35 : 21 : 7 : 1							

- A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits Rs. 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is: (Rs. 121)
- What is the difference between the compound interests on Rs. 5000 for $1\frac{1}{2}$ years at 4% per annum compounded yearly and half-yearly? (Rs. 5306.04 - Rs. 5304 = Rs. 2.04)
- The least number of complete years in which a sum of money put out at 20% compound interest will be more than doubled is: (4 years)
- The effective annual rate of interest corresponding to a nominal rate of 6% per annum payable half-yearly is: (6.09%)
- The difference between simple interest and compound on Rs. 1200 for one year at 10% per annum reckoned half-yearly is: (Rs. 3 → Put for S.I. T = 1 and for C.I. T = 2)

Sub-chapter 8.1 - Data Sufficiency from Compound Interest

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- I alone sufficient while II alone not sufficient to answer
- II alone sufficient while I alone not sufficient to answer
- Either I or II alone sufficient to answer
- Both I and II are not sufficient to answer
- Both I and II are necessary to answer

- An amount of money was lent for 3 years. What will be the difference between the simple and the compound interest earned on it at the same rate? → E

- The rate of interest was 8 p.c.p.a.
- The total amount of simple interest was Rs. 1200.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- What is the rate of interest p.c.p.a.? → E

- An amount doubles itself in 5 years on simple interest.
- Difference between the compound interest and the simple interest earned on a certain amount in 2 years is Rs. 400.
- Simple interest earned per annum is Rs. 2000.

- I only
- II and III only
- All I, II and III
- Any two of the three
- I only or II and III only

- What will be the compound interest earned on an amount of Rs. 5000 in 2 years? → A

- The simple interest on the same amount at the same rate of interest in 5 years is Rs. 2000.

2. The compound interest and the simple interest earned in one year is the same.
3. The amount became more than double on compound interest in 10 years.

- A. I only
- B. I and II only
- C. II and III only
- D. I and III only
- E. None of these

- **Mr. Gupta borrowed a sum of money on compound interest. What will be the amount to be repaid if he is repaying the entire amount at the end of 2 years? → D**

1. The rate of interest is 5 p.c.p.a.
2. Simple interest fetched on the same amount in one year is Rs. 600.
3. The amount borrowed is 10 times the simple interest in 2 years.

- A. I only
- B. III only
- C. I or II only
- D. II and Either I or III only
- E. All I, II and III are required

- **What is the compound interest earned at the end of 3 years? → D**

1. Simple interest earned on that amount at the same rate and for the same period is Rs. 4500.
2. The rate of interest is 10 p.c.p.a.
3. Compound interest for 3 years is more than the simple interest for that period by Rs. 465.

- A. I and II only
- B. II and III only
- C. I and III only
- D. I and Either II or III only
- E. Any two of the three

Sub-chapter 9 - Population Growth and depression

- **Population Growth**

Population Growth	Population depression
• $P^T = P \left(1 + \frac{R}{100}\right)^T$ – P^T = After growth population – p = Before growth population – R = Growth rate – T = time in year	• $D^T = D \left(1 - \frac{R}{100}\right)^T$ – P^T = After depression – D = Before depression – R = Depression rate – T = time in year
• $P^T = P \left(1 \pm \frac{x}{100}\right) \left(1 \pm \frac{y}{100}\right) \left(1 \pm \frac{z}{100}\right) \dots\dots\dots$	

- **The population of a town increased from 1,75,000 to 2,62,500 in a decade. The average percent increase of population per year is → 5% [$P^T = P \left(1 + \frac{R}{100}\right)^T$]**

Increase in 10 years = (262500 - 175000) = 87500

Increase% = 50% So, Increase% per year = 5%

Sub-chapter 10 - Partnership

- When investments of all the partners are for the same time, the gain or loss is distributed among the partners in the ratio of their investments.
- Profit ratio = Investment × month (Here, Investment = investment × x)
- Profit ratio = Investment ratio × x (For not given month or same months of multiple investors)
- If ratio of investment of A : B = 3 : 2 The, part of investment of A = $\frac{3}{5}$ and B = $\frac{2}{5}$
- Share of A denote (investment + shared profit)
- **A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is → Rs. 1500**

Let total profit = x.

Since 5% goes to charity so.

Now total profit = $\frac{95x}{100}$

Part of A = $\frac{3}{5}$ and Part of B = $\frac{2}{5}$

Share of A = $\frac{95x}{100} \times \frac{3}{5} = 855 \rightarrow x = \text{Rs. } 1500$

- A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest Rs. 6500 for 6 months, B, Rs. 8400 for 5 months and C, Rs. 10,000 for 3 months. A wants to be the working member for which, he was to receive 5% of the profits. The profit earned was Rs. 7400. Calculate the share of B in the profit. $\rightarrow \text{Rs. } 2660$

Profit = Rs. 7400

A received = $7400 \times 5\% = \text{Rs. } 370$

Remaining profit = Rs. 7030

	A	B	C
Ratio of their investments	$6500 \times 6 = 13 = \frac{13}{37}$	$8400 \times 5 = 14 = \frac{14}{37}$	$10000 \times 3 = 10 = \frac{10}{37}$

B's share = $\text{Rs. } 7030 \times \frac{14}{37} = \text{Rs. } 2660$

- A, B and C enter into a partnership in the ratio $\frac{7}{2} : \frac{4}{3} : \frac{6}{5}$. After 4 months, A increases his share 50%. If the total profit at the end of one year be Rs. 21,600, then B's share in the profit is : $\rightarrow (\text{Rs. } 4000)$

Ratio of initial investments = $\frac{7}{2} \times 30 = 105$,

$\frac{4}{3} \times 30 = 40$,

$\frac{6}{5} \times 30 = 36$

Let the initial investments be 105x, 40x and 36x.

A : B : C = $(105x \times 4 + \frac{150}{100} \times 105x \times 8) : 40x \times 12 : 36x \times 12 = 35 : 10 : 9$

Hence, B's share = $\text{Rs. } 21,600 \times \frac{10}{54} = \text{Rs. } 4000$

- A, B, C subscribe Rs. 50,000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35,000, A receives: (Rs. 14,700)

Total investment = Rs. 50,000

	A	B	C	Calculation
Investment	$x + 9000$	$x + 5000$	x	$3x = 5000 - 14000 = 36000 \rightarrow x = 12000$
Calculated investment	21000	17000	12000	A receives = $\text{Rs. } 35,000 \times \frac{21}{50} = 14,700$
Ratio of investment	21	17	12	

- Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments? $\rightarrow (20 : 49 : 64)$

Let initial A's, B's, C's investment is Rs x, y, z respectively.

And, Let the Profit = P

Ratio of investment of A : B : C = $14x : 8y : 7z = 5 : 7 : 8$

$x = \frac{5}{14}$, $y = \frac{7}{8}$, $z = \frac{8}{7}$ (Taking LCM, $x = \frac{5}{14} \times 56$, $y = \frac{7}{8} \times 56$, $z = \frac{8}{7} \times 56$)

Therefore, Ratio = $x : y : z = 20 : 49 : 64$

- A starts business with Rs. 3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2 : 3. What is B's contribution in the capital? $\rightarrow (9000)$

Let, B's contribution in the capital = x

	A	B	Ratio of profit = 2 : 3
Contribution	3500	x	Profit ratio = investment ratio $\times$ month
Month	12	7	$\frac{2}{3} = \frac{3500 \times 12}{7 \times x} \rightarrow x = \frac{3500 \times 12 \times 3}{14} \rightarrow x = 9000$

- A and B entered into partnership with capitals in the ratio 4 : 5. After 3 months, A withdrew $\frac{1}{4}$ of his capital and B withdrew $\frac{1}{5}$ of his capital. The gain at the end of 10 months was Rs. 760. A's share in this profit is $\rightarrow (\text{Rs } 330)$
- A and B started a partnership business investing some amount in the ratio of 3 : 5. C joined then after six months with an amount equal to that of B. In what proportion should the profit at the end of one year be distributed among A, B and C? $\rightarrow (6 : 10 : 5.)$

	A	B	C	Let the initial investments of A and B be 3x and 5x.
Month	12	12	6	Then, initial investments of C = 5x
Investment Ratio	3	5		A : B : C = $3x \times 12 : 5x \times 12 : 5x \times 6 = 6 : 10 : 5.$
Investment Ratio	3x	5x	5x	

- A and B started a business in partnership investing Rs. 20,000 and Rs. 15,000 respectively. After six months, C joined them with Rs. 20,000. What will be B's share in total profit of Rs. 25,000 earned at the end of 2 years from the starting of the business? $\rightarrow (\text{Rs. } 7500)$

	A	B	C	Investment Ratio = 4 : 3 : 3 $4x + 3x + 3x = 25,000$ $x = 2500$ B's share = $2500 \times 3 = 7500$
Investment	20,000	15,000	20,000	
Month	24	24	18	
Investment Ratio	$20,000 \times 24$	$15,000 \times 24$	$20,000 \times 18$	

- **A began a business with Rs. 85,000. He was joined afterwards by B with Rs. 42,500. For how much period does B join, if the profits at the end of the year are divided in the ratio of 3 : 1? → (8 month)**

	A	B	Here, Profit ratio = 3 : 1 Formula, Profit ratio = Investment $\times$ month $\frac{3}{1} = \frac{12 \times 85000}{x \times 42500} \rightarrow x = \frac{12 \times 85000}{3 \times 42500} = 8 \text{ month}$
Investment	85,000	42,500	
Month	12	x	

- **Arun, Kamal and Vinay invested Rs. 8000, Rs. 4000 and Rs. 8000 respectively in a business. Arun left after six months. If after eight months, there was a gain of Rs. 4005, then what will be the share of Kamal? → (Rs. 890) (Month for Arun, Kamal and Vinay is 6, 8, 8 month)**

Sub-chapter 10.1 Data Sufficiency from Partnership

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- I alone sufficient while II alone not sufficient to answer
 - II alone sufficient while I alone not sufficient to answer
 - Either I or II alone sufficient to answer
 - Both I and II are not sufficient to answer
 - Both I and II are necessary to answer
- **Ravi, Gagan and Nitin are running a business firm in partnership. What is Gagan's share in the profit earned by them? → E**
 - Ravi, Gagan and Nitin invested the amounts in the ratio of 2 : 4 : 7.
 - Nitin's share in the profit is Rs. 8750.
 - **Rahul, Anurag and Vivek started a business together. In what proportion would the annual profit be distributed among them? → E**
 - Rahul got one-fourth of the profit.
 - Rahul and Vivek contributed 75% of the total investment.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- **How much did Rohit get as profit at the year-end in the business done by Nitin, Rohit and Kunal? → D**
 - Kunal invested Rs. 8000 for nine months, his profit was $\frac{3}{2}$ times that of Rohit's and his investment was four times that of Nitin.
 - Nitin and Rohit invested for one year in the proportion 1 : 2 respectively.
 - The three together got Rs. 1000 as profit at the year end.
 - Only I and II
 - Only I and III
 - Question cannot be answered even with the information in all the three statements.
 - All I, II and III
 - None of these
- **What is R's share of profit in a joint venture? → D**
 - Q started business investing Rs. 80,000.
 - R joined him after 3 months.
 - P joined after 4 months with a capital of Rs. 1,20,000 and got Rs. 6000 as his share profit.
 - All I, II and III
 - I and III only
 - II and III only
 - Even with all I, II and III, the answer cannot be arrived at
 - None of these

- Three friends, P, Q and R started a partnership business investing money in the ratio of 5 : 4 : 2 respectively for a period of 3 years. What is the amount received by P as his share profit? →
B
- A. Total amount invested in the business in Rs. 22,000.
- B. Profit earned at the end of 3 years is $\frac{3}{8}$ of the total investment.
- C. The average amount of profit earned per year is Rs. 2750.
- A. I or II or III
- B. Either III only, or I and II together
- C. Any two of the three
- D. All I, II and III are required.
- E. None of these

Sub-chapter 11 - Number system Natural number and Prime Number

❖ Prime Number (Having only two factors : 1 and itself)

• Testing whether any number is prime or not :

1. Even Number cannot be prime number.
 2. The smallest 3 digit prime number is: 101
- $(a^2 - b^2) = (a + b)(a - b)$
 - $(a + b)^2 = (a^2 + b^2 + 2ab)$
 - $(a - b)^2 = (a^2 + b^2 - 2ab)$
 - $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
 - $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
 - $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
 - $(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ac)$
 - When $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$
 - $(112 \times 5^4) = 112 \times (\frac{10}{2})^4 = 10000$
 - $(x^n - a^n)$ is completely divisible by $(x - a)$, when : n is any natural number

Number	Division Formula
2	All even number (2, 4, 6, 8....
3/9	If sum of all digits are perfectly divisible by 3/9 (126 : 1 + 2 + 6 = 9
4	If last two digits in a number is divisible by 4 (116.....
5	If the last digit is either 0 or 5 (125, 250.....
6	If both an even number and divisible by 3 (246....
7	If difference between double of last digit and rest number of digit is divisible by 0 or 7 (483 → 48 - 3 × 2 = 42)
8	If last three digits in a number is divisible by 8 or having at last 000(4120, 3246.....
10	If the last digit is either 0 (10, 20, 100.....
11	If the sum of its digits at odd places and the sum of its digits at even places is equal or difference is divisible by 11 (2121, 1342.....
12	If divisible by both 3 and 4 (144, 288.....
13	
14	
15	If divisible by both 3 and 5 (225, 450.....
16	
17	
18	
19	
20	
21	
22	
23	

- It is being given that $(2^{32} + 1)$ is completely divisible by a whole number. Which of the following numbers is completely divisible by this number? $\rightarrow (2^{96} + 1)$
- What least number must be added to 1056, so that the sum is completely divisible by 23 ? $\rightarrow (2)$
- $1397 \times 1397 = ? \rightarrow (1397)^2 = (1400 - 3)^2$
- $(935421 \times 625) = 935421 \times 5^4 = 935421 \times (\frac{10}{2})^4$
- If the number $517*324$ is completely divisible by 3, then the smallest whole number in the place of * will be $\rightarrow (2)$
- If the number $481 * 673$ is completely divisible by 9, then the smallest whole number in place of * will be $\rightarrow (7)$
Sum of digits = $(4 + 8 + 1 + x + 6 + 7 + 3) = (29 + x)$, which must be divisible by 9.
- The difference between the local value and the face value of 7 in the numeral 32675149 is $\rightarrow 69993$ (Local value of 7) - (Face value of 7) = $(70000 - 7) = 69993$
- On dividing a number by 56, we get 29 as remainder. On dividing the same number by 8, what will be the remainder ? $\rightarrow (R = 5)$
(Divisor $\times$ Quotient + Remainder = Dividend)
- If n is a natural number, then $(6n^2 + 6n)$ is always divisible by: $\rightarrow 6$ and 12 both
- A 3-digit number $4a3$ is added to another 3-digit number 984 to give a 4-digit number $13b7$, which is divisible by 11. Then, $(a + b) = ? \rightarrow 10$
- On dividing a number by 357, we get 39 as remainder. On dividing the same number 17, what will be the remainder ? $\rightarrow (5)$
Divisor = 357, Remainder = 39, So Quotient = $357 + 39 = 396$
On the other hand, Dividing by 17, We get Remainder = 5
- In dividing a number by 585, a student employed the method of short division. He divided the number successively by 5, 9 and 13 (factors 585) and got the remainders 4, 8, 12 respectively. If he had divided the number by 585, the remainder would have been $\rightarrow (584)$
- In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 46, what is the dividend ? $\rightarrow (5336)$
- Which of the following numbers will completely divide $(49^{15} - 1)$? $\rightarrow 50$
 $(x^n - 1)$ will be divisibly by $(x + 1)$ only when n is even
- On dividing 2272 as well as 875 by 3-digit number N , we get the same remainder. The sum of the digits of N is: $\rightarrow (10)$
- When a number is divided by 13, the remainder is 11. When the same number is divided by 17, then remainder is 9. What is the number ? $\rightarrow (349)$
- n is a whole number which when divided by 4 gives 3 as remainder. What will be the remainder when $2n$ is divided by 4 ? $\rightarrow (2)$
- The sum of the two numbers is 12 and their product is 35. What is the sum of the reciprocals of these numbers ? $\rightarrow (\frac{12}{35})$ (Numbers = 5, 7 So, sum of the reciprocals of these numbers = $\frac{1}{5} + \frac{1}{7} = \frac{12}{35}$)
- What will be remainder when 17^{200} is divided by 18 ? $\rightarrow (1)$
When n is even. $(x^n - a^n)$ is completely divisibly by $(x + a)$
 $(17^{200} - 1^{200})$ is completely divisible by $(17 + 1)$, i.e., 18, So $R = 1$
- The difference of the squares of two consecutive even integers is divisible by which of the following integers ? (4)
Let the two consecutive even integers be $2n$ and $(2n + 2)$. Then, $(2n + 2)^2 = (2n + 2 + 2n)(2n + 2 - 2n) = 2(4n + 2) = 4(2n + 1)$, which is divisible by 4.
- If a and b are odd numbers, then which of the following is even ? $(a + b)$
- If x and y are positive integers such that $(3x + 7y)$ is a multiple of 11, then which of the following will be divisible by 11 ? $(4x - 9y)$
- In a division sum, the remainder is 0. As student mistook the divisor by 12 instead of 21 and obtained 35 as quotient. What is the correct quotient ? (20)
- What is the unit digit in $(4137)^{754}$? (9)
Unit digit in $(4137)^{754} = \text{Unit digit in } \{[(4137)^4]^{188} \times (4137)^2\}$
 $= \text{Unit digit in } \{ 292915317923361 \times 17114769 \} = (1 \times 9) = 9$

- A number was divided successively in order by 4, 5 and 6. The remainders were respectively 2, 3 and 4. The number is: (214)
- On dividing a number by 68, we get 269 as quotient and 0 as remainder. On dividing the same number by 67, what will the remainder ? (R = 1) $(x = 68 \times 269 + 0 = 18292 \text{ So, } 18292/67)$
- What is the unit digit in the product $(3^{65} \times 6^{59} \times 7^{71})$? (4)
Unit digit in $3^4 = 1$ (Searching on option to 1)
Unit digit in $(3^4)^{16} = 1$ Unit digit in $3^{65} = \text{Unit digit in } [(3^4)^{16} \times 3] = (1 \times 3) = 3$
Unit digit in $6^{59} = 6$
Unit digit in $7^4 \Rightarrow \text{Unit digit in } (7^4)^{17} \text{ is } 1.$
Unit digit in $7^{71} = \text{Unit digit in } [(7^4)^{17} \times 7^3] = (1 \times 3) = 3$
Required digit = Unit digit in $(3 \times 6 \times 3) = 4.$
- If the number 653xy is divisible by 90, then $(x + y) = ?$ (4)
Clearly, 653xy is divisible by 10, so $y = 0$
Now, 653x0 is divisible by 9.
So, $(6 + 5 + 3 + x + 0) = (14 + x)$ is divisible by 9. So, $x = 4.$
Hence, $(x + y) = (4 + 0) = 4$
- What is the unit digit in 7^{105} ? (7)
Unit digit in $7^{105} = \text{Unit digit in } [(7^4)^{26} \times 7]$
But, unit digit in $(7^4)^{26} = 1$
Unit digit in $7^{105} = (1 \times 7) = 7$
- Which of the following numbers will completely divide $(4^{61} + 4^{62} + 4^{63} + 4^{64})$? (10)
 $(4^{61} + 4^{62} + 4^{63} + 4^{64}) = 4^{61} \times (1 + 4 + 4^2 + 4^3) = 4^{61} \times 85 = 4^{60} \times (4 \times 85) = (4^{60} \times 340)$, which is divisible by 10.
- A number when divided successively by 4 and 5 leaves remainders 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be (2,3)
- Three times the first of three consecutive odd integers is 3 more than twice the third. The third integer is: (15)
Let the three integers be $x, x + 2$ and $x + 4.$
Then, $3x = 2(x + 4) + 3 \Leftrightarrow x = 11$ Third integer = $x + 4 = 15.$
- Find a positive number which when increased by 17 is equal to 60 times the reciprocal of the number. (3)
Let the number be $x.$ Then, $x + 17 = \frac{60}{x} \Rightarrow x^2 + 17x - 60 = 0 \Rightarrow x = 3.$

Sub-chapter 12 - Square Number/Root and Cube Number/Root

- The square of natural number only ends with in : 0, 1, 4, 5, 6, 9

Square number ending	0	1	4	5	6	9
Square root ending	0	1	2	3	4	5

- The square of natural number doesnot ends with in : 2, 3, 7, 8
- The square of odd number must be odd and even number must be even.
- Square numbers and cube numbers

Number	Square	Cube	Number	Square	Cube	Number	Square	Cube
1	1	1	11	121	1331	21	441	9261
2	4	8	12	144	1728	22	484	10648
3	9	27	13	169	2197	23	529	12167
4	16	64	14	196	2744	24	576	13824
5	25	125	15	225	3375	25	625	15625
6	36	216	16	256	4096	26	676	17576
7	49	343	17	289	4913	27	792	19683
8	64	512	18	324	5832	28	784	21952
9	81	729	19	361	6859	29	841	24389
10	100	1000	20	400	8000	30	900	27000

• **Finding approximate square root of non-perfect square number**

$\sqrt{12}$	$\sqrt{33}$	$\sqrt{95}$	$\sqrt{17}$	$\sqrt{105}$	Trick
$= \sqrt{(16 - 4)}$	$= \sqrt{(36 - 3)}$	$= \sqrt{(100 - 5)}$	$= \sqrt{(16 + 1)}$	$= \sqrt{(100 + 5)}$	Both (+) or (-) Valid Multiply by 2
$= 4 - \frac{4}{8}$	$= 6 - \frac{3}{12}$	$= 10 - \frac{5}{20}$	$= 4 + \frac{1}{8}$	$= 10 + \frac{5}{20}$	
$= 4 - 0.5$	$= 6 - 0.25$	$= 10 - 0.25$	$= 4 + 0.13$	$= 10 + 0.25$	
$= 3.5$	$= 5.75$	$= 9.75$	$= 4.13$	$= 10.25$	

• **The cube root of .000216 is: (0.06)**

$$(0.000216)^{1/3} = \left(\frac{216}{1000}\right)^{1/3} = \left(\frac{6}{100}\right)^{1/3 \times 3} = 0.06$$

• **The least perfect square, which is divisible by each of 21, 36 and 66 is: (213444)**

L.C.M. of 21, 36, 66 = 2772.

Now, $2772 = 2 \times 2 \times 3 \times 3 \times 7 \times 11$

To make it a perfect square, it must be multiplied by 7×11 .

So, required number = $2^2 \times 3^2 \times 7^2 \times 11^2 = 213444$

• **A group of students decided to collect as many paise from each member of group as is the number of members. If the total collection amounts to Rs. 59.29, the number of the member is the group is: (77)**

Money collected = (59.29×100) paise = 5929 paise.

Number of members = $\sqrt{5929} = 77$.

• **How many two-digit numbers satisfy this property.: The last digit (unit's digit) of the square of the two-digit number is 8 ? (None of these)**

A number ending in 8 can never be a perfect square.

Sub-chapter 13 - Alpha numerical system

• **Mixed of number and alphabets**

• **Insert the correct mathematical signs**

– First of all solve from the brackets

Sub-chapter 14 - Time and work

• If A can do a piece of work in n days, then A's 1 day's work = $\frac{1}{n}$

• If A's 1 day's work = $\frac{1}{n}$, then A can finish the work in n days

• If A is thrice as good a workman as B, then: Ratio of work done by A and B = 3 : 1.
Ratio of times taken by A and B to finish a work = 1 : 3.

• If $A + B + C = \frac{1}{4}$ $A = \frac{1}{16}$ $B = \frac{1}{12}$ Then, $C = \frac{1}{4} - \left(\frac{1}{16} + \frac{1}{12}\right) = 9 \frac{3}{5}$

• **A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day? (Make for all days factor)**

Only A works in every two days = $\left(\frac{1}{20} + \frac{1}{20}\right) = \frac{1}{10}$

(A + B + C)'s 1 day's work = $\frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{1}{20} = \frac{1}{10}$

Work done in 3 days = $\frac{1}{10} + \frac{1}{10} = \frac{1}{5}$ So, Whole work will be done in $(3 \times 5) = 15$ days.

• **A is 30% more efficient than B. How much time will they, working together, take to complete a job which A alone could have done in 23 days?**

A can complete the work in 23 days.

Here, A can complete the work in $23 + 23 \times \frac{30}{100}$ days. (A is 30% more efficient than B) = $\frac{299}{10}$ days

(A + B)'s 1 day's work = $\frac{1}{23} + \frac{10}{299} = \frac{1}{13}$ So, (A + B) can complete the work in 13 days

• **A is thrice as good a workman as B and therefore is able to finish a job in 60 days less than B. Working together, they can do it in:**

Here, A takes 1 day B take 3 days

Let A can do it in x days so, B can do it in $\frac{x}{3}$ days Hence, $x - \frac{x}{3} = 60 \Rightarrow x = 90$

A can do in 90 days. B can do in 30 days. Therefore Working together, they can do it in $22\frac{1}{2}$

- **A and B can do a job together in 7 days. A is $1\frac{3}{4}$ times as efficient as B. The same job can be done by A alone in :**

Ratio of 1 day's work of A and B = $\frac{7}{4} : 1$

So, $\frac{7}{4}x + x = \frac{1}{7} \Rightarrow x = \frac{1}{11} \Rightarrow 11$ days.

- **If 6 men and 8 boys can do a piece of work in 10 days while 26 men and 48 boys can do the same in 2 days, the time taken by 15 men and 20 boys in doing the same type of work will be:**

Let 1 man's 1 day's work = x and 1 boy's 1 day's work = y.

$6x + 8y = \frac{1}{10}$ i $26x + 48y = \frac{1}{2}$ ii

solving : $x = \frac{1}{100}$ and $y = \frac{1}{200}$ So, $15 \times \frac{1}{100} + 20 \times \frac{1}{200} = 4$ days

- **A can do a certain work in the same time in which B and C together can do it. If A and B together could do it in 10 days and C alone in 50 days, then B alone could do it in:**

Here, A $\rightarrow$ B + C A + B $\rightarrow$ 10 days (A + B $\rightarrow \frac{1}{10}$ days) C $\rightarrow$ 50 days (C $\rightarrow \frac{1}{50}$ days)

Putting value of A, So B + C + B $\rightarrow \frac{1}{10}$ days

$\Rightarrow 2B + C \rightarrow \frac{1}{10}$ days $\Rightarrow 2B + \frac{1}{50} = \frac{1}{10} \Rightarrow 2B = \frac{4}{50} \Rightarrow B = \frac{2}{25}$

Hence, B alone could do it in 25 days.

- **P can complete a work in 12 days working 8 hours a day. Q can complete the same work in 8 days working 10 hours a day. If both P and Q work together, working 8 hours a day, in how many days can they complete the work?**

P can complete the work in (12 x 8) hrs. = 96 hrs. work In 1 hour = $\frac{1}{96}$

Q can complete the work in (8 x 10) hrs. = 80 hrs. work In 1 hour = $\frac{1}{80}$

P + Q work In 1 hour = $\frac{1}{96} + \frac{1}{80} = \frac{11}{480}$ So, P + Q work In 8 hour = $\frac{11}{480} \times 8 = \frac{11}{60} \Rightarrow \frac{60}{11}$ days

- **X and Y can do a piece of work in 20 days and 12 days respectively. X started the work alone and then after 4 days Y joined him till the completion of the work. How long did the work last?**

The total time taken = (6 + 4) days = 10 days.

- **Ravi and Kumar are working on an assignment. Ravi takes 6 hours to type 32 pages on a computer, while Kumar takes 5 hours to type 40 pages. How much time will they take, working together on two different computers to type an assignment of 110 pages? (Time taken before pages) $\rightarrow$ 8 hours 15 minutes**

- **Sakshi can do a piece of work in 20 days. Tanya is 25% more efficient than Sakshi. The number of days taken by Tanya to do the same piece of work is: (Use Ratio of efficiency)**

	Sakshi	Tanya	$x \times 125 = 20 \times 100 \Rightarrow x = 16$ Tanya to do the same piece of work is: 16 days
Efficiency rate	100	125	
Days to do work	20	x	

- **A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in:**

A, B, C can finish a piece of work in $x, \frac{x}{2}, \frac{x}{3}$ days respectively

In 1 day A + B + C can do $(\frac{1}{x} + \frac{2}{x} + \frac{3}{x})$ parts of work. = $\frac{1}{2}$ parts of work

$\Rightarrow \frac{6}{x} = \frac{1}{2} \Rightarrow x = 12$ So, B can do the work alone in 6 days.

- **A can complete (1) a work in 15 days. So, A can do $\frac{1}{3}$ parts of a work in $\frac{1}{3} \times 15$ days = 5 days (Take care for required minimum and maximum value)**

- **20 women can do a work in 16 days. 16 men can complete the same work in 15 days. What is the ratio between the capacity of a man and a woman?**

1 woman's 1 day's work = $\frac{1}{20} \times \frac{1}{16} = \frac{1}{320}$ 1 man's 1 day's work = $\frac{1}{16} \times \frac{1}{15} = \frac{1}{240}$

Ratio between the capacity of a man and a woman = $\frac{1}{240} : \frac{1}{320} \Rightarrow \frac{1}{3} : \frac{1}{4} \Rightarrow 4 : 3$

- **A and B together can do a piece of work in 30 days. A having worked for 16 days, B finishes the remaining work alone in 44 days. In how many days shall B finish the whole work alone?**

Let A's 1 day's work = x

B's 1 day's work = y.

$$\begin{array}{rcl} x + y = 30 \text{ days} & \dots\dots\dots & \text{i} \\ \hline 16x + 44y = 1 & \dots\dots\dots & \text{ii} \\ \hline \end{array}$$

$$A = 16 \text{ days} \quad B = 44 \text{ days}$$

Then, $x + y = \frac{1}{30} \dots\dots\dots \text{i}$ $16x + 44y = 1 \dots\dots\dots \text{ii}$ Solving $x = \frac{1}{60}$ and $y = \frac{1}{60}$ Ans

- A ले कुनै काम गर्न A र B मिलेर गर्न भन्दा x दिन अधिक समय लगाउँछ। B ले कुनै काम गर्न A र B मिलेर गर्न भन्दा y दिन अधिक समय लगाउँछ। भने A र B मिलेर सो काम $x = \sqrt{xy}$ दिनमा गर्न सक्छन्।
- ५ वटा पेन र ८ वटा कपीको मूल्य रु. 140 पर्छ भने 15 वटा पेन र 24 वटा कपीको मूल्य कति पर्छ ?
 $5p + 8c = 140 \rightarrow 5p \times 3 + 8c \times 3 = 140 \times 3 \rightarrow 15p + 24c = 420$
- 3 पुरुष र ४ केटाले कुनै काम 12 दिनमा गर्छन्, 4 पुरुष र ३ केटाले सो काम 10 दिनमा गर्न सक्छन् भने 2 पुरुष र 3 केटाले सो काम कति दिनमा गर्न सक्छन् ?

Men	Boy	Day
3 (A1)	4 (B1)	12 (C1)
4 (A2)	3 (B2)	10 (C2)
2 (A3)	3 (B3)	X (C3)
Formula : $C1C2 (A1B2 - A2B1) = C1C3 (A1B3 - A3B1) - C2C3 (A2B3 - A3B2)$		

Sub-chapter 14.1- Data Sufficiency from Time and work

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- A. I alone sufficient while II alone not sufficient to answer
- B. II alone sufficient while I alone not sufficient to answer
- C. Either I or II alone sufficient to answer
- D. Both I and II are not sufficient to answer
- E. Both I and II are necessary to answer
- A and B together can complete a task in 7 days. B alone can do it in 20 days. What part of the work was carried out by A? $\rightarrow A$
 1. A completed the job alone after A and B worked together for 5 days
 2. Part of the work done by A could have been done by B and C together in 6 days
- How long will Machine Y, working alone, take to produce x candles? $\rightarrow E$
 1. Machine X produces x candles in 5 minutes.
 2. Machine X and Machine Y working at the same time produce x candles in 2 minutes.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- In how many days can 10 women finish a work? $\rightarrow A$
 1. 10 men can complete the work in 6 days.
 2. 10 men and 10 women together can complete the work in $3\frac{3}{7}$ days.
 3. If 10 men work for 3 days and thereafter 10 women replace them, the remaining work is completed in 4 days.

A. Any two of the three
B. I and II only
C. II and III only
D. I and III only
E. None of these
- How many workers are required for completing the construction work in 10 days? $\rightarrow E$
 1. 20% of the work can be completed by 8 workers in 8 days.
 2. 20 workers can complete the work in 16 days.
 3. One-eighth of the work can be completed by 8 workers in 5 days.

A. I only
B. II and III only

- C. III only
- D. I and III only
- E. Any one of the three

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

- 8 men and 14 women are working together in a field. After working for 3 days, 5 men and 8 women leave the work. How many more days will be required to complete the work? → D
- 1. 19 men and 12 women together can complete the work in 18 days.
- 2. 16 men can complete two-third of the work in 16 days.
- 3. In 1 day, the work done by three men is equal to the work done by four women.
- A. I only
- B. II only
- C. III only
- D. I or II or III
- E. II or III only

Sub-chapter 15 - Chain Rule

- Two quantities are said to be directly proportional, if on the increase (or decrease) of the one, the other increases (or decreases) to the same extent.
- Two quantities are said to be indirectly proportional, if on the increase of the one, the other decreases to the same extent and vice-versa.
- 3 pumps, working 8 hours a day, can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day? (12)

pumps	working hour	day	Let the required number of working hours per day be x.
3	8	2	With checking relations, $\frac{3}{4} = \frac{x}{8} \times \frac{1}{2} \rightarrow x = 12$
4	x	1	

- If the cost of x metres of wire is d rupees, then what is the cost of y metres of wire at the same rate? ($\frac{dy}{x}$)
- A fort had provision of food for 150 men for 45 days. After 10 days, 25 men left the fort. The number of days for which the remaining food will last, is: (42)

	Men	Days	Unti 10 days 150 men eat that food. rest for them = 35 days
In Initial	150	45	$\frac{150}{125} = \frac{x}{35} \rightarrow x = 42$
Remaining	150	35	
At final	125	x	

- A wheel that has 6 cogs is meshed with a larger wheel of 14 cogs. When the smaller wheel has made 21 revolutions, then the number of revolutions made by the larger wheel is: (49)
- A flagstaff 17.5 m high casts a shadow of length 40.25 m. The height of the building, which casts a shadow of length 28.75 m under similar conditions will be: (12.5)
- In a camp, there is a meal for 120 men or 200 children. If 150 children have taken the meal, how many men will be catered to with remaining meal? (30)
- An industrial loom weaves 0.128 metres of cloth every second. Approximately, how many seconds will it take for the loom to weave 25 metres of cloth? (195.31= Approximately 195)

Sub-chapter 16 - Problems on Pipes and Cistern

- A pipe connected with a tank or a cistern or a reservoir, that fills it, is known as an inlet.
- A pipe connected with a tank or cistern or reservoir, emptying it, is known as an outlet.
- If a pipe can fill a tank in x hours, then: part filled in 1 hour = $\frac{1}{x}$
- If a pipe can empty a tank in y hours, then: part emptied in 1 hour = $\frac{1}{y}$

- If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $y > x$), then on opening both the pipes, then the net part filled in 1 hour = $\frac{1}{x} - \frac{1}{y}$
- If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours (where $x > y$), then on opening both the pipes, then the net part emptied in 1 hour = $\frac{1}{y} - \frac{1}{x}$

• **पानी तान्ने क्षमता**

पानी तान्न लाग्ने समय	पानी तान्ने क्षमता	पानी फाल्न लाग्ने समय	पानी फाल्ने क्षमता
20 मिनेटमा	200 लि.	25 मिनेट	200 लि.
१ मिनेटमा	200/20 लि.	१ मिनेट	200/25 लि.
१ मिनेटमा वचत पानी - २ लि. So, 20 मिनेटमा बचत पानी - 40 लि.			

- Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes, and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P, Q and R respectively. What is the proportion of the solution R in the liquid in the tank after 3 minutes?** ($\frac{6}{11}$)

Part filled by (A + B + C) in 3 minutes = $\frac{11}{20}$

Part filled by C in 3 minutes = $\frac{3}{10}$ Hence, Required ratio = $\frac{3}{10} \times \frac{20}{11} = \frac{6}{11}$

- A pump can fill a tank with water in 2 hours. Because of a leak, it took $2\frac{1}{3}$ hours to fill the tank. The leak can drain all the water of the tank in:** (14 hours)

Work done by the leak in 1 hour = $\frac{1}{2} - \frac{3}{7} = \frac{1}{14}$

Hence, Leak will empty the tank in 14 hrs.

- Two pipes A and B can fill a cistern in $37\frac{1}{2}$ minutes and 45 minutes respectively. Both pipes are opened. The cistern will be filled in just half an hour, if the B is turned off after:** (9 min)

Let B be turned off after x minutes. Then,

Part filled by (A + B) in x min. + Part filled by A in $(30 - x)$ min. = 1. or full

$$x\left(\frac{1}{45} + \frac{2}{74}\right) + \frac{2}{74} \times (30 - x) = 1 \rightarrow x = 9$$

- A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank in the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 5 hours faster than the first pipe and 4 hours slower than the third pipe. The time required by the first pipe is:** (15 hr.)

Suppose, first pipe alone takes x hours to fill the tank .

Then, second and third pipes will take $(x - 5)$ and $(x - 9)$ hours respectively to fill the tank.

$$\frac{1}{x} + \frac{1}{x-5} = \frac{1}{x-9} \rightarrow x = 15$$

- Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is:** (120 gallons)

Work done by the waste pipe in 1 minute = $\frac{1}{15} - \left(\frac{1}{20} + \frac{1}{24}\right) = \frac{1}{40}$

The capacity of the tank = (3×40) gallons = 120 gallons.

- A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank?** (35 hrs)

Suppose pipe A alone takes x hours to fill the tank.

Then, pipes B and C will take $\frac{x}{2}$ and $\frac{x}{4}$ hours respectively to fill the tank

$$\frac{1}{x} + \frac{x}{2} + \frac{x}{4} = \frac{1}{5} \rightarrow x = 35$$

- A large tanker can be filled by two pipes A and B in 60 minutes and 40 minutes respectively. How many minutes will it take to fill the tanker from empty state if B is used for half the time and A and B fill it together for the other half? (30 min)

$$\text{Part filled by (A + B) in 1 minute} = \frac{1}{60} + \frac{1}{40} = \frac{1}{24}$$

$$\text{Suppose the tank is filled in } x \text{ minutes. } \frac{x}{2} \left(\frac{1}{24} + \frac{1}{40} \right) = 1 \rightarrow \frac{x}{2} \times \frac{1}{15} = 1 \rightarrow x = 30 \text{ min.}$$

- A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely? (3 hrs. 45 mins.)

Time taken by one tap to fill **half of the tank** = 3 hrs.

$$\text{Part filled by the four taps in 1 hour} = \frac{1}{6} \times 4 = \frac{2}{3}$$

$$\text{So, total time taken} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4} \text{ hr.} = 45 \text{ min.} + 3 \text{ hr.} = 3 \text{ hrs. 45 mins}$$

- Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, the tank will be full in: (7 hrs.)

$$\text{Work done by A + B in 1 hr.} = \frac{1}{12} + \frac{1}{15} = \frac{3}{20}$$

$$\text{Work done by A + C in 1 hr.} = \frac{1}{12} + \frac{1}{20} = \frac{2}{15}$$

$$\text{Part filled in 2 hrs where A + B and A + C are involved once} = \frac{3}{20} + \frac{2}{15} = \frac{17}{60}$$

A can do full work in 12 days and every two days is done by all three.

$$\text{Part filled in 6 hrs} = \frac{17}{60} \times 3 = \frac{17}{20} \text{ So, Remaining part of work} = 1 - \frac{17}{20} = \frac{3}{20}$$

Now turn is of A + B

$$\text{Now, it is the turn of A and B done } \frac{3}{20} \text{ in } \frac{3}{20} \times \frac{20}{3} \text{ hr.} = 1 \text{ hr.}$$

Total time taken to fill the tank = (6 + 1) hrs = 7 hrs.

Sub-chapter 16.1 Data Sufficiency from Pipes and Cistern

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- A. I alone sufficient while II alone not sufficient to answer
- B. II alone sufficient while I alone not sufficient to answer
- C. Either I or II alone sufficient to answer
- D. Both I and II are not sufficient to answer
- E. Both I and II are necessary to answer

- How much time will the leak take to empty the full cistern? → E

1. The cistern is normally filled in 9 hours.
2. It takes one hour more than the usual time to fill the cistern because of la leak in the bottom.

- How long will it take to empty the tank if both the inlet pipe A and the outlet pipe B are opened simultaneously? → E

1. A can fill the tank in 16 minutes.
2. B can empty the full tank in 8 minutes.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- If both the pipes are opened, how many hours will be taken to fill the tank? → B

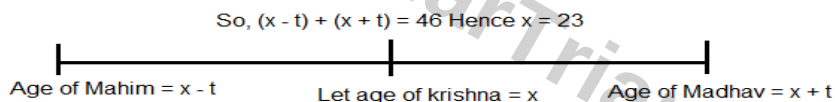
1. The capacity of the tank is 400 litres.
2. The pipe A fills the tank in 4 hours.
3. The pipe B fills the tank in 6 hours.

- A. Only I and II
- B. Only II and III

- C. All I, II and III
- D. Any two of the three
- E. Even with all the three statements, answer cannot be given.

Sub-chapter 17 - Problems on Ages

- If the current age is x , then n times the age is nx .
- If the current age is x , then age n years later/hence $= x + n$.
- If the current age is x , then age n years ago $= x - n$.
- The ages in a ratio $a : b$ will be ax and bx .
- If the current age is x , then $\frac{1}{n}$ of the age is $\frac{x}{n}$.
- **Father is aged three times more than his son Ronit. After 8 years, he would be two and a half times of Ronit's age. After further 8 years, how many times would he be of Ronit's age?** $\rightarrow$ (2 times)
Let Ronit's present age be x years. Then, father's present age $= (x + 3x)$ years $= 4x$ years.
- **A is two years older than B who is twice as old as C. If the total of the ages of A, B and C be 27, then how old is B?** $\rightarrow$ (10 years) $(x + x - 2 + \frac{x-2}{2} = 27 \rightarrow x = 10)$
- **Mahim is younger than Krishna as Madhav is elder than Krishna. Sum of age of Mahim and madhav is 46 years. What is the age of Krishna ?**



- **The age of a person one year before is a square number and one year from now it will be a cube number. What is the present age of the person?**
 1. First 10 squares, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100.
 2. First 5 cubes, 1, 8, 27, 64, 125.

Difference between cube and square is 2, only (So, 27 and 25 are visibly)
 $\rightarrow$ Present age of the person : 26 years.
- **Ram's age was square of a number last year and it will be cube of a number next year. How long must he wait before his age is again the cube of a number?**
 Difference between cube and square is 2, only.
 Present age of the person : 26 years.
 $\rightarrow$ This cube number is 27 and another will be 64. So, he waits $(64 - 26 = 38)$ years to be again the cube of a number.

Sub-chapter 17.1 Data Sufficiency from Problems on Ages

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- A. I alone sufficient while II alone not sufficient to answer
- B. II alone sufficient while I alone not sufficient to answer
- C. Either I or II alone sufficient to answer
- D. Both I and II are not sufficient to answer
- E. Both I and II are necessary to answer
- **What is Sonia's present age?** $\rightarrow$ E
 1. Sonia's present age is five times Deepak's present age.
 2. Five years ago her age was twenty-five times Deepak's age at that time.
- **Average age of employees working in a department is 30 years. In the next year, ten workers will retire. What will be the average age in the next year?** $\rightarrow$ E
 1. Retirement age is 60 years.
 2. There are 50 employees in the department.
- **Divya is twice as old as Shruti. What is the difference in their ages?** $\rightarrow$ C
 1. Five years hence, the ratio of their ages would be 9 : 5.
 2. Ten years back, the ratio of their ages was 3 : 1.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

• **What is Arun's present age?** → D

- Five years ago, Arun's age was double that of his son's age at that time.
- Present ages of Arun and his son are in the ratio of 11 : 6 respectively.
- Five years hence, the respective ratio of Arun's age and his son's age will become 12 : 7. →

- Only I and II
- Only II and III
- Only I and III
- Any two of the three
- None of these

• **What is Ravi's present age?** → A

- The present age of Ravi is half of that of his father.
- After 5 years, the ratio of Ravi's age to that of his father's age will be 6 : 11.
- Ravi is 5 years younger than his brother.

- I and II only
- II and III only
- I and III only
- All I, II and III
- Even with all the three statements answer cannot be determined.

• **What is the present age of Tanya?** → E

- The ratio between the present ages of Tanya and her brother Rahul is 3 : 4 respectively.
- After 5 years the ratio between the ages of Tanya and Rahul will be 4 : 5.
- Rahul is 5 years older than Tanya.

- I and II only
- II and III only
- I and III only
- All I, II and III
- Any two of the three

• **What will be the ratio between ages of Sam and Albert after 5 years?** → A

- Sam's present age is more than Albert's present age by 4 years.
- Albert's present age is 20 years.
- The ratio of Albert's present age to Sam's present age is 5 : 6.

- Any two of I, II and III
- II only
- III only
- I or III only
- II or III only

• **What is the difference between the present ages of Ayush and Deepak?** → A

- The ratio between Ayush's present age and his age after 8 years 4 : 5.
- The ratio between the present ages of Ayush and Deepak is 4 : 3.
- The ratio between Deepak's present age and his age four years ago is 6 : 5.

- Any two of I, II and III
- I or III only
- Any one of the three
- All I, II and III are required
- Even with all I, II and III, the answer cannot be obtained.

Sub-chapter 18 - Permutation and Combination

- And (×), Or (+)
- Let n be a positive integer. Then, factorial n , denoted $n!$ is defined as: $n! = n(n - 1)(n - 2) \dots 3.2.1$.
We define $0! = 1$.
 $4! = (4 \times 3 \times 2 \times 1) = 24$.
 $5! = (5 \times 4 \times 3 \times 2 \times 1) = 120$.

- **All permutations** (or arrangements) made with the letters a, b, c by taking **two at a time** are (ab, ba, ac, ca, bc, cb) .
- **All permutations** made with the letters a, b, c taking **all at a time** are: $(abc, acb, bac, bca, cab, cba)$
- Number of all permutations of n things, taken r at a time, is given by ${}^n P_r = \frac{n!}{(n-r)!}$
 ${}^6 P_2 = (6 \times 5) = 30$ ${}^7 P_3 = (7 \times 6 \times 5) = 210$
- If there are n subjects of which p_1 are alike of one kind; p_2 are alike of another kind; p_3 are alike of third kind and so on and p_r are alike of r^{th} kind, such that $(p_1 + p_2 + \dots + p_r) = n$.

Then, number of permutations of these n objects is $= \frac{n!}{(p_1!)(p_2!) \dots (p_r!)}$

- Suppose we want to select two out of three boys A, B, C . Then, possible selections are AB, BC and CA . (AB and BA represent the same selection.)
- All the combinations formed by a, b, c taking ab, bc, ca .
- The only combination that can be formed of three letters a, b, c taken all at a time is abc
- Various groups of 2 out of four persons A, B, C, D are: AB, AC, AD, BC, BD, CD .
- Note that ab ba are two different permutations but they represent the same combination.
- The number of all combinations of n things, taken r at a time is: ${}^n C_r = \frac{n!}{r!(n-r)!}$

$${}^{11} C_4 = \frac{(11 \times 10 \times 9 \times 8)}{(4 \times 3 \times 2 \times 1)} = 330 \quad {}^{16} C_{13} = \frac{(16 \times 15 \times 14)}{3!} = 560$$

- **Making group of member 2 from total 5 people -**

$$\text{Total groups} = \frac{n!}{(n-r)! r!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{(5-2)! \times 2!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 2 \times 1} = 10$$

- Writing in different types of a word : $\text{SUNDAY} = N 1 = 6! = 720$

- Writing 1, 2, 3, 4 in different 3 membered groups,
Total number of groups $= 4^3 = 64$ (With number repetition)

$$\text{Total number of groups} = \frac{n!}{(n-r)!} = \frac{4!}{(4-3)!} = 24 \text{ (Without number repetition)}$$

- Writing in different types of a word : TOMORROW (Without repetition of O and R)

• Condition	• Calculation
– Repetition of O = 3 Times – Repetition of R = 2 Times	So, Total arrangements methods $= \frac{n!}{O! R!} = \frac{8!}{3! 2!} = 336$

- We cannot put $(n!)$ in round table, but can only in $(n-1)!$
– 5 peoples can be arranged in round table differently $= (n-1)! = (5-1)! = 4! = 24$
- Ways of selection of 4 peoples out of 8 $= \frac{n!}{(n-r)! r!} = \frac{8!}{(8-4)! 4!} = 70$
- 5 can be selected in 10 people where, two are selected previously, Then, $n = 10 - 2 = 8$ and $r = 5 - 2 = 3$

$$\text{Ways of selection of peoples} = \frac{n!}{(n-r)! r!} = \frac{8!}{(8-3)! 3!} = 56$$

- **In how many ways can the letters of the word 'LEADER' be arranged? (360)**

LEADER in which E occurs 2 times can be arranged in $\frac{6!}{2!} = 360$ ways

- **Writing in different types of a word : EDUCATION** (vowels always come together)
we have to arrange the letters DCTN (EUAIO)

Now, $4 + 1 = 5$ letters can be arranged in $5! = 120$ ways

The vowels (EUAIO) can be arranged among themselves in $5! = 120$ ways

Total required number of ways $= (120 \times 120) = 14400$

- **In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together? (720)**

we have to arrange the letters LNDG (EAI)

Now, $4 + 1 = 5$ letters can be arranged in $5! = 120$ ways

The vowels (EAI) can be arranged among themselves in $3! = 6$ ways.

Total required number of ways = $(120 \times 6) = 720$.

- **In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together?** (50400)

we have to arrange the letters CRPRTN (OOAIO)

This has $6 + 1 = 7$ letters of which R occurs 2 times can be arranged in $\frac{7!}{2!} = 2520$ ways.

The vowels (OOAIO) in which O occurs 3 times can be arranged in $\frac{5!}{3!} = 20$ ways

Total required number of ways = $(2520 \times 20) = 50400$.

- **In how many different ways can the letters of the word 'MATHEMATICS' be arranged so that the vowels always come together?** (120960)

we have to arrange the letters MTHMTCS (AEAI)

This has $7 + 1 = 8$ letters of which M and T occurs 2 times can be arranged in $\frac{8!}{2!2!} = 10080$ ways.

The vowels (AEAI) in which A occurs 2 times can be arranged in $\frac{4!}{2!} = 12$ ways

Total required number of ways = $(10080 \times 12) = 120960$.

- **Ways of arrangements to 4 men and 4 women out of 7 men and 8 women**

Required number of ways = $({}^7C_4 \times {}^8C_4) = \left(\frac{7 \times 6 \times 5 \times 4}{4 \times 3 \times 2 \times 1}\right) \times \left(\frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1}\right) = 2450$

- **In how many ways a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?** (11760)

Required number of ways = $({}^8C_5 \times {}^{10}C_6) = \left(\frac{8 \times 7 \times 6 \times 5 \times 4}{5 \times 4 \times 3 \times 2 \times 1}\right) \times \left(\frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{6 \times 5 \times 4 \times 3 \times 2 \times 1}\right) = 11760$

- **From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there on the committee. In how many ways can it be done?** (756)

We may have (3 men and 2 women) or (4 men and 1 woman) or (5 men only).

Required number of ways = $({}^7C_3 \times {}^6C_2) + ({}^7C_4 \times {}^6C_1) + ({}^7C_5) = 756$ (Where ${}^7C_3 = \frac{7 \times 6 \times 5}{3 \times 2 \times 1}$)

- **Out of 7 consonants and 4 vowels, how many words of 3 consonants and 2 vowels can be formed?** (25200)

Ways of selecting (3 consonants out of 7) and (2 vowels out of 4) = $({}^7C_3 \times {}^4C_2) = 210$

Ways of forming new group of (3 consonants and 2 vowels) = 5 is $5! = 210$

Required number of ways = $210 \times 210 = 25200$

- **In a group of 6 boys and 4 girls, four children are to be selected. In how many different ways can they be selected such that at least one boy should be there?** (209)

We may have (1 boy and 3 girls) or (2 boys and 2 girls) or (3 boys and 1 girl) or (4 boys).

Required number of ways = $({}^6C_1 \times {}^4C_3) + ({}^6C_2 \times {}^4C_2) + ({}^6C_3 \times {}^4C_1) + ({}^6C_4) = 209$

- **How many 3-digit numbers can be formed from the digits 2, 3, 5, 6, 7 and 9, which are divisible by 5 and none of the digits is repeated?** (20)

- **A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?** (64)

We may have (1 black and 2 non-black) or (2 black and 1 non-black) or (3 black).

Required number of ways = $({}^3C_1 \times {}^6C_2) + ({}^3C_2 \times {}^6C_1) + ({}^3C_3) = 45 + 18 + 1 = 64$

- **In how many different ways can the letters of the word 'DETAIL' be arranged in such a way that the vowels occupy only the odd positions?** (36)

All 6 letters of which there are 3 vowels and 3 consonants. Vowel and consonant can be put in $3/3$ ways

So, ways of arranging the vowels $\times$ ways of arranging the consonants = ${}^3C_3 \times {}^3C_3 = 36$

- **How many 4-letter words with or without meaning, can be formed out of the letters of the word, 'LOGARITHMS', if repetition of letters is not allowed?** (5040)

Sub-chapter 19 - Probability

- **Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5? (9/20)**
- **A bag contains 2 red, 3 green and 2 blue balls. Two balls are drawn at random. What is the probability that none of the balls drawn is blue? (10/21)**
Total number of balls = $(2 + 3 + 2) = 7$.
Number of ways of drawing 2 balls out of 7 = ${}^7C_2 = 21$
Event of drawing 2 balls, none of which is blue. = ${}^5C_2 = 10$
probability that none of the balls drawn is blue = $10/21$
- **In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green? (1/3)**
- **What is the probability of getting a sum 9 from two throws of a dice? (1/9)**
In two throws of a dice, $n(S) = (6 \times 6) = 36$.
Let E = event of getting a sum = $\{(3, 6), (4, 5), (5, 4), (6, 3)\}$.
the probability of getting a sum 9 from two throws of a dice = $4/36 = 1/9$
- **Three unbiased coins are tossed. What is the probability of getting at most two heads? (7/8)**
Here $S = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}$
Let E = event of getting at most two heads.
Then $E = \{TTT, TTH, THT, HTT, THH, HTH, HHT\}$.
The probability of getting at most two heads? = $7/8$
- **Two dice are thrown simultaneously. What is the probability of getting two numbers whose product is even? (3/4)**
In a simultaneous throw of two dice, we have $n(S) = (6 \times 6) = 36$.
Then, $E = \{(1, 2), (1, 4), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 2), (3, 4), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 4), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$
The probability of getting two numbers whose product is even = $3/4$
- **In a class, there are 15 boys and 10 girls. Three students are selected at random. The probability that 1 girl and 2 boys are selected, is:**
Number ways of selecting 3 students out of 25 = ${}^{25}C_3 = 2300$
 $n(E) = ({}^{10}C_1 \times {}^{15}C_2)$
The probability that 1 girl and 2 boys are selected, is: $1050/2300 = 21/46$
- **In a lottery, there are 10 prizes and 25 blanks. A lottery is drawn at random. What is the probability of getting a prize? (2/7)**
- **From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings? (1/221)**
Let S be the sample space. Then, $n(S) = {}^{52}C_2 = 1326$
Let E = event of getting 2 kings out of 4.
 $n(E) = {}^4C_2 = 6$
the probability of both the cards being kings = $6/1326 = 1/221$
- **Two dice are tossed. The probability that the total score is a prime number is: (5/12)**
Clearly, $n(S) = (6 \times 6) = 36$.
Let E = Event that the sum is a prime number.
Then $E = \{(1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 2), (5, 6), (6, 1), (6, 5)\}$
The probability that the total score is a prime number is: $5/12$
- **A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is: (1/26)**
- **A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is: (2/91)**
number of ways of drawing 3 balls out of 15 = ${}^{15}C_3 = 455$
event of getting all the 3 red balls. = ${}^5C_3 = 10$
The probability that all of them are red, is: $2/91$

- Two cards are drawn together from a pack of 52 cards. The probability that one is a spade and one is a heart, is: $(13/102)$
sample space. $= {}^{52}C_2 = 1326$.
number of ways of choosing 1 spade out of 13 and 1 heart out of 13 $= ({}^{13}C_1 \times {}^{13}C_1)$
The probability that one is a spade and one is a heart, is: $13/102$
- One card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is a face card (Jack, Queen and King only)? $(3/13)$
- A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white? $(4/7)$

Sub-chapter 20 - Game and Race or Time and Distance

- $1 \text{ km/hr.} = \frac{5}{18} \text{ m/s}$ $1 \text{ m/s} = \frac{18}{5} \text{ km/hr.}$
- If the ratio of the speeds of A and B is $\frac{v_1}{v_2}$, Then the ratio of the times taken $= \frac{T_2}{T_1}$ So, $\frac{v_1}{v_2} = \frac{T_2}{T_1}$
- For equal distance :** $\frac{v_1}{v_2} = \frac{T_2}{T_1}$
- झापादेखि गाईघाटसम्म कुनै गाडीलाई 75 km/hr. लाग्छ, सोही बाटो 60 km/hr. को गतिमा फर्कदा 40 मिनेट अधिक समय लाग्छ भने दुरी कति होला ?
 $\frac{v_1}{v_2} = \frac{T_2}{T_1} \rightarrow \frac{75}{60} = \frac{x+40/60}{x} \rightarrow x = 40/15 \rightarrow \text{So, } d = 200 \text{ km.}$
- For equal distance** Let, speed of first part $= x \text{ km/hr}$ and speed of second part $= y \text{ km/hr}$.
Then, the average speed during the whole journey $= \frac{2xy}{x+y} \text{ km/hr.}$ Or, $= \frac{3xyz}{xy+yz+xz}$
- कुनै मानिसले आफ्नो कुल यात्राको P भाग $x \text{ km/hr.}$ को वेगले, Q भाग $y \text{ km/hr.}$ को वेगले र R भाग $z \text{ km/hr.}$ को वेगले पार गर्छ भने, औसत वेग $= \frac{1}{\frac{P}{x} + \frac{Q}{y} + \frac{R}{z}}$
- If a person walks speed at 14 km/hr instead of 10 km/hr , he would have walked 20 km more.**
The actual distance travelled by him is: $(v = \frac{d}{t})$
 $t = \frac{d}{10} \dots\dots\dots(i)$ $t = \frac{d+20}{14} \dots\dots\dots(ii)$ $\Rightarrow 14d = 10d + 200 \Rightarrow d = 50 \text{ km}$
- A train can travel 50% faster than a car. Both start from point A at the same time and reach point B 75 kms away from A at the same time. On the way, however, the train lost about 12.5 minutes while stopping at the stations. The speed of the car is:**
Let, speed of the car be $v \text{ km/hr.}$ speed of the train be $\frac{3v}{2} \text{ km/hr.}$
Time difference $= 12.5 \text{ min} = \frac{12.5}{60} \text{ hr.}$
speed difference $= \frac{3v}{2} - v \text{ km/hr.} = \frac{v}{2} \text{ km/hr.}$
For same distance, $d = \frac{v}{t} \Rightarrow 75 \text{ km} = \frac{v}{\frac{12.5}{60}} \Rightarrow v = 75 \times \frac{60}{12.5} = 120 \text{ km/hr.}$
- Excluding stoppages, the speed of a bus is 54 km/hr. and including stoppages, it is 45 km/hr.**
For how many minutes does the bus stop per hour? (Faster speed $= v_1$, slower speed $= v_2$)
difference in speed $= 54 - 45 = 9 \text{ km/hr.}$
Stoppage time per hour $= \frac{v_1 - v_2}{v_1} = \frac{54 - 45}{54} = \frac{9}{54} = \frac{1}{6} \text{ hr.} = 10 \text{ minutes.}$
- In a flight of 600 km , an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/hr and the time of flight increased by 30 minutes. The duration of the flight is: (Use $v = \frac{d}{t}$ for both cases)**

Case - I	Case - II	
$d = 600 \text{ km}$	$d' = 600 \text{ km}$	$\frac{600}{t} - 200 = \frac{600}{t+1/2} \Rightarrow \frac{600-200}{t} = \frac{600}{t+1/2}$ $\Rightarrow 400t + 200 = 600t$ $\Rightarrow 200t = 200$ $\Rightarrow t = 1 \text{ hr.} \Rightarrow \text{Duration of flight} = 1 \text{ hr.}$
$t = t$	$t' = t + 30 \text{ min} = t + 1/2$	
$v = v$	$v' = v - 200$	
$v = \frac{600}{t} \dots\dots\dots i$	$v - 200 = \frac{600}{t+1/2} \dots\dots\dots ii$	

- A man complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr . Find the total journey in km . (Find first average velocity by using $\frac{2xy}{x+y}$ and then use $v = \frac{d}{t}$)**

- In covering a distance of 30 km, Abhay takes 2 hours more than Sameer. If Abhay doubles his speed, then he would take 1 hour less than Sameer. Abhay's speed is:

For Sameer	For Avay Case-1	For Avay Case-2	For same distance :
$d = 30\text{km}$ $t = t$ $v = \frac{30}{t} \dots\dots\dots i$	$d' = 30\text{km}$ $t' = t + 2\text{hr.}$ $v' = \frac{30}{t+2} \dots\dots\dots ii$	$d' = 30\text{km}$ $t' = t - 1\text{hr.}$ $v' = 2(\frac{30}{t+2}) \dots\dots\dots ii$	$(t + 2) \times (\frac{30}{t+2}) = (t - 1) \times 2(\frac{30}{t+2})$ Therefore, $t = 4$ $v' = \frac{30}{t+2} = 5\text{km/hr}$

- Robert is travelling on his cycle and has calculated to reach point A at 2 P.M. if he travels at 10 kmph, he will reach there at 12 noon if he travels at 15 kmph. At what speed must he travel to reach A at 1 P.M.?

Case I	Case II	Case III	
$d = d$ $t = t$ $v = 10\text{km/hr.}$	$d' = d$ $t' = t - 2$ $v' = 15\text{km/hr.}$	$d'' = d$ $t'' = t - 1$ $v' = ?$	$10 \times t = 15 \times (t-2) \Rightarrow t = 6$ So, $t'' = 6-1 = 5\text{hr.}$ and $d = vt = 60\text{km/hr}$ Therefore, $v'' = \frac{d}{t''} = 12\text{km/hr}$

- It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the cars is : (Combine rule for time)

Total distnace = 600km Let, speed of train = v and car = v'	
For time $t = 8$ hr.	For time $t = 8$ hr. 20 min = $\frac{25}{3}$ hr.
$t = \frac{d_1}{v} + \frac{d_2}{v'} \Rightarrow \frac{120}{v} + \frac{480}{v'} = 8 \dots\dots\dots i$	$t = \frac{d_1}{v} + \frac{d_2}{v'} \Rightarrow \frac{200}{v} + \frac{400}{v'} = \frac{25}{3} \dots\dots\dots ii$
Then, by calculating $v : v' = 3 : 4$	

- A farmer travelled a distance of 61 km in 9 hours. He travelled partly on foot @ 4 km/hr and partly on bicycle at 9 km/hr. The distance travelled on foot is:

Total distance = 61km Total time = 9 hr. Let, distance travelled by foot = d and bicycle = 61-d
$T = \frac{d_1}{v} + \frac{d_2}{v'} \Rightarrow \frac{d}{4} + \frac{61-d}{9} = 9 \Rightarrow d = 16 \text{ km}$

- A man covered a certain distance at some speed. Had he moved 3 kmph faster, he would have taken 40 minutes less. If he had moved 2 kmph slower, he would have taken 40 minutes more. The distance (in km) is: (40 km)

Let distance = d and usual rate = v

$$\frac{d}{v} - \frac{d}{v+3} = \frac{40}{60} \Rightarrow 2v(v+3) = 9d \dots\dots\dots i \quad \quad \quad \frac{d}{v-2} - \frac{d}{v} = \frac{40}{60} \Rightarrow v(v-2) = 3d \dots\dots\dots ii$$

Dividing, So $d = 40 \text{ km}$

- In a 100 m race, A can give B 10 m and C 28 m. In the same race B can give C: $\rightarrow 20 \text{ m}$

$$A : B = 100 : 90. \quad A : C = 100 : 72. \quad \text{So, } \frac{A}{B} = \frac{100}{90}, \frac{A}{C} = \frac{100}{72} \rightarrow \frac{B}{A} \times \frac{A}{C} = \frac{90}{100} \times \frac{100}{72} = \frac{90}{72}$$

In respect to 100 m, In 9072 Hence, In 100.....80

- A and B take part in 100 m race. A runs at 5 kmph. A gives B a start of 8 m and still beats him by 8 seconds. The speed of B is: $\rightarrow (4.14)$

$$A's \text{ speed} = \frac{25m}{18s} \quad \quad \quad \text{Time taken by A to cover 100 m} = \frac{18}{25} \times 100 = 72 \text{ s}$$

$$\text{Time taken by B to cover 92 m} = (72 + 8) = 80 \text{ sec.} \quad \quad \quad B's \text{ speed} = \frac{92}{80} \times \frac{18}{5} = 4.14 \text{ m/s}$$

- In a 500 m race, the ratio of the speeds of two contestants A and B is 3 : 4. A has a start of 140 m. Then, A wins by $\rightarrow (20 \text{ m})$

To reach the winning post A will have to cover a distance of $(500 - 140)\text{m}$, i.e., 360 m.

While A covers 3 m, B covers 4 m. So, While A covers 360 m, B covers $\frac{4}{3} \times 360 = 480 \text{ m}$

A wins by 20 m.

- In a 100 m race, A beats B by 10 m and C by 13 m. In a race of 180 m, B will beat C by:

$$A : B = 100 : 90. \quad A : C = 100 : 87. \quad \text{So, } B : C = 90 : 87$$

B90C87

B.....1C $\frac{87}{90}$

B180C $\frac{87}{90} \times 180 = 174$ So, B will beat C by 6 m.

- At a game of billiards, A can give B 15 points in 60 and A can give C to 20 points in 60. How many points can B give C in a game of 90? (10 points)

$$A : B = 60 : 45.$$

$$A : C = 60 : 40.$$

$$\text{So, } B : C = 90 : 80$$

B can give C 10 points

- **In 100 m race, A covers the distance in 36 seconds and B in 45 seconds. In this race A beats B by → (20 m)**

$$\text{Distance covered by B in 9 sec.} = \frac{100}{45} \times 9 \text{ m.} = 20 \text{ m.}$$

A beats B by 20 metres.

- **In a game of 100 points, A can give B 20 points and C 28 points. Then, B can give C: (10 points)**

$$A : B = 100 : 80. \quad A : C = 100 : 72. \quad \text{So, } B : C = 80 : 72$$

In respect to 100 m, In 8072 Hence In 100.....90

- **In a 200 metres race A beats B by 35 m or 7 seconds. A's time over the course is: (33 Sec.)**

$$B \text{ covers } 200 \text{ m in } \frac{7}{35} \times 200 = 40 \text{ sec.}$$

B's time over the course = 40 sec.

A's time over the course (40 - 7) sec = 33 sec.

- **In a 300 m race A beats B by 22.5 m or 6 seconds. B's time over the course is: (80 sec.)**

$$B \text{ runs } 22.5 \text{ m in } 6 \text{ sec. So, } B \text{ covers } 300 \text{ m in } \frac{1}{22.5} \times 6 \times 300 = 80 \text{ sec.}$$

- **A can run 22.5 m while B runs 25 m. In a kilometre race B beats A by: (100m)**

$$B \text{} 25 \text{} A \text{} 22.5$$

$$B \text{} 1 \text{} A \text{} 25/22.5$$

$$B \text{} 1000 \text{} A \text{} \frac{25}{22.5} \times 1000 = 900 \text{ m. Hence, B beats A by } 100 \text{ m.}$$

- **A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80 m, how far must the winning post be so that A and B might reach it at the same time? (200 m)**

$$\text{Ratio of the speeds of A and B} = 1\frac{2}{3} : 1 = 5 : 3$$

Thus, in race of 5 m, A gains 2 m over B.

2 m are gained by A in a race of 5 m.

$$80 \text{ m will be gained by A in race of } \frac{5}{2} \times 80 \text{ m} = 200 \text{ m.}$$

Winning post is 200 m away from the starting point.

- **A clerk walks from his house at 5km/hr and reaches his office 10 mins late. If his speed is 6km/hr, he will reach 10 mins early. How far in km is his office? (10 km) $(T_2 - T_1) = \frac{d}{V_1} - \frac{d}{V_2}$**

$$\text{Let the distance be } d \text{ km, Difference in time : } (T_2 - T_1) = \frac{d}{V_1} - \frac{d}{V_2} = \frac{d}{5} - \frac{d}{6} = \frac{20}{60} \text{ hr. So, } d = 10 \text{ km}$$

- **A man performs $\frac{4}{15}$ of the total journey by train, $\frac{3}{20}$ by bus, and the remaining 12 km on foot. What is his total journey in km?**

Here, Total distance = x,

$$\text{Then, Rest distance} = 1 - \left(\frac{4}{15} + \frac{3}{20}\right) = \frac{7}{12} \rightarrow x \times \frac{7}{12} = 12 \text{ km} \rightarrow x = 144/7 \text{ km} = 20.57 \text{ km}$$

- **For race of two participants : $\frac{\text{Winner's Time}}{\text{Looser's distance}} = \frac{\text{Looser's Time}}{\text{Winner's distance}} = \frac{\text{Start Time+Beat Time}}{\text{Start distance+Beat distance}}$**

$$\frac{WT}{LD} = \frac{LT}{WD} = \frac{ST+BT}{SD+BD}$$

Eg. In 1 km race, A beats B by 36 meters or 9 seconds. How long does A take to finish the course ?

$$\frac{LT}{WD} = \frac{ST+BT}{SD+BD} \rightarrow \frac{A's \text{ Time}}{1000-36} = \frac{9+0}{36+0} \rightarrow A's \text{ time} = 241 \text{ sec.}$$

- **Race of three participants. Distance between 1st - 2nd = AB, 1st - 3rd = AC, 2nd - 3rd = BC,**

$$\text{Then, Length of Race} = \frac{AB \cdot BC}{AB+BC-AC}$$

Eg. In a race, A beats B by 12 m, and C by 18 m. B beats C by 8 m., What is the Length of race?

$$\text{Length of Race} = \frac{AB \cdot BC}{AB+BC-AC} \rightarrow \text{Length of Race} = \frac{12 \times 8}{12+8-18} = 48 \text{ m.}$$

Sub-chapter 20.1 - Data Sufficiency from Game and Race or Time and Distance

Apply the condition from the given statements :

- 1 alone sufficient while 2 alone not sufficient to answer
- 2 alone sufficient while 1 alone not sufficient to answer
- Either 1 or 2 alone sufficient to answer

- D. Both 1 and 2 are not sufficient to answer
 E. Both 1 and 2 are necessary to answer
- **Two towns are connected by railway. Can you find the distance between them? → D**
 1. The speed of the mail train is 12 km/hr more than that of an express train.
 2. A mail train takes 40 minutes less than an express train to cover the distance.
 - **The towns A, B and C are on a straight line. Town C is between A and B. The distance from A to B is 100 km. How far is A from C? → C**
 1. The distance from A to B is 25% more than the distance from C to B.
 2. The distance from A to C is $\frac{1}{4}$ of the distance C to B.
 - **Two cars pass each other in opposite direction. How long would they take to be 500 km apart? → A**
 1. The sum of their speeds is 135 km/hr.
 2. The difference of their speed is 25 km/hr
 - **How much time did X take to reach the destination? → E**
 1. The ratio between the speed of X and Y is 3 : 4.
 2. Y takes 36 minutes to reach the same destination.

Sub-chapter 21 - Height and Distance

	0°	30°	45°	60°	90°	
Sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	• Angle of Elevation • Angle of Depression • $\sin^2 \theta + \cos^2 \theta = 1$. • $1 + \tan^2 \theta = \sec^2 \theta$. • $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.
Cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	
Tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	

- **From a point P on a level ground, the angle of elevation of the top tower is 30°. If the tower is 100 m high, the distance of point P from the foot of the tower is:** ($\tan 30^\circ = \frac{1}{\sqrt{3}}$, So, $b = 100 \sqrt{3} = 173$)
- **Angle of elevation**
- **Angle of depression**

Sub-chapter 22 - Problems on Trains and speed

- $\frac{km}{hr}$ to $\frac{m}{s}$ conversion: $a \frac{km}{hr} = a \times \frac{5}{18} \frac{m}{s}$
- $\frac{m}{s}$ to $\frac{km}{hr}$ conversion: $a \frac{m}{s} = a \times \frac{18}{5} \frac{km}{hr}$
- **Formulas for finding Speed, Time and Distance**
 1. Use ($d = v \times t$) in normal condition.
 2. Time taken by a train of **length/meters** to pass a pole or standing man or a signal post = Time taken by the train to cover/meters.
 3. Time taken by a train of length l metres to pass a stationary object of length b metres = Time taken by the train to cover $(l + b)$ metres.
 4. If two bodies are moving at u m/s and v m/s, where $u > v$, relative speed is
 For same direction R.V. = $(u - v)$ m/s.
 For opposite direction R.V. = $(u + v)$ m/s.
 5. If two bodies of length a metres and b metres are moving, Time taken by the body to cross each other
 For opposite directions, Time taken = $\frac{a + b}{u + v}$ second.
 For same direction, Time taken = $\frac{a + b}{u - v}$ second.
 6. If two trains/bodies start at the same time from points A and B towards each other and after crossing they take a and b sec in reaching B and A respectively,
 Then, Speed of A : Speed of B = $b : a$

- A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train? (Find $d = ?$ where, $d = v \times t$)
- A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed of the train is : (Use $d = v/t$ for both train and man)

Length of train = 125 m So, Relative speed = $\frac{125}{10} \text{ m/s} = \frac{125}{10} \times \frac{18}{5} \text{ km/hr} = 45 \text{ km/hr}$	Time = 10sec. speed of man = 5km/hr For same direction, R.V. = $V_1 - V_2$ So, $V_1 = 45 + 5 = 50 \text{ km/hr}$.
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- The length of the bridge, which a train 130 metres long and travelling at 45 km/hr can cross in 30 seconds, is ($v = d \times t$)

For Train, for only train $d = 130 \text{ m}$ $V = 45 \text{ km/hr} = 12.5 \text{ m/s}$ $T = 30 \text{ s}$	For Bridge, For relative to train $v = 12.5 \text{ m/s}$ $T = 30 \text{ s}$ So, $d = 12.5 \times 30 = 375$ Therefore, Length of only bridge = $375 - 130 = 245 \text{ m}$
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- Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds.

The ratio of their speeds ($T = 23 \text{ s}$, $t_1 = 27 \text{ s}$, $t_2 = 17 \text{ s}$)

$$V_1 : V_2 = \frac{T - T_2}{T - T_1} = \frac{23 - 17}{23 - 27} = 3 : 2 \quad (\text{Note : Ignore negative sign})$$

- A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/hr, what is the length of the platform?

For only train	
For station = 36s	For man = 20 s So, Platform is smaller than train so, $T = 36 - 20 = 16 \text{ s}$
$V = 54 \text{ km/hr} = 15 \text{ m/s}$	So, length of the platform (d) = $V \times T = 15 \times 16 = 240 \text{ m}$

Sub-chapter 22.1 - Data Sufficiency from Problems on Trains

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- (A) - 1 alone sufficient while 2 alone not sufficient.
 - (B) - 2 alone sufficient while 1 alone not sufficient.
 - (C) - Either 1 or 2 alone sufficient.
 - (D) - Both 1 and 2 are not sufficient.
 - (E) - Both 1 and 2 are necessary.
- What is the speed of the train whose length is 210 metres? → E
 - The train crosses another train of 300 metres length running in opposite direction in 10 seconds.
 - The train crosses another train running in the same direction at the speed of 60 km/hr in 30 seconds.
 - What is the length of a running train crossing another 180 metre long train running in the opposite direction? → E
 - The relative speed of the two trains was 150 kmph.
 - The trains took 9 seconds to cross each other.
 - What is the length of a running train? → E
 - The train crosses a man in 9 seconds.
 - The train crosses a 240 metre long platform in 24 seconds.

Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.

- What is the speed of the train? → D
 - The train crosses a signal pole in 18 seconds.
 - The train crosses a platform of equal length in 36 seconds.
 - Length of the train is 330 metres.

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 3 and either 1 or 2 only
E. Any two of the three
- What is the speed of the train? → A
 - The train crosses a tree in 13 seconds.

2. The train crosses a platform of length 250 metres in 27 seconds.
3. The train crosses another train running in the same direction in 32 seconds.
 A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. Any two of the three
 E. None of these

Each of these questions is followed by three statements. You have to study the question and all the three statements given to decide whether any information provided in the statement(s) is redundant and can be dispensed with while answering the given question.

• **At what time will the train reach city X from city Y? → D**

1. The train crosses another train of equal length of 200 metres and running in opposite directions in 15 seconds.
2. The train leaves city Y at 7.15 a.m. for city X situated at a distance of 558 km.
3. The 200 metres long train crosses a signal pole in 10 seconds.
 A. 1 only B. 2 only C. 3 only D. 2 and 3 only E. All 1, 2 and 3 are required.

• **What is the length of a running train P crossing another running train Q? → E**

1. These two trains take 18 seconds to cross each other.
2. These trains are running in opposite directions.
3. The length of the train Q is 180 metres.
 A. 1 only B. 2 only C. 3 only D. All I, II and III are required
 E. Even with I, II and III, the answer cannot be obtained.

Sub-chapter 23 - Boats and Streams

- In water, the **direction along the stream** is called **downstream**. And, the **direction against the stream** is called **upstream**.
- If the **speed of a boat** in still water is **u km/hr** and the **speed of the stream** is **v km/hr**, then
 Speed downstream = $(u + v)$ km/hr. Speed upstream = $(u - v)$ km/hr.
- If the **speed downstream** is **a km/hr** and the **speed upstream** is **b km/hr**, then
 Speed in still water = $\frac{1}{2}(a + b)$ km/hr. Rate of stream = $\frac{1}{2}(a - b)$ km/hr.
- **A boat can travel with a speed of 13 km/hr in still water. If the speed of the stream is 4 km/hr, find the time taken by the boat to go 68 km downstream.** (4 hrs.)
 Speed downstream = $(13 + 4)$ km/hr = 17 km/hr.
 Time taken to travel 68 km downstream = $\frac{68}{17}$ hr. = 4 hr.
- **A man's speed with the current is 15 km/hr and the speed of the current is 2.5 km/hr. The man's speed against the current is:** (10 km/hr)
 Man's rate in still water = $(15 - 2.5)$ km/hr = 12.5 km/hr.
 Man's rate against the current = $(12.5 - 2.5)$ km/hr = 10 km/hr.
- **A boat running upstream takes 8 hours 48 minutes to cover a certain distance, while it takes 4 hours to cover the same distance running downstream. What is the ratio between the speed of the boat and speed of the water current respectively?** (8 : 3)
 Let the man's rate upstream be x kmph and that downstream be y kmph.
 Then, distance covered upstream in 8 hrs 48 min = Distance covered downstream in 4 hrs.
 $x \times \frac{44}{5} = y \times x$ So, $y = \frac{11x}{5}$ Required ratio = $\frac{y+x}{2}, \frac{y-x}{2} = 8 : 3$
- **A motorboat, whose speed in 15 km/hr in still water goes 30 km downstream and comes back in a total of 4 hours 30 minutes. The speed of the stream (in km/hr) is:** (5hr)
 Let the speed of the stream be x km/hr. Then,
 Speed downstream = $(15 + x)$ km/hr,
 Speed upstream = $(15 - x)$ km/hr.
 $\frac{30}{15+x} + \frac{30}{15-x} = \frac{9}{2}$ So, $x = 5$ hr.
- **In one hour, a boat goes 11 km/hr along the stream and 5 km/hr against the stream. The speed of the boat in still water (in km/hr) is:** 8 kmph.
 Speed in still water = $\frac{1}{2}(11 + 5)$ kmph = 8 kmph.

- A boat running downstream covers a distance of 16 km in 2 hours while for covering the same distance upstream, it takes 4 hours. What is the speed of the boat in still water? (6 km/hr)**

Rate downstream = $\frac{16}{2} = 8$ kmph Rate upstream = $\frac{16}{4} = 4$ kmph

Speed in still water = $\frac{1}{2}(8 + 4)$ kmph = 6 kmph.
- The speed of a boat in still water is 15 km/hr and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is:**

Speed downstream = $(15 + 3)$ kmph = 18 kmph.

Distance travelled = $(18 \times \frac{12}{60})$ km = 3.6 km.
- A boat takes 90 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 mph, the speed of the stream is: (2 mph)**

Let the speed of the stream x mph. Then,

Speed downstream = $(10 + x)$ mph,

Speed upstream = $(10 - x)$ mph.

$\frac{36}{10-x} - \frac{36}{10+x} = \frac{90}{60}$ So, $x = 2$ mph
- A man can row at 5 kmph in still water. If the velocity of current is 1 kmph and it takes him 1 hour to row to a place and come back, how far is the place? (2.4 km)**

Speed downstream = $(5 + 1)$ kmph = 6 kmph.

Speed upstream = $(5 - 1)$ kmph = 4 kmph.

Let the required distance be x km.

$\frac{x}{6} + \frac{x}{4} = 1 \rightarrow x = 2.4$ km.
- A boat covers a certain distance downstream in 1 hour, while it comes back in $1\frac{1}{2}$ hours. If the speed of the stream be 3 kmph, what is the speed of the boat in still water? (15 kmph)**

Let the speed of the boat in still water be x kmph. Then,

Speed downstream = $(x + 3)$ kmph,

Speed upstream = $(x - 3)$ kmph.

$(x + 3) \times 1 = (x - 3) \times \frac{3}{2} \rightarrow x = 15$ kmph.
- A boatman goes 2 km against the current in 1 hour and goes 1 km along the current in 10 minutes. How long will it take to go 5 km in stationary water? (1 hr 15 min)**

Rate downstream = $(\frac{1}{6} \times 60)$ km/hr = 6 km/hr

Rate upstream = 2 km/hr.

Speed in still water = km/hr = $\frac{1}{2}(6 + 2)$ km/hr = 4 km/hr.

Required time = $\frac{5}{4}$ hrs = 1 hr 15 min.
- A man can row three-quarters of a kilometre against the stream in $11\frac{1}{4}$ minutes and down the stream in $7\frac{1}{2}$ minutes. The speed (in km/hr) of the man in still water is: (5 km/hr.)**

We can write three-quarters of a kilometre as 750 metres, and $11\frac{1}{4}$ minutes as 675 seconds.

Rate upstream = $\frac{750}{675}$ m/sec = $10/9$ m/sec Rate downstream = $\frac{750}{450}$ m/sec = $5/3$ m/sec

Rate in still water = $\frac{1}{2}(\frac{10}{9} + \frac{5}{3})$ = m/sec = 5 km/hr.
- Speed of a boat in standing water is 9 kmph and the speed of the stream is 1.5 kmph. A man rows to a place at a distance of 105 km and comes back to the starting point. The total time taken by him is: (24 hours)**

Speed upstream = 7.5 kmph.

Speed downstream = 10.5 kmph.

Total time taken = $(\frac{105}{7.5} + \frac{105}{10.5})$ hrs. = 24 hours.

- **A man takes twice as long to row a distance against the stream as to row the same distance in favour of the stream. The ratio of the speed of the boat (in still water) and the stream is: (3 : 1)**
Let man's rate upstream be x kmph.
Then, his rate downstream = $2x$ kmph.
(Speed in still water) : (Speed of stream) = $\frac{2x+x}{2} : \frac{2x-x}{2} = 3 : 1$
- **A man rows to a place 48 km distant and come back in 14 hours. He finds that he can row 4 km with the stream in the same time as 3 km against the stream. The rate of the stream is: (1 km/hr.)**
Suppose he move 4 km downstream in x hours. Then,
Speed downstream = $\frac{4}{x}$ km/hr. Speed upstream = $\frac{3}{x}$ km/hr. $\frac{48}{\frac{4}{x}} + \frac{48}{\frac{3}{x}} = 14 \rightarrow x = \frac{1}{2}$
So, Speed downstream = 8 km/hr, Speed upstream = 6 km/hr.
Rate of the stream = $\frac{1}{2}(8 - 6)$ km/hr = 1 km/hr.

Sub-chapter 23.1 - Data Sufficiency from Boats and Streams

Each of the questions given below consists of a statement and / or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is / are sufficient to answer the given question.

- I alone sufficient while II alone not sufficient to answer
 - II alone sufficient while I alone not sufficient to answer
 - Either I or II alone sufficient to answer
 - Both I and II are not sufficient to answer
 - Both I and II are necessary to answer
- **A boat takes a total time of three hours to travel downstream from P and Q and upstream back from Q to P. What is the speed of the boat in still water? → E**
 - The speed of the river current is 1 km per hour.
 - The distance between P and Q is 4 km.
 - **What is the speed of the boat in still water? → D**
 - It takes 2 hours to cover the distance between A and B downstream.
 - It takes 4 hours to cover the distance between A and B upstream.
 - **What is the speed of the boat in still water? → E**
 - The boat covers a distance of 48 kms in 6 hours while running upstream
 - The boat covers the same distance in 4 hours while running downstream.
- Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the question.
- **What is the speed of the boat in still water? → D**
 - The speed downstream is 12 kmph.
 - The speed upstream is 4 kmph.
 - In a to and fro journey between two points, the average speed of the boat was 6 kmph.
 - I and II only
 - All I, II and III
 - III, and either I or II
 - Any two of the three
 - None of these

Chapter 4 - General mental ability, logical reasoning and analytical ability

Sub-chapter 1 - Data Verification, Checking and Truth

- Be careful with negative/ positive for alternatives of relative subject (Eg. Book cannot be without pages)
- My first is in **blue**, second is in **odd**, third is in **look**, fourth is in **folk** → BOOK
- Finding the same Mixed words and arranging.
- Finding the different words from given two or more texts.

Sub-chapter 2 - Substitution

- Substitute one **given word** to **appropriate another word** with logic.
- Result may be factual or not.
- Meaningful conclusion of words may or may not be identical with given words.

Sub-chapter 3 - Confusing Instruction Test

- Read through the question's instructions to identify potential points of ambiguity or lack of clarity. Think the questions what the question is asking.
- Use elimination method of other choices by evaluating each choice in reverse order.
- **In case $\frac{1}{2}$ when added to $\frac{1}{4}$ makes $\frac{3}{4}$ write "wrong" unless $\frac{3}{4}$ when divided by $\frac{1}{4}$ gives $\frac{1}{2}$, in which case write "right".** → Wrong
- **In a case if $4 + 12 = 18$ is 'right' then state 'right' or 'wrong' for $8 + 24 = 37$** → Right
- **In case $\frac{1}{2} + \frac{1}{4} = \frac{3}{4}$ is wrong, then state 'right' or 'wrong' for $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$** → Wrong

Sub-chapter 4 - Jumble words/ Ordering of words

- **Trick : Check by options.**

In each question below, there is a sentence of which some parts have been jumbled up. Rearrange these parts which are labelled P, Q, R and S to produce the correct sentence. Choose the proper sequence.

- **When he**

P : did not know

Q : he was nervous and

R : heard the hue and cry at midnight

S : what to do

The Proper sequence should be: A. RQPS B. QSPR C. SQPR D. PQRS → A

- **It has been established that**

P : Einstein was

Q : although a great scientist

R : weak in arithmetic

S : right from his school days

The Proper sequence should be: A. SRPQ B. QPRS C. QPSR D. RQPS → B

- **Then**

P : it struck me

Q : of course

R : suitable it was

S : how eminently

The Proper sequence should be: A. SPQR B. QSRP C. PSRQ D. QPSR → C

- **I read an advertisement that said**

P : posh, air-conditioned

Q : gentleman of taste

R : are available for

S : fully furnished rooms

The Proper sequence should be: A. PQRS B. PSRQ C. PSQR D. SRPQ → B

- **Since the beginning of history**

P : have managed to catch

Q : the Eskimos and Red Indians

R : by a very difficult method

S : a few specimens of this aquatic animal

The Proper sequence should be: A. QRPS B. SQPR C. SQRQ D. QPSR → D

- **A French woman**

P : committed suicide

Q : where she had put up

R : who had come to Calcutta

S : by jumping from the first floor balcony of the hotel

The Proper sequence should be: A. PRQS B. QSRP C. RPSQ D. SRQP → C

• **The national unity of a free people**

P : to make it impracticable

Q : for there to be an arbitrary administration

R : depends upon a sufficiently even balance of political power

S : against a revolutionary opposition that is irreconcilably opposed to it

The Proper sequence should be: A. QRPS B. QRSP C. RPQS D. RSPQ → D

• **The grocer**

P : did not listen to the protests of customer

Q : who was in the habit of weighing less

R : whom he had cheated

S : with great audacity

The Proper sequence should be: A. PRSQ B. QSPR C. QPRS D. PQSR → C

• **They felt safer**

P : to watch the mountain

Q : of more than five miles

R : as they settled down

S : from a distance

The Proper sequence should be: A. RPSQ B. RSQP C. PQSR D. PRSQ → A

• **People**

P : at his dispensary

Q : went to him

R : of all professions

S : for medicine and treatment

The Proper sequence should be: A. QPRS B. RPQS C. RQSP D. QRPS → C

• **If you need help**

P : promptly and politely

Q : ask for attendants

R : to help our customers

S : who have instructions

The Proper sequence should be: A. SQPR B. QPSR C. QSRP D. SQRP → C

• **It is easier**

P : to venture into space

Q : for men

R : beneath their feet

S : than to explore

The Proper sequence should be: A. QRPS B. QPSR C. PSRQ D. PQSR → B

• **It was to be**

P : before their school examination

Q : which was due to start

R : the last expedition

S : in a month's

The Proper sequence should be: A. SRQP B. RQSP C. RPQS D. SPRQ → C

• **This time**

P : exactly what he had been told

Q : the young man did

R : beyond his dreams

S : and the plan succeeded

The Proper sequence should be: A. QPRS B. QPSR C. PQSR D. QSRP → B

• **In the darkness**

P : the long, narrow beard

Q : was clearly visible with

R : the tall stooping figure of the doctor

S : and the aquiline nose

The Proper sequence should be: A. RQPS B. PSQR C. RSQP D. QPRS → A

• **We have to**

P : as we see it

Q : speak the truth

R : there is falsehood and weakness

S : even if all around us

The Proper sequence should be: A. RQSP B. QRPS C. RSQP D. QPSR → D

• **When it began to rain suddenly on the first of January**

P : to celebrate the new year

Q : we ran for shelter

R : to the neighbouring house

S : where many people had gathered

The Proper sequence should be: A. QRPS B. PSQR C. PRSQ D. QRSP → D

• **He told us that**

P : and enjoyed it immensely

Q : in a prose translation

R : he had read Milton

S : which he had borrowed from his teacher

The Proper sequence should be: A. RSQP B. QRPS C. RQSP D. RQPS → C

• **He knew that**

P : and then to save himself

Q : was to save all the lives

R : entrusted to his care

S : the duty of a captain

The Proper sequence should be: A. PQRS B. SQRP C. SPRQ D. QSRP → B

• **It is not, therefore**

- P : that I pay a tribute
R : with any feeling of unfriendliness
The Proper sequence should be: A. PQRS B. RQPS C. RSPQ D. PSRQ → C
- **Of many artists**
P : those who impressed me the most
R : but those with unique personalities off stage
The Proper sequence should be: A. SRQP B. QRSP C. RSPQ D. QPSR → D
- **As a disease**
P : and breaks up marriages
R : alcoholism leads to
The Proper sequence should be: A. SRPQ B. RPSQ C. SRQP D. RQPS → C
- **It is easy to excuse**
P : but it is hard
R : the mischief of early childhood
The Proper sequence should be: A. RPQS B. QRSP C. QRPS D. RPSQ → D
- **The majestic mahogany table**
P : belongs to an old prince
R : who is no impoverished
The Proper sequence should be: A. PQSR B. QRSP C. PRSQ D. QPRS → D
- **Education is**
P : of the proper sense of responsibilities
R : in a citizen
The Proper sequence should be: A. SQPR B. QSRP C. QSPR D. PQRS → C
- **It would**
P : appear from his statement
R : in dealing with the strike
The Proper sequence should be: A. RPSQ B. PSQR C. RQPS D. PRQS → B
- **He was so kind and generous that**
P : he not only
R : but also
The Proper sequence should be: A. PSRQ B. SPQR C. PRSQ D. QPRS → A
- **We went**
P : along the railway line
R : where other people not allowed to go
The Proper sequence should be: A. RPQS B. PRSQ C. RSQP D. PRQS → B
- **There was**
P : needed for it everyday life
R : for itself most of the things it
The Proper sequence should be: A. QRSP B. RQPS C. RSPQ D. QSRP → D
- **Little**
P : that he had been let down
R : did he realise
The Proper sequence should be: A. RPSQ B. RSQP C. QSRP D. QSPR → A
- **I saw that**
P : but seeing my host in this mood
R : as I had frequently done before
The Proper sequence should be: A. QPSR B. QRPS C. SPQR D. SRPQ → D
- **It is very easy**
P : a great deal more than one realises
R : that a phrase that one does not quite understand
The Proper sequence should be: A. RSQP B. SPQR C. SRQP D. RQPS → B
- **I was**
P : and stay for a few days in Delhi
R : very excited
The Proper sequence should be: A. PQRS B. RQSP C. QRSP D. SPQR → B
- Q : to conductors as a class
S : to a particular member of that class
- Q : I was a child
S : were not always the successful ones
- Q : accidents and suicides
S : affecting all ages
- Q : in a boy of fourteen
S : to tolerate even unavoidable faults
- Q : which has one leg missing
S : but not without some pride
- Q : the first need
S : for the development
- Q : about the policy of management
S : that he was quite in the dark
- Q : made others do so
S : helped them himself
- Q : and had a right to
S : but daddy belonged to the railway
- Q : a time when each family
S : actually produced
- Q : stood by all these years
S : by a colleague whom he had
- Q : I deemed it proper to take leave
S : it had been my intention to pass the night there
- Q : may mean
S : to persuade oneself
- Q : when my father told me
S : that I could go with him

• **work is the one thing**

P : and without it

R : to keep the world going

Q : that is necessary

S : we should all die

The Proper sequence should be: A. QPSR B. RPQS C. SRPQ D. QRPS → D

• **The appearance**

P : this dinosaurs were at their peak

R : at the time when

Q : of the first mammals on the earth

S : went almost unnoticed

The Proper sequence should be: A. SRPQ B. QSRP C. QRPS D. RPQS → C

• **By this time**

P : at the railway station

R : reports of violence were flooding in

Q : reported mass looting

S : which police dispatches

The Proper sequence should be: A. RSPQ B. SPRQ C. SQRP D. RSQP → C

• **Then the women**

P : lamenting their evil desire

R : wept loudly

Q : that had brought

S : this sorrow upon them

The Proper sequence should be: A. RPQS B. RQPS C. PQSR D. PRQS → A

• **Women**

P : till the other day

R : about spending their time cooking

Q : who were content being housewives

S : now sound apologetic

The Proper sequence should be: A. PQRS B. RSPQ C. QPSR D. RQPS → C

• **Though he dialled frequently**

P : on telephone

R : and had left no information

Q : my brother could not contact me

S : as I had gone out of office

The Proper sequence should be: A. QPRS B. SQRQ C. QPSR D. SPQR → C

• **He said**

P : here is the cat

R : which had been paying havoc with things

Q : throughout the night

S : that ate the rat

The Proper sequence should be: A. PSQR B. PSRQ C. QPRS D. QRPS → B

• **He sat**

P : through the Town Hall Park

R : under the boughs

Q : which flanked a path running

S : of a spreading tamarind tree

The Proper sequence should be: A. PQSR B. RSQP C. RSPQ D. PRSQ → B

• **I don't remember**

P : I saw a man dying in front of a hospital

R : hit apparently by a fast moving car

Q : but when I left lucknow in 1984

S : the exact date

The Proper sequence should be: A. SQRP B. SQPR C. QRPS D. SPRQ → B

• **It was**

P : in keeping with my mood

R : as I walked sedately

Q : a soft summer evening

S : in the direction of new house

The Proper sequence should be: A. SRPQ B. QRPS C. QPRS D. SQPR → C

• **All such students**

P : get success

R : and recognition

Q : who are honest and hard-working

S : in very walk of life

The Proper sequence should be: A. SPRQ B. PRQS C. QPRS D. PSRQ → C

• **The master**

P : who was very loyal to him

R : without giving any valid reason

Q : punished the servant

S : when he left the work unfinished

The Proper sequence should be: A. RQPS B. RQSP C. QPSR D. QRPS → C

• **With her body**

P : dragging her unwilling feet

R : doubled with age

Q : weak and infirm

S : she persisted in her mission

The Proper sequence should be: A. PQRS B. QPRS C. RQPS D. SRPQ → C

• **We must read**

P : if we want to absorb the fruits of great literature

R : them not as we do cricket stories

The Proper sequence should be: A. QPSR

B. PSQR

Q : but with concentration

S : undefined

C. PRSQ

D. PRQS → A

• **I enclose**

P : and the postage

Q : a postal order

R : the price of books

S : which will cover

The Proper sequence should be:

A. RPSQ

B. QSPR

C. QSRP

D. QPSR → C

Sub-chapter 5 - Logical Sequence of Words

• **In this type of question, some words are given. we have to arrange these words in a meaningful order. The order may be according to age, size and need etc.**

- 1. Death 2. Marriage 3. Education 4. Birth 5. Funeral → 4, 3, 2, 1, 5
- 1. Grass 2. Curd 3. Milk 4. Cow 5. Butter → 4, 1, 3, 2, 5
- 1. Key 2. Door 3. Lock 4. Room 5. Switch on → 1, 3, 2, 4, 5
- 1. Word 2. Paragraph 3. Sentence 4. Letters 5. Phrase → 4, 1, 5, 3, 2
- 1. Police 2. Punishment 3. Crime 4. Judge 5. Judgement → 3, 1, 4, 5, 2
- 1. Family 2. Community 3. Member 4. Locality 5. Country → 3, 1, 2, 4, 5
- 1. Poverty 2. Population 3. Death 4. Unemployment 5. Disease → 2, 4, 1, 5, 3
- 1. Leaf 2. Fruit 3. Stem 4. Root 5. Flower → 4, 3, 1, 5, 2
- 1. Nation 2. Village 3. City 4. District 5. State → 2, 3, 4, 5, 1
- 1. Caste 2. Family 3. Newly married Couple 4. Clan 5. Species → 3, 2, 1, 4, 5
- 1. Cut 2. Put on 3. Mark 4. Measure 5. Tailor → 4, 3, 1, 5, 2
- 1. House 2. Street 3. Room 4. Town 5. District → 3, 1, 2, 4, 5
- 1. Wall 2. Clay 3. House 4. Room 5. Bricks → 2, 5, 1, 4, 3
- 1. Probation 2. Interview 3. Selection 4. Appointment 5. Advertisement
- 6. Application → 5, 6, 2, 3, 4, 1
- 1. Elephant 2. Cat 3. Mosquito 4. Tiger 5. Whale → 3, 2, 4, 1, 5
- 1. Yarn 2. Plant 3. Saree 4. Cotton 5. Cloth → 2, 4, 1, 5, 3
- 1. Infant 2. Old 3. Adult 4. Adolescent 5. Child → 1, 5, 4, 3, 2
- 1. Leaves 2. Branch 3. Flower 4. Tree 5. Fruit → 4, 2, 1, 3, 5
- 1. Tree 2. Seed 3. Flowers 4. Fruit 5. Plant → 2, 5, 1, 3, 4
- 1. Adult 2. Child 3. Infant 4. Boy 5. Adolescent → 3, 2, 4, 5, 1
- 1. Protect 2. Pressure 3. Relief 4. Rain 5. Flood → 2, 4, 5, 1, 3
- 1. Presentation 2. Recommendation 3. Arrival 4. Discussion 5. Introduction → 3, 5, 1, 4, 2
- 1. Income 2. Status 3. Education 4. Well-being 5. Job → 3, 5, 1, 2, 4
- 1. Heel 2. Shoulder 3. Skull 4. Neck 5. Knee 6. Chest
- 7. Thigh 8. Stomach 9. Face 10. Hand → 3, 9, 4, 2, 10, 6, 8, 7, 5, 1
- 1. Rainbow 2. Rain 3. Sun 4. Happy 5. Child → 2, 3, 1, 5, 4
- 1. Never 2. Sometimes 3. Generally 4. Seldom 5. Always → 5, 3, 2, 4, 1
- 1. Index 2. Contents 3. Title 4. Chapters 5. Introduction → 3, 2, 5, 4, 1
- 1. Table 2. Tree 3. Wood 4. Seed 5. Plant → 4, 5, 2, 3, 1
- 1. Windows 2. Walls 3. Floor 4. Foundation 5. Roof 6. Room → 4, 2, 1, 5, 3, 6
- 1. Rain 2. Monsoon 3. Rescue 4. Flood 5. Shelter 6. Relief → 2, 1, 4, 3, 5, 6
- 1. Foetus 2. Child 3. Baby 4. Adult 5. Youth → 1, 3, 2, 5, 4
- 1. Sea 2. Rivulet 3. Ocean 4. River 5. Glacier → 5, 2, 4, 1, 3
- 1. Reading 2. Composing 3. Writing 4. Printing → 3, 2, 4, 1
- 1. Hecto 2. Centi 3. Deca 4. Kilo 5. Deci → 2, 5, 3, 1, 4
- 1. Honey 2. Flower 3. Bee 4. Wax → 2, 3, 1, 4
- 1. Country 2. Furniture 3. Forest 4. Wood 5. Trees → 1, 3, 5, 4, 2
- 1. Site 2. Plan 3. Rent 4. Money 5. Building 6. Construction → 4, 1, 2, 6, 5, 3
- 1. Cutting 2. Dish 3. Vegetable 4. Market 5. Cooking → 4, 3, 1, 5, 2
- 1. Study 2. Job 3. Examination 4. Earn 5. Apply → 1, 3, 5, 2, 4
- 1. Atomic Age 2. Metallic Age 3. Stone Age 4. Alloy Age → 3, 2, 4, 1
- 1. Mother 2. Child 3. Milk 4. Cry 5. Smile → 2, 4, 1, 3, 5

- 1. Post-box 2. Letter 3. Envelope 4. Delivery 5. Clearance → 3, 2, 1, 5, 4

Sub-chapter 6 - Essentia Parts

Find the word that names a necessary part of the given word.

- Harvest** A. autumn B. stockpile C. tractor D. crop → D
- Desert** A. cactus B. arid C. oasis D. flat → B
- Language** A. tongue B. slang C. writing D. words → D
- School** A. student B. report card C. test D. learning → A
- Gala** A. celebration B. tuxedo C. appetizer D. orator → A
- Monopoly** A. corrupt B. exclusive C. rich D. gigantic → B
- Lightning** A. electricity B. thunder C. brightness D. rain → A
- Ovation** A. outburst B. bravo C. applause D. encore → C
- Bonus** A. reward B. raise C. cash D. employe → A
- Antique** A. rarity B. artifact C. aged D. prehistoric → C
- Domicile** A. tenant B. dwelling C. kitchen D. house → B
- Itinerary** A. map B. route C. travel D. guidebook → B
- Orchestra** A. violin B. stage C. musician D. soloist → C
- Facsimile** A. picture B. image C. mimeograph D. copier → B
- Provisions** A. groceries B. supplies C. gear D. caterers → B
- Sustenance** A. nourishment B. water C. grains D. menu → A
- Infirmity** A. surgery B. disease C. patient D. receptionist → C
- Purchase** A. trade B. money C. bank D. acquisition → D
- Wedding** A. love B. church C. ring D. marriage → D
- Faculty** A. buildings B. textbooks C. teachers D. meetings → C
- Recipe** A. desserts B. directions C. cookbook D. utensils → B
- Saddle** A. horse B. seat C. stirrups D. horn → B
- Dome** A. rounded B. geodesic C. governmental D. coppery → A
- Directory** A. telephone B. listing C. computer D. names → B

Sub-chapter 7 - Word Formation Test

- Word formation using letters of given word. Careful on which letter is not there.
- Cannot be use one letter into more times.
- If A = 1, P = 2, L = 3, E = 4, T = 5 and S = 6, then a series of the following letters does not make a meaningful word. Find it? → 25461

Sub-chapter 8 - Blood relations

Father of grandfather or grandmother	Great grandfather	Daughter of father or mother	Sister
Mother of grandfather or grandmother	Great grandmother	Son of second wife of father	Step brother
Father of father or mother	Grandfather	Daughter of second wife of father	Step sister
Mother of father or mother	Grandmother	Son/daughter of uncle/aunt	Cousin
Wife of grandfather	Grandmother	Brother of husband or wife	Brother-in-law
Husband of grandmother	Grandfather	Sister of husband or wife	Sister-in-law
Father-in-law of father/mother	Grandfather	Husband of sister/sister-in-law	Brother-in-law
Mother-in-law of father/mother	Grandmother	Son of father	Oneself/Brother
Father's father/mother only	Father	Mother of son/daughter	Oneself/Wife

Only daughter-in-law of father's father/father's mother	Mother	Father of daughter/son	Oneself/husband
Husband of mother	Father	Son of son of grandmother/grandfather	Brother/Oneself /Cousin
Wife of father	Mother	Daughter of son of grandmother/grandfather	Cousin/Oneself /Sister
Second wife of father	Step mother	Son of brother or sister	Nephew
Brother of father	Uncle	Daughter of brother/sister	Niece
Brother of mother	Maternal Uncle	Grandson of father/mother	Son/Nephew
Sister of father	Aunt	Granddaughter of father/mother	Daughter or Niece
Sister of mother	Maternal Aunt	Husband of daughter	Son-in-law
Husband of aunt	Uncle	Wife of brother/brother-in-law	Sister-in-law
Wife of uncle	Aunt	Wife of son	Daughter-in-law
Son of grandfather/grandmother	Father/Uncle	Son of son/Daughter	Grandson
Daughter of father-in-law/mother-in-law of father	Mother/Aunt	Daughter of son/Daughter	Granddaughter
Father of wife/husband	Father-in-law	Son's/Daughter's grandson	Great Grandson
Mother of wife/husband	Mother-in-law	Son's/Daughter's granddaughter	Great granddaughter
Children of same parents	Siblings		
Father's/Mother's only son/daughter	Oneself		
Son of father or mother	Brother		

- Indirect Relations
- Direct Relations
- Coded Blood Relations